

Chapter 1

Representation of Discrete Signals and Systems



Learning Objectives

After completing this chapter, you should be able to:

- ✕ define various terminologies related to signals and systems.
- ✕ classify signals and systems.
- ✕ give mathematical description and representation of signals and systems.
- ✕ perform basic operations on DT signals.
- ✕ classify DT signals as periodic and non-periodic, odd and even and power and energy signals.
- ✕ classify DT systems.
- ✕ obtain the step response, causality and stability of DT systems.
- ✕ describe sampling techniques, quantization, Nyquist rate and aliasing effect.

1.1 Introduction

Most of the signals encountered in science and engineering are analog in nature. That is, the signals are functions of a continuous variable, such as time or space, and usually take on values in a continuous range. Such signals may be processed directly by appropriate analog systems (such as filters or frequency analyzers or frequency multipliers) for the purpose of changing their characteristics or extracting some desired information. In such a case, the signal has been processed directly in its analog form. Both the input signal and the output signal are in analog form and are shown in Fig. 1.1a.

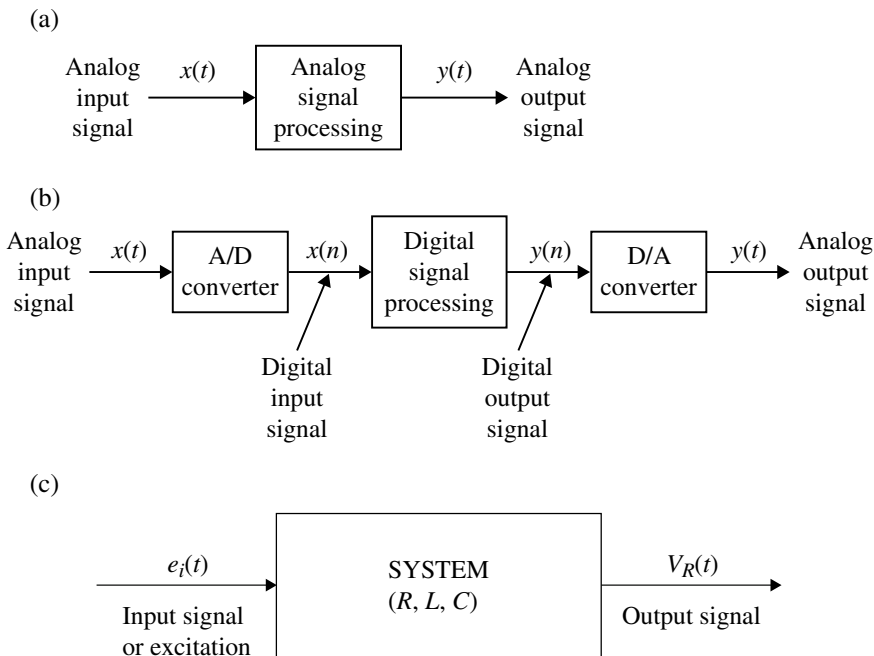


Fig. 1.1 **a** Analog signal processing **b** Digital signal processing **c** Block diagram representation of signals and systems

Digital signal processing provides an alternative method for processing the analog signal and is shown in Fig. 1.1b. To perform processing digitally, there is a need for an interface between the analog signal and the digital processor. This interface is called an analog to digital (A/D) converter. The output of A/D converter is a digital signal that is applied as an input to the digital processor.

The digital signal processor may be a large programmable digital computer or a small microprocessor programmed to perform the desired operations on the input signal. Programmable machines provide the flexibility to change the signal processing operation through a change in the software, whereas hardwired machines are difficult to reconfigure. In an application where the digital output from the digital signal processor is to be given to the user in analog form, we must provide an interface and this is called a digital to analog (D/A) converter. Thus, the signal is provided to the user in analog form.

Digital signal processing has developed very rapidly over the past five decades mainly due to the advances in digital computer technology and very large-scale integrated electronic circuits. These inexpensive smaller but faster and more powerful digital computers are capable of performing very complex signal processing func-

tions which are usually too difficult to perform by analog circuitry. The following are the advantages of digital signal processing (DSP) over analog processing:

1. Flexibility : Digital programmable systems allow flexibility in reconfiguring the DSP operations by simply changing the program.
2. Accuracy : DSP provides better control of accuracy requirements, while tolerance limits have to be met in the analog counterpart.
3. Easy storage : Digital signals can be easily stored in magnetic media without deterioration or loss of signal fidelity. They can also be easily transportable and can be processed off-time in remote laboratories.
4. Processing : DSP allows for the implementation of more sophisticated signal processing than its analog counterpart.
5. Cost effective : With advancement in VLSI technology digital implementation of the signal process system is cheaper.

The limitation of DSP is that the conversion speed of ADC and the processing speed of signal processors should be very high to perform real time processing. Signals of high bandwidth require fast sampling rate ADCs and fast processors.

Some of the applications of the digital signal processors are speech processing, signal transmission on telephone channels, image processing, biomedical, seismology and consumer electronics.

1. Speech processing : Speech compression and decompression for voice storage system and for transmission and reception of voice signals.
2. Communication : Elimination of noise by filtering and echo cancellation by adaptive filtering in transmission channels.
3. Biomedical : Spectrum analysis of ECG signals to identify various disorders in the heart.
Spectrum analysis of EEG signals to study the malfunction or disorders in the brain.
4. Consumer electronics : Music synthesis, digital audio and video.
5. Seismology : Spectrum analysis of seismic signals can be used to predict the earthquake, nuclear explosions and earth movement.
6. Image processing : Two dimensional filtering on images for image enhancement, finger print matching, identifying hidden images in the signals received by radars, etc.

The concepts of signals and systems play a very important role in many areas of science and technology. These concepts are very extensively applied in the field of circuit analysis and design, long-distance communication, power system generation and distribution, electron devices, electrical machines, biomedical engineering, aeronautics, process control, speech and image processing to mention a few. **Signals represent some independent variables that contain some information about the behaviour of some natural phenomenon.** Voltages and currents in electrical and electronic circuits, electromagnetic radio waves, human speech and sounds produced

by animals are some of the examples of signals. **When these signals are operated on some objects, they give out signals in the same or modified form. These objects are called systems.** A system is, therefore, defined as the interconnection of objects with a definite relationship between objects and attributes. Signals appearing at various stages of the system are attributes. R , L , C components, spring, dash-pots, mass, etc. are the objects. The electrical and electronic circuits comprising of R , L , C components and amplifiers, the transmitter and receiver in a communication system, the petrol and diesel engines in an automobile, chemical plants, nuclear reactor, human beings, animals, a government establishment, etc. are all examples of systems. **In this book, we deal with only discrete signals and systems.**

1.2 Terminologies Related to Signals and Systems

Before we give mathematical descriptions and representations of various terminologies related to signals and systems, the following terminologies which are very frequently used are defined as follows.

1.2.1 *Signal*

A signal is defined as a physical phenomenon that carries some information or data. The signals are usually functions of independent variable time. There are some cases where the signals are not functions of time. The electrical charge distributed in a body is a signal which is a function of space and not time.

1.2.2 *System*

A system is defined as a set of interconnected objects with a definite relationship between objects and attributes. The interconnected components provide the desired function.

Objects are parts or components of a system. For example, switches, springs, masses, dash-pots, etc. in a mechanical systems, and inductors, capacitors and resistors in an electrical system are the objects. The displacement of mass, spring and dash pot and the current flow and the voltage across the inductor, capacitor and resistor are the attributes. There is a definite relationship between the objects and attributes. The voltages across R , L , C series components can be expressed as $V_R = iR$; $V_L = L \frac{di}{dt}$ and $V_C = \frac{1}{C} \int i dt$. If this series circuit is excited by the voltage source $e_i(t)$, the $e_i(t)$ is the input attribute or the input signal. If the voltage across any of the objects R , L and C is taken, then such an attribute is called the output signal. The block diagram

representation of input and output (voltage across the resistor) signals and the system is shown in Fig. 1.1c.

1.3 Continuous and Discrete Time Signals

Signals are broadly classified as:

1. Continuous time signal (CT signal).
2. Discrete time signal (DT signal).
3. Digital signal.

The signal that is specified for every value of time t is called continuous time signal and is denoted by $x(t)$. On the other hand, the signal that is specified at the discrete value of time is called the discrete time signal. The discrete time signal is represented as a sequence of numbers and is denoted by $x[n]$, where n is an integer. Here time t is divided into n discrete time intervals. The continuous time signal (CT) and discrete time signal (DT) are represented in Figs. 1.2 and 1.3, respectively.

It is to be noted that, in continuous time signal representation, the independent variable t , which has unit as s, is put in the parenthesis ($\cdot$), and in discrete time signal, the independent variable n , which is an integer, is put inside the square parenthesis $[\cdot]$. Accordingly, the dependent variables of the continuous time signal/system are denoted as $x(t)$, $g(t)$, $u(t)$, etc. Similarly, the dependent variables of discrete time signals/systems are denoted as $x[n]$, $g[n]$, $u[n]$, etc.

A discrete time signal $x[n]$ is represented by the following two methods:

1.

$$x[n] = \begin{cases} \left(\frac{1}{a}\right)^n & n \geq 0 \\ 0 & n < 0 \end{cases} \quad (1.1)$$

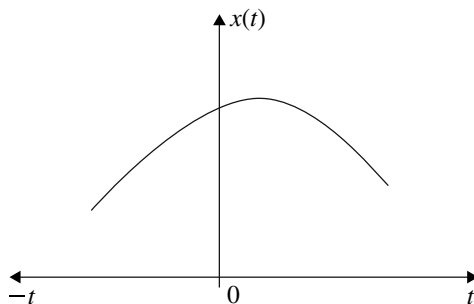


Fig. 1.2 CT signal

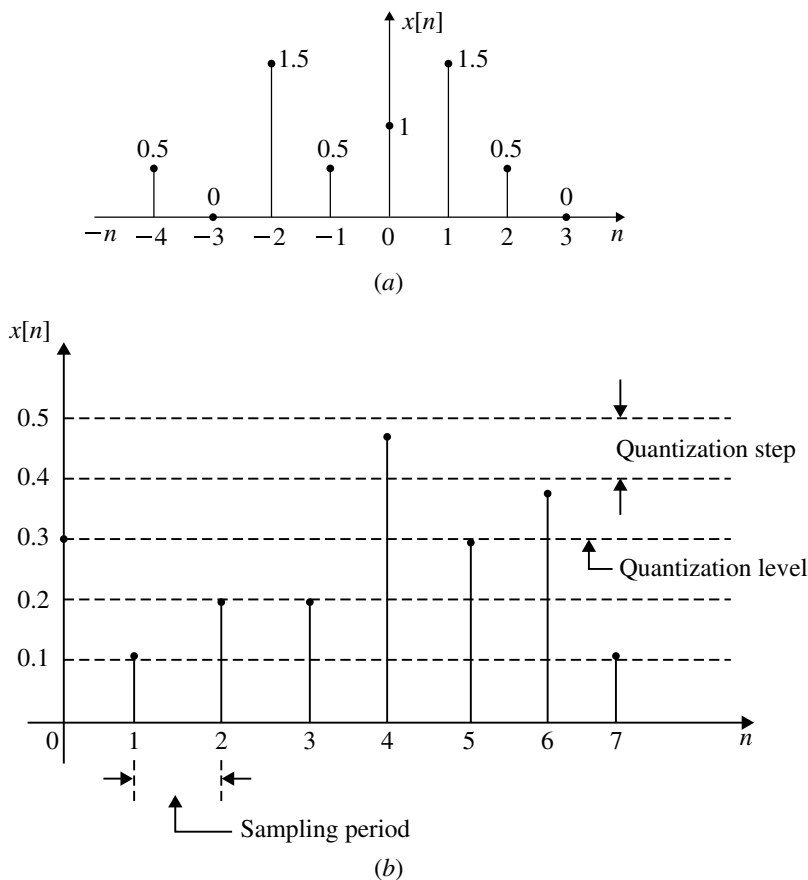


Fig. 1.3 **a** Discrete time (DT) signal **b** Digital signal

Substituting various values of n where $n \geq 0$ in Equation (1.1), the sequence for $x[n]$, which is denoted by $x\{n\}$, is written as follows:

$$x\{n\} = \left\{ 1, \frac{1}{a}, \frac{1}{a^2}, \dots, \frac{1}{a^n} \right\}$$

2. The sequence is also represented as

$$x[n] = \{3, 2, 5, 4, 6, 8, 2\}$$

↑

The arrow indicates the value of $x[n]$ at $n = 0$, which is 5 in this case. The numbers to the left of the arrow indicate the negative sequence $n = -1, -2$, etc.

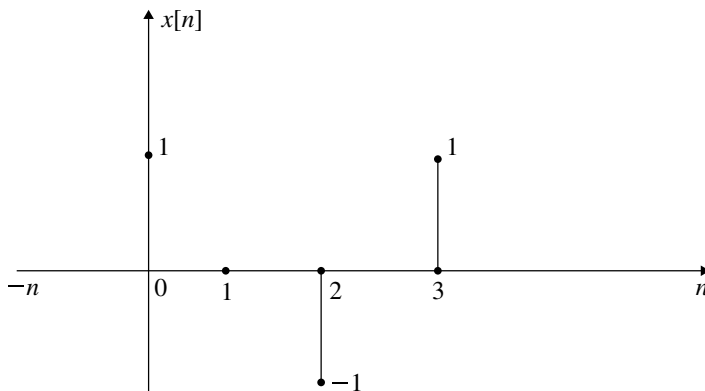


Fig. 1.4 Graphical representation of $x[n]$

The numbers to the right of the arrow correspond to $n = 1, 2, 3, 4$, etc. Thus, for the above sequence $x[-1] = 2$; $x[-2] = 3$; $x[0] = 5$; $x[1] = 4$; $x[2] = 6$; $x[3] = 8$ and $x[4] = 2$. If no arrow is marked for a sequence, the sequence starts from the first term in the extreme left. Consider the sequence

$$x\{n\} = \{5, 3, 4, 2\}$$

Here $x[0] = 5$; $x[1] = 3$; $x[2] = 4$ and $x[3] = 2$. There is no negative sequence here.

A digital signal is not very much different from a discrete signal except that a digital signal amplitude is quantized at a certain specific level. This is because a digital computer can accept only sequences of numbers that are expressed in terms of bits. The representation of the digital signal is shown in Fig. 1.3b. The amplitude at any discrete time interval can be only one of quantization levels. If the amplitude is chosen from a finite set of numbers, the amplitude is said to be discretized. The quantization level is equidistance and is called quantization step. An analog signal can be converted into a digital signal by means of sampling and quantizing. It is then digitized by rounding off its value to the closest permissible level. Thus, error exists when an analog signal is discretized and quantized. This is called quantization error. The digital signal is, therefore, discretized in time and quantized in amplitude.

Example 1.1 Graphically represent the following sequence:

$$x\{n\} = \{1, 0, -1, 1\}$$

Solution The graphical representation $x\{n\} = \{1, 0, -1, 1\}$ is shown in Fig. 1.4.

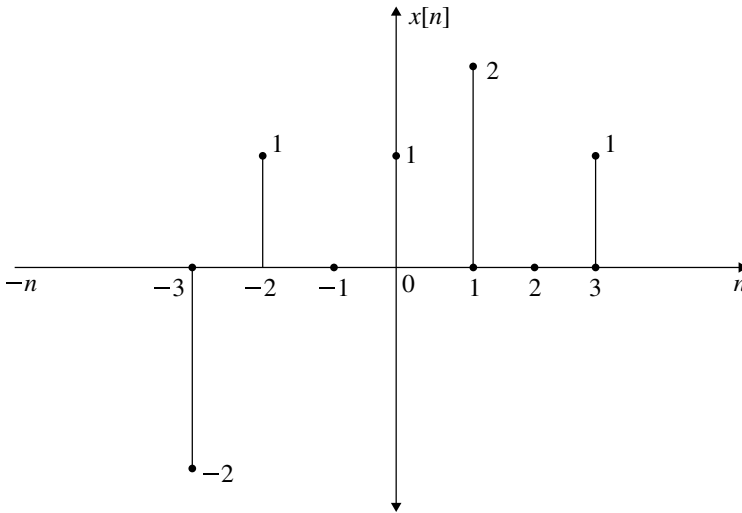


Fig. 1.5 Graphical representation of $x[n]$

Example 1.2 Graphically represent the following sequence:

$$x\{n\} = \{-2, 1, 0, 1, 2, 0, 1\}$$

↑

Solution The sequence

$$x\{n\} = \{-2, 1, 0, 1, 2, 0, 1\}$$

↑

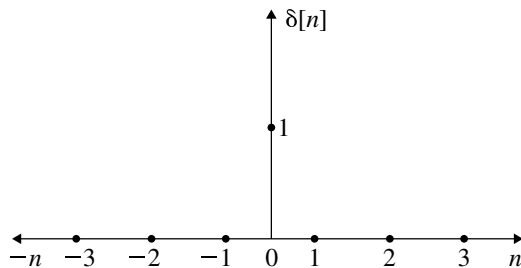
is represented in Fig. 1.5.

1.4 Basic Discrete Time Signals

Similar to continuous time signals, basic discrete signals are available. However, these signals are represented at discrete intervals of time ' n ' where n is an integer. Representation of basic discrete time signals is discussed below.

1.4.1 The Unit Impulse Sequence

The basic impulse sequence is shown in Fig. 1.6. The unit impulse sequence also called the sample is defined as

**Fig. 1.6** Basic unit impulse sequence

$$\delta[n] = \begin{cases} 1 & n = 0 \\ 0 & n \neq 0 \end{cases} \quad (1.2)$$

1.4.2 The Basic Unit Step Sequence

The basic unit step sequence is represented in Fig. 1.7. It is denoted by $u(n)$. It is defined as

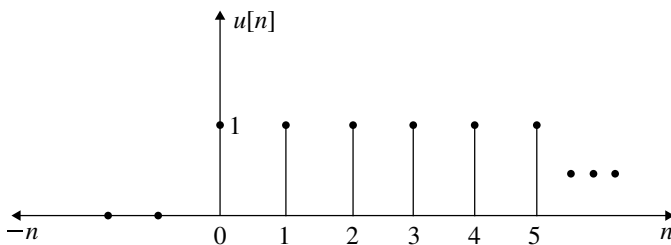
$$u[n] = \begin{cases} 1 & n \geq 0 \\ 0 & n < 0 \end{cases} \quad (1.3)$$

Any discrete sequences $x[n]$ for $n \geq 0$ is expressed as $x[n]u[n]$. For $n < 0$, it is expressed as $x[n]u[-n]$. It is noted that at $n = 0$, the value of $u[n] = 1$.

1.4.3 The Basic Unit Ramp Sequence

The basic unit ramp sequence which is denoted by $r[n]$ is represented in Fig. 1.8. It is defined as

$$r[n] = \begin{cases} n & n \geq 0 \\ 0 & n < 0 \end{cases} \quad (1.4)$$

**Fig. 1.7** Basic unit step sequence

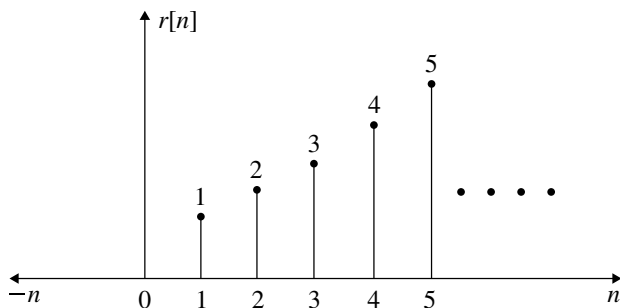


Fig. 1.8 Basic unit ramp sequence

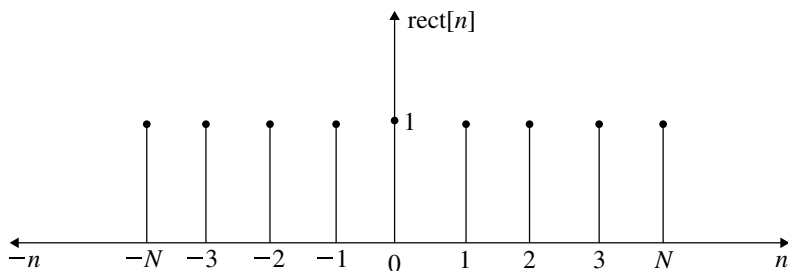


Fig. 1.9 Unit rectangular sequence

1.4.4 Unit Rectangular Sequence

The discrete time unit rectangular sequence is shown in Fig. 1.9. It is defined as

$$\text{rect}[n] = \begin{cases} 1 & |n| \leq N \\ 0 & |n| > N \end{cases} \quad (1.5)$$

The above equation can also be expressed as

$$\text{rect}[n] = 1 \quad -N \leq n \leq N.$$

1.4.5 Sinusoidal Sequence

The discrete time sinusoidal signal is defined by the following mathematical expression:

$$x[n] = Ae^{-an} \sin(\omega_0 n + \phi) \quad (1.6)$$

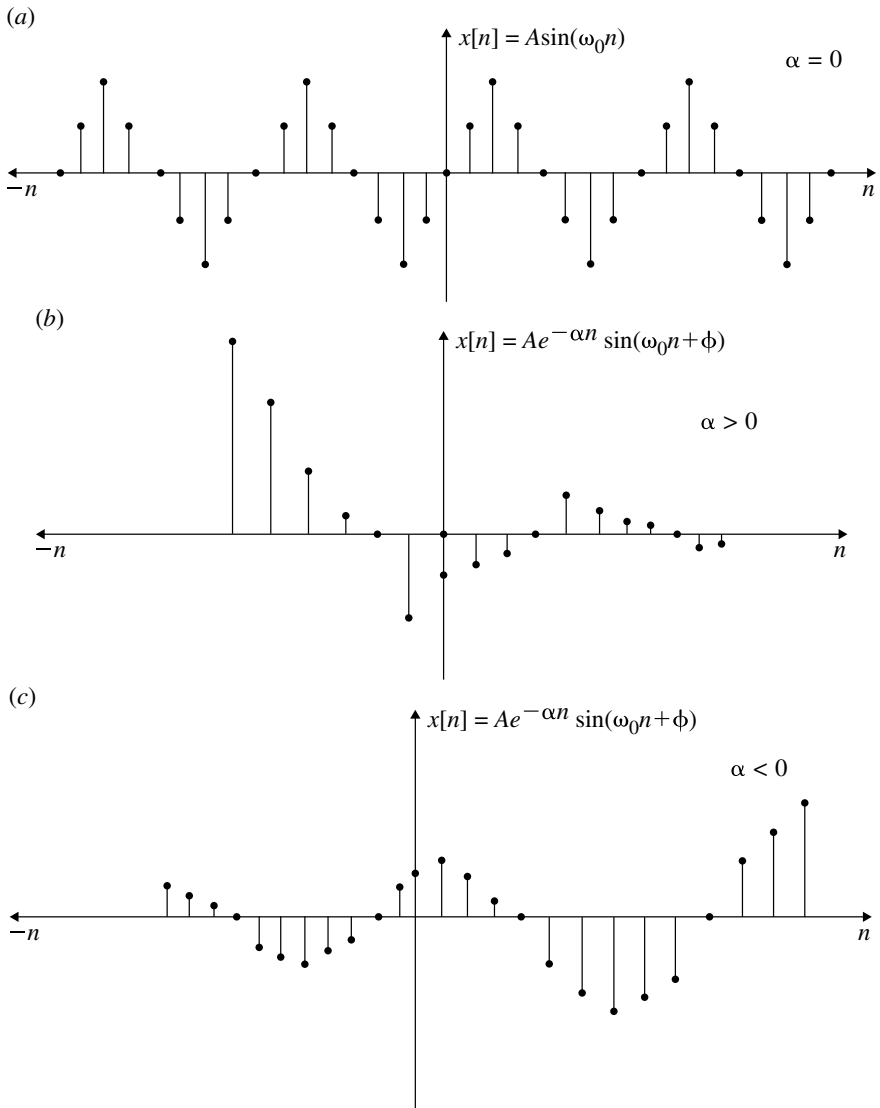


Fig. 1.10 Discrete time sinusoidal signal. **a** Purely sinusoidal; **b** Decaying sinusoidal; **c** Growing sinusoidal

where A and α are real numbers and ϕ is the phase shift. Depending on the value of α , the sinusoidal sequence is divided into the following categories:

- A purely sinusoidal sequence ($\alpha = 0$).
- Decaying sinusoidal sequence ($\alpha > 0$).
- Growing sinusoidal sequence ($\alpha < 0$).

The above sinusoidal sequences are illustrated in Fig. 1.10a, b and c, respectively.

1.4.6 Discrete Time Real Exponential Sequence

The general complex exponential sequence is defined as

$$x[n] = A\alpha^n \quad (1.7)$$

where A and α are in general complex numbers.

In Eq. (1.7), if A and α are real, the sequence is called real exponential. These sequences for various values of α are shown in Fig. 1.11. Depending on the value of α , the sequence is classified as:

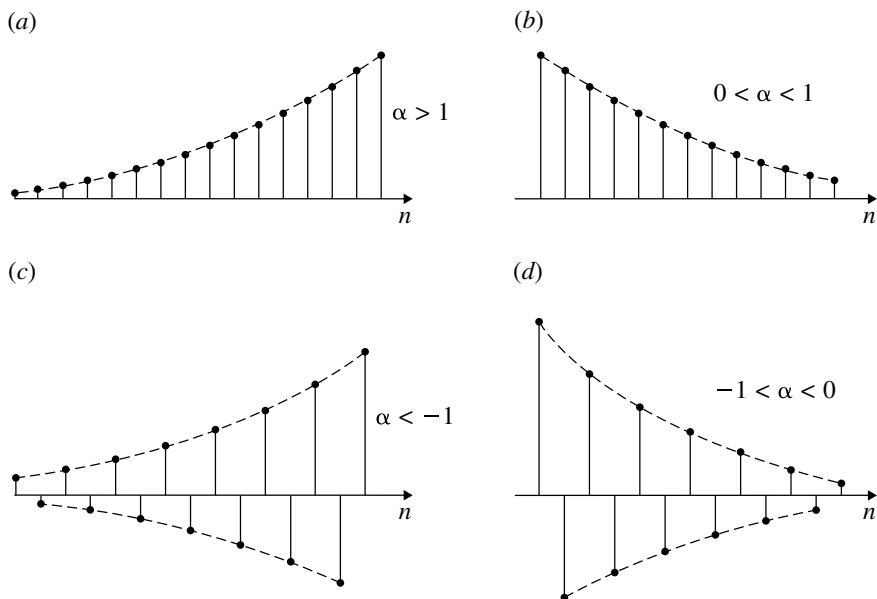


Fig. 1.11 Discrete time real exponential sequences. **a** $\alpha > 1$; **b** $0 < \alpha < 1$; **c** $\alpha < -1$; **d** $-1 < \alpha < 0$

1. Exponentially growing signal ($\alpha > 1$, Fig. 1.11a).
2. Exponentially decaying signal ($0 < \alpha < 1$, Fig. 1.11b).
3. Exponentially growing for the alternate value of n ($\alpha < -1$, Fig. 1.11c).
4. Exponentially decaying for the alternate value of n ($-1 < \alpha < 0$, Fig. 1.11d).

1.5 Basic Operations on Discrete Time Signals

The basic operations that are applied to continuous time signals are also applicable to discrete time signals. The time t in CT signal is replaced by n in DT signals. The basic operations as applied to DT signals are explained below.

1.5.1 Addition of Discrete Time Sequence

Addition of discrete time sequence is done by adding the signals at every instant of time. Consider the signals $x_1[n]$ and $x_2[n]$ shown in Fig. 1.12a and b respectively. The addition of these signals at every n is done and represented as $y[n] = x_1[n] + x_2[n]$. This is shown in Fig. 1.12c.

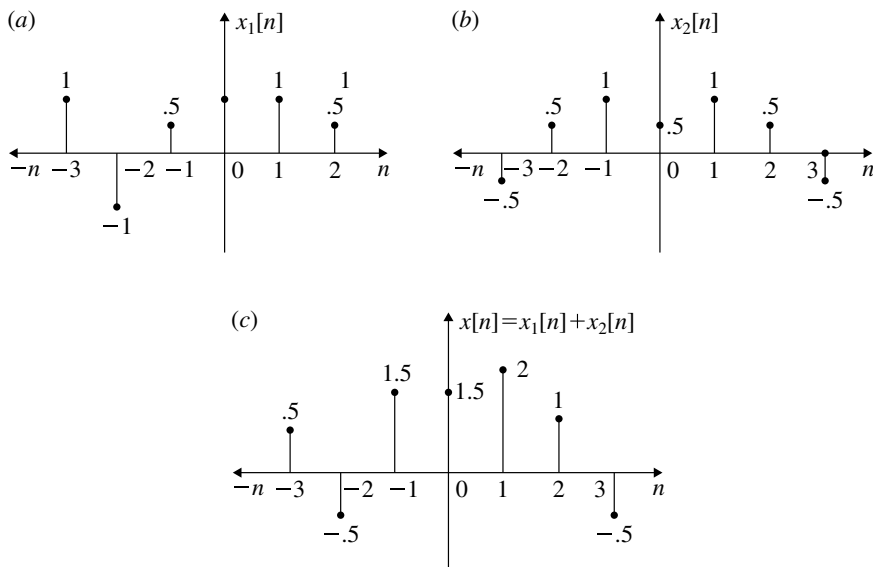


Fig. 1.12 Addition of DT signals

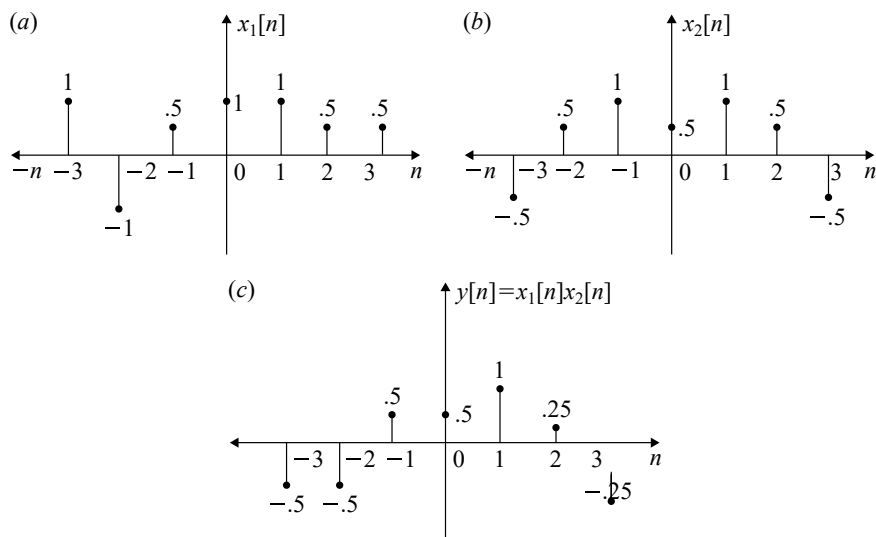


Fig. 1.13 Multiplications of two DT signals

1.5.2 Multiplication of DT Signals

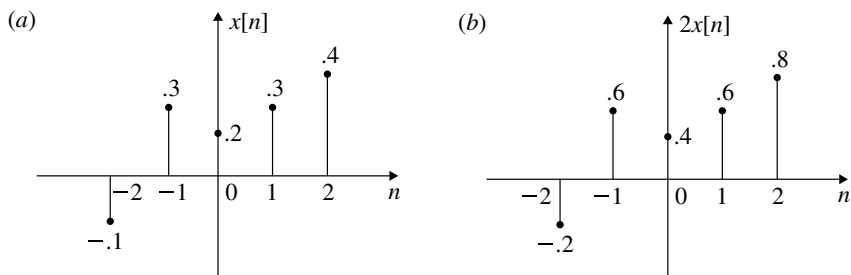
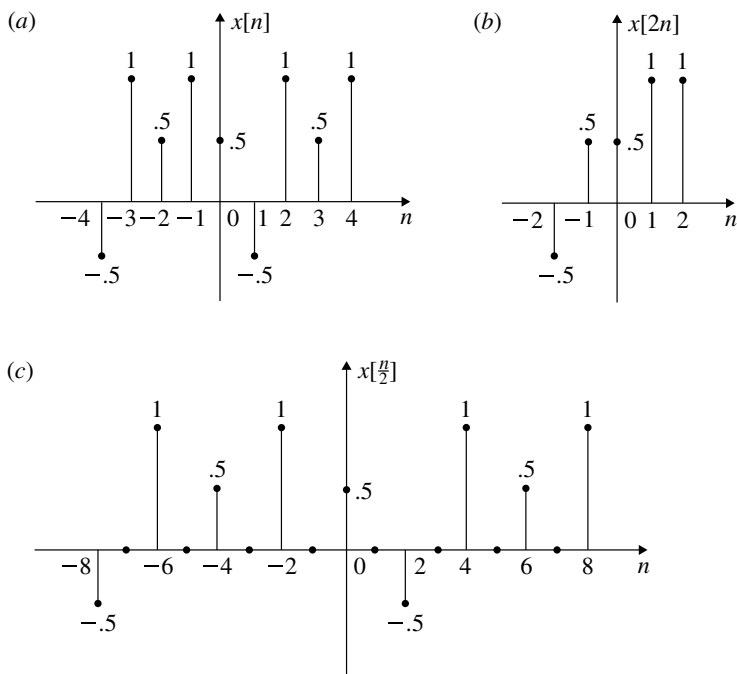
The multiplication of two DT signals $x_1[n]$ and $x_2[n]$ is obtained by multiplying the signal values at each instant of time n . Consider the signal $x_1[n]$ and $x_2[n]$ represented in Fig. 1.13a and b. At each instant of time n , the samples of $x_1[n]$ and $x_2[n]$ are multiplied and represented as shown in Fig. 1.13c.

1.5.3 Amplitude Scaling of DT Signal

Let $x[n]$ be a discrete time signal. The signal $Ax[n]$ is represented by multiplying the amplitude of the sequence by A at each instant of time n . Consider the signal $x[n]$ shown in Fig. 1.14a. The signal $2x[n]$ is represented and shown in Fig. 1.14b.

1.5.4 Time Scaling of DT Signal

The time compression or expansion of a DT signal in time is known as time scaling. Consider the signal $x[n]$ shown in Fig. 1.15a. The time compressed signal $x[2n]$ and time expanded signal $x[\frac{n}{2}]$ are shown in Fig. 1.15b and c, respectively. One should note that while doing compression and expansion of DT signal, **only for integer**

**Fig. 1.14** Amplitude scaling of DT signals**Fig. 1.15** Time scaling of DT signal

value of n the samples exist. For non-integer value of n , the samples do not exist.

Time Compression

Let

$$\begin{aligned}
 y[n] &= x[2n] \\
 y[-2] &= x[-4] = -0.5 \\
 y[-1] &= x[-2] = 0.5 \\
 y[0] &= x[0] = 0.5
 \end{aligned}$$

$$y[1] = x[2] = 1$$

$$y[2] = x[4] = 1$$

The plot of $x[2n]$ is shown in Fig. 1.15b.

Time Expansion

Let

$$\begin{aligned} y[n] &= x\left[\frac{n}{2}\right] \\ y[-8] &= x[-4] = -0.5 \\ y[-6] &= x[-3] = 1 \\ y[-4] &= x[-2] = 0.5 \\ y[-2] &= x[-1] = 1 \\ y[0] &= x[0] = 0.5 \\ y[2] &= x[1] = -0.5 \\ y[4] &= x[2] = 1 \\ y[6] &= x[3] = 0.5 \\ y[8] &= x[4] = 1 \end{aligned}$$

The plot of $x[\frac{n}{2}]$ is shown in Fig. 1.15c.

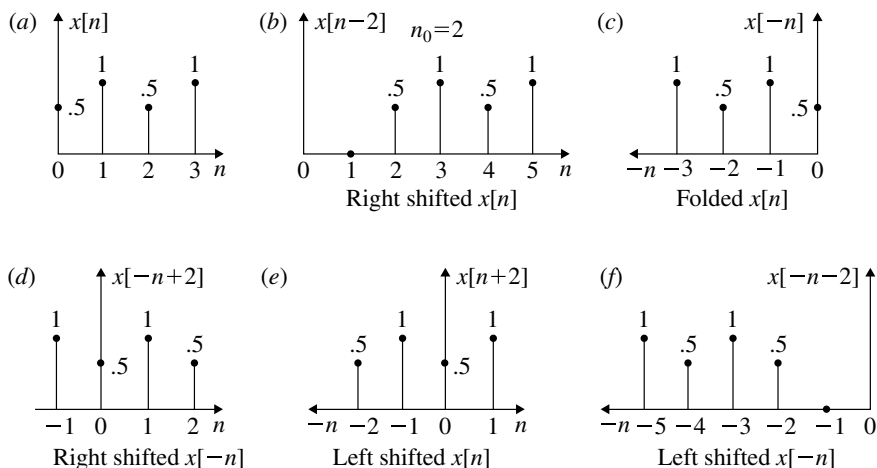
1.5.5 Time Shifting of DT Signal

As in the case of CT signal, time shifting property is applied to DT signal also. Let $x[n]$ be the DT signal. Let n_0 be the time by which $x[n]$ is time shifted. Since n is an integer, n_0 is also an integer. The following points are applicable while DT signal is time shifted:

- For the DT signals $x[-n - n_0]$ and $x[n + n_0]$, the signals $x[-n]$ and $x[n]$ are to be left shifted by n_0 .
- For the DT signals $x[n - n_0]$ and $x[-n + n_0]$, the signals $x[n]$ and $x[-n]$ are to be right shifted by n_0 .

Figure 1.16 shows the time shifting of the DT signal.

In Fig. 1.16a, the sequence $x[n]$ is shown. The sequence $x[n - 2]$ which is right shifted by 2 samples is shown in Fig. 1.16b. $x[-n]$ which is the folded signal is shown in Fig. 1.16c. $x[-n + 2]$ which is left shifted of $x[-n]$ is shown in Fig. 1.16d. $x[n + 2]$ which is right shifted of $x[n]$ is shown in Fig. 1.16e. $x[-n - 2]$ which is left shifted of $x[-n]$ is shown in Fig. 1.16f.

**Fig. 1.16** Time shifting of DT signal

1.5.6 Multiple Transformation

The transformations, namely amplitude scaling, time reversal, time shifting, time scaling, etc. are applied to represent DT sequence. The sequence of operation of these transformations is important and followed as described below.

Consider the following DT signal:

$$y[n] = Ax \left[-\frac{n}{a} + n_0 \right]$$

1. Plot $x[n]$ sequence and obtain $Ax[n]$ by amplitude scaling.
2. Using time reversal (folding), plot $Ax[-n]$.
3. Using time shifting, plot $Ax[-n + n_0]$ where $n_0 > 0$. The time shift is to be right of $x[-n]$ by n_0 samples.
4. Using time scaling, plot $Ax[-\frac{n}{a} + n_0]$ where a is in integer. In the above case, keeping amplitude constant, time is expanded by a .

The following examples illustrate the above operations:

Example 1.3 Let $x[n]$ and $y[n]$ be as given in Fig. 1.17a and b, respectively. Plot

- | | |
|-----------------|---------------------------|
| (a) $x[2n]$ | (c) $x[n - 2] + y[n - 2]$ |
| (b) $x[3n - 1]$ | (d) $y[1 - n]$ |

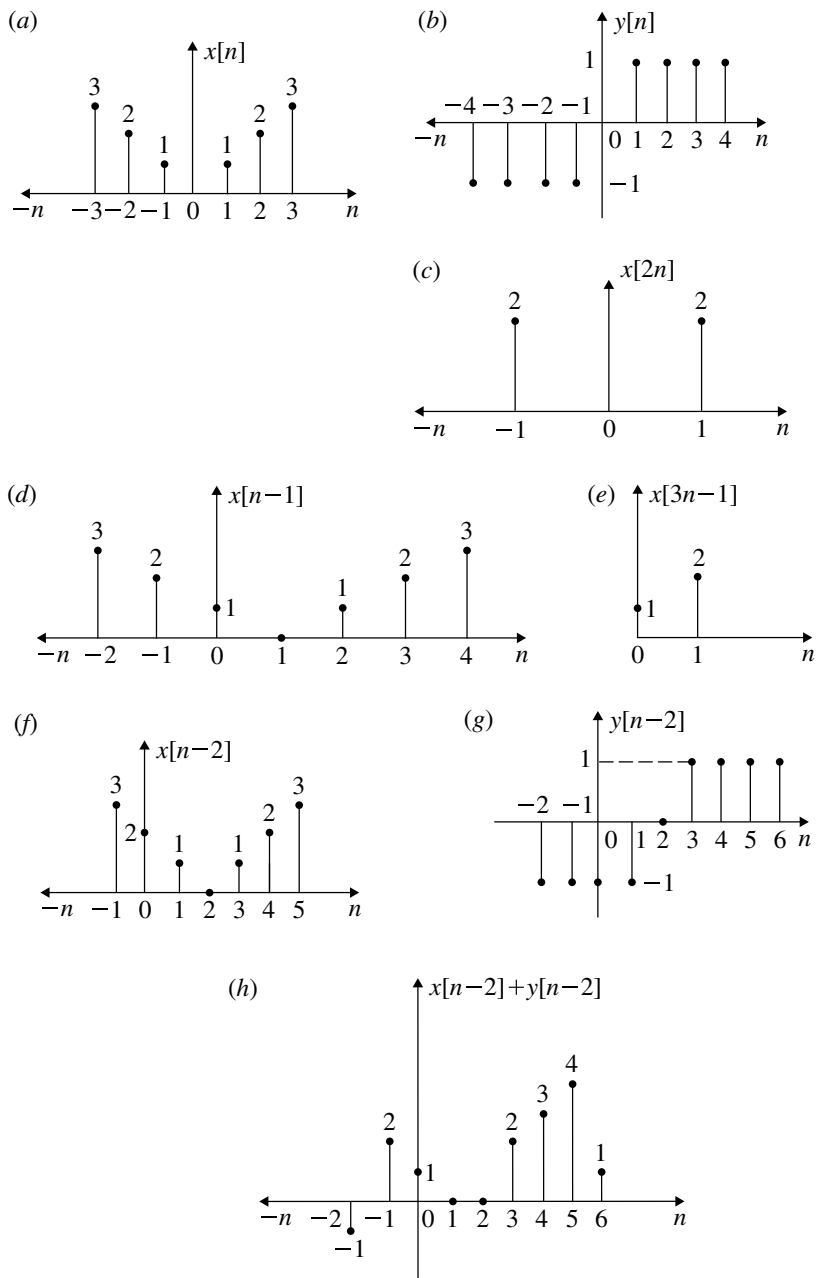
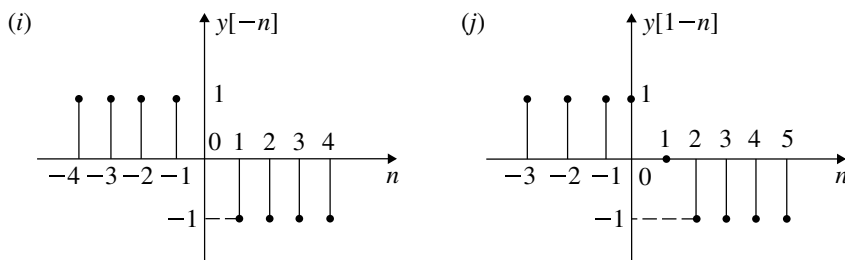
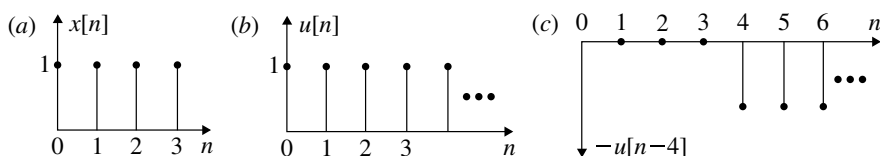


Fig. 1.17 Two discrete sequences

**Fig. 1.17** (continued)**Fig. 1.18** Sequences expressed in terms of step sequences**Solution (a) To plot $x[2n]$**

Here the DT sequence is time compressed by a factor of 2. Hence, the samples only with even numbers are divided by a factor of 2 and the corresponding amplitudes marked and shown in Fig. 1.17c. When odd values of n are divided by the factor 2, it becomes a fraction and they are skipped.

(b) To plot $x[3n - 1]$

The plot of $x[n - 1]$ is obtained by right shifting of $x[n]$ by $n_0 = 1$. This is shown in Fig. 1.17d. When $x[n - 1]$ is time compressed by a factor 3, $x[3n - 1]$ is obtained. Only integers which are divisible by 3 in the sequence $x[n - 1]$ are to be taken to plot $x[3n - 1]$. Thus, samples for $n = 0$ and $n = 3$, will be plotted as shown in Fig. 1.17e.

(c) To plot $x[n - 2] + y[n - 2]$

The sequence $x[n - 2]$ is obtained by right shifting of $x[n]$ by 2 and is shown in Fig. 1.17f. Similarly, the sequence $y[n - 2]$ is obtained by right shifting of $y[n]$ by 2 and is shown in Fig. 1.17(g). The sequence $x[n - 2] + y[n - 2]$ is obtained by summing up the sequences in Fig. 1.17f and (g) for all n and is shown in Fig. 1.17h.

(d) To plot $y[1 - n]$

The sequence $y[-n]$ is obtained by folding $y[n]$ and is shown in Fig. 1.17(i). $y[-n]$ is right shifted by 1 sample to get the sequence $y[1 - n]$. This is shown in Fig. 1.17(j).

Example 1.4 Consider the sequence shown in Fig. 1.18a. Express the sequence in terms of step function.

Solution The unit step sequence $u[n]$ is shown in Fig. 1.18b. The unit negative step sequence with a time delay of $n_0 = 4$ is shown in Fig. 1.18b. It is evident from Fig.

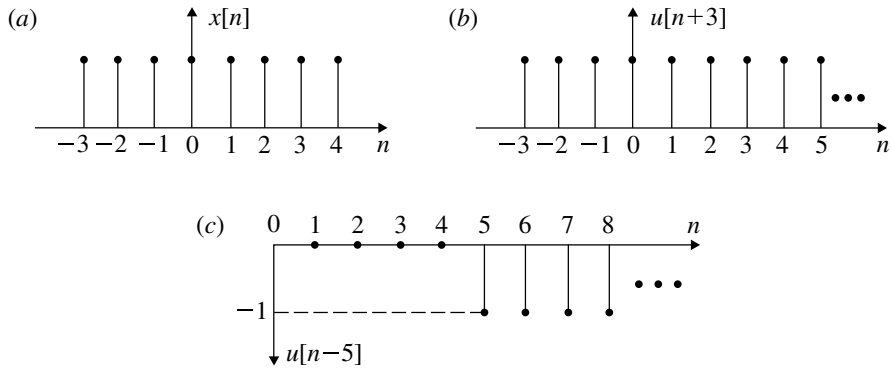


Fig. 1.19 DT sequences expressed in terms of step sequences

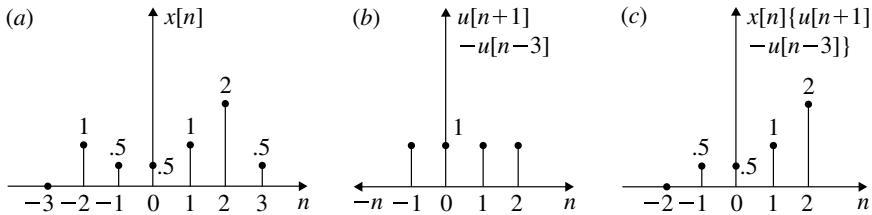


Fig. 1.20 Multiplication of DT sequences

1.18 that $\{u[n] - u[n - 4]\}$ gives the required $x[n]$ sequence which is represented in Fig. 1.18a. Thus, $x[n] = \{u[n] - u[n - 4]\}$.

Example 1.5 Consider the sequence shown in Fig. 1.19a. Express the sequence in terms of step function.

- Solution**
- Figure 1.19a represents the sequence $x[n]$ in the interval $-3 \leq n \leq 4$.
 - Consider $u[n + 3]$ which is represented in Fig. 1.19b. The sequence interval is $-3 \leq n < \infty$.
 - Consider the step sequence with a time delay of $n_0 = 5$ and inverted. This can be written as $-u[n - 5]$ for the interval $5 \leq n < \infty$. This is represented in Fig. 1.19c.
 - Now consider the sum of the sequences $u[n + 3]$ and $-u[n - 5]$. This is nothing but $x[n]$. Thus

$$x[n] = u[n + 3] - u[n - 5]$$

Example 1.6 A discrete time sequence $x[n]$ is shown in Fig. 1.20a. Find

$$x[n]\{u[n + 1] - u[n - 3]\}$$

Solution 1. $x[n]$ sequence is represented in Fig. 1.20a.

2. $\{u[n+1] - u[n-3]\}$ sequence is nothing but the time delayed unit step sequence with $n_0 = 3$, being subtracted from the time advanced unit step sequence with $n_0 = 1$. This sequence is represented in Fig. 1.20b.
3. Multiplying sample wise of Fig. 1.20a and b, the required sequence $x[n]\{u[n+1] - u[n-3]\}$ is obtained and represented in Fig. 1.20c.

Example 1.7 Sketch $x[n] = a^n$ where $-2 \leq n \leq 2$ for the two cases shown below:

$$(1) \quad a = \left(-\frac{1}{4}\right)$$

$$(2) \quad a = -4$$

Solution For $x[n] = \left(-\frac{1}{4}\right)^n$ and $x[n] = (-4)^n$ where $-2 \leq n \leq 2$, $x[n]$ is found and tabulated below:

n	-2	-1	0	1	2
$x[n] = \left(-\frac{1}{4}\right)^n$	16	-4	1	$-\frac{1}{4}$	$\frac{1}{16}$
$x[n] = (-4)^n$	$\frac{1}{16}$	$-\frac{1}{4}$	1	-4	16

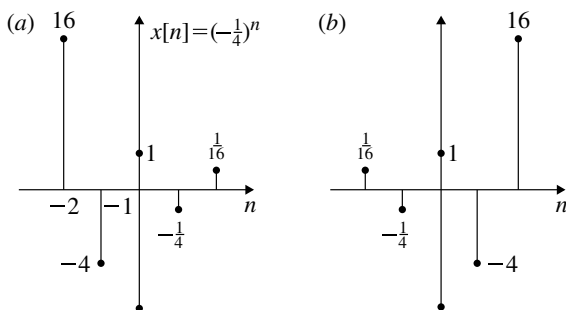
The samples of $x[n]$ are plotted and shown in Fig. 1.21. $x[n] = \left(-\frac{1}{4}\right)^n$ is represented in Fig. 1.21a and $x[n] = (-4)^n$ is represented in Fig. 1.21b.

Example 1.8 Express

$$x[n] = (-1)^n \quad -2 \leq n \leq 2$$

as a sum of scaled and shifted step function.

Fig. 1.21 DT sequences of Example 1.7



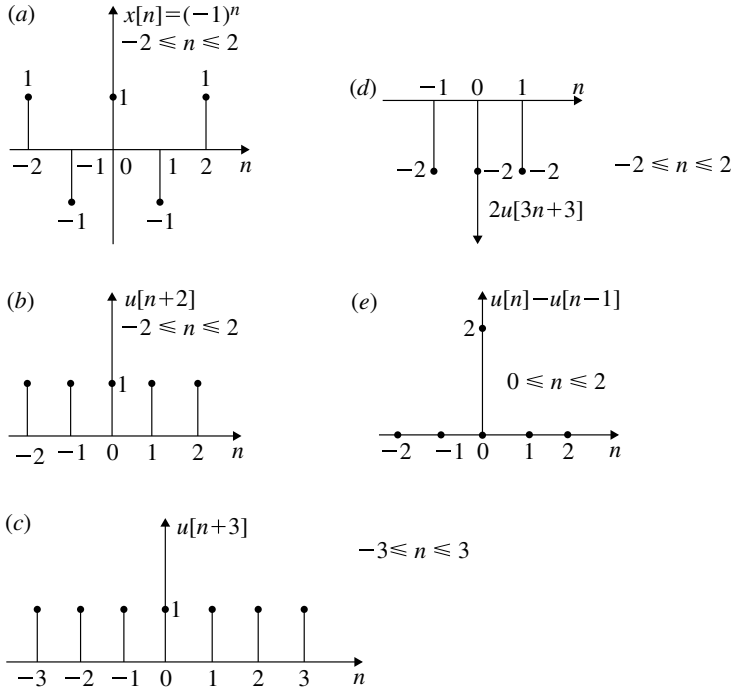


Fig. 1.22 DT sequences of Example 1.8

Solution (1) $x[n] = (-1)^n$ is tabulated for $-2 \leq n \leq 2$.

n	-2	-1	0	1	2
$x[n]$	1	-1	1	-1	1

The samples corresponding to the above table are sketched and shown as $x[n]$ in Fig. 1.22a.

- (2) Consider the step sequence $u[n+2]$ for $-2 \leq n \leq 2$. The samples are shown in Fig. 1.22b.
- (3) Consider unit step sequence $u[n+3]$ for $-3 \leq n \leq 3$. This is represented in Fig. 1.22c. From Fig. 1.22c, $-2u[3n+3]$ is obtained by amplitude inversion and multiplication and time scaling (compression). This is represented in Fig. 1.22d for $-2 \leq n \leq 2$.
- (4) Consider the step sequence $2\{u[n] - u[n-1]\}$ for $n \geq 0$. This is represented in Fig. 1.22e. This is nothing but the sample of strength 2 at $n = 0$.
- (5) Now, by adding the samples in Fig. 1.22b, d and e, it can be easily verified that

$$x[n] = u[n+2] - 2u[3n+3] + 2[u[n] - u[n-1]] \quad -2 \leq n \leq 2$$

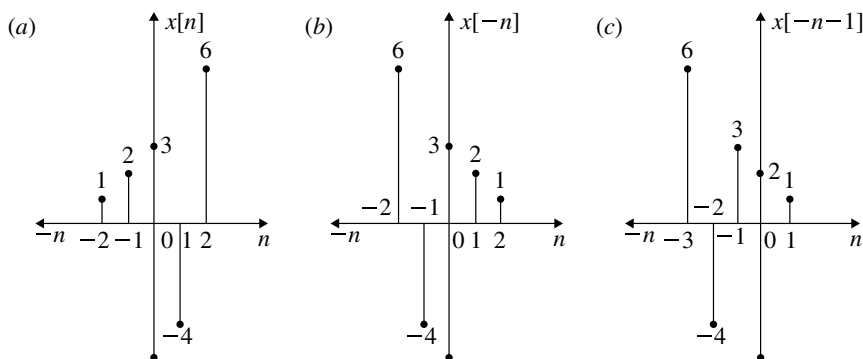


Fig. 1.23 DT sequences of Example 1.9

Example 1.9 Given

$$x[n] = \{1, 2, 3, -4, 6\}$$

↑

Plot the signal $x[-n - 1]$.

Solution 1. The sequence $x[n]$ is represented in Fig. 1.23a.

2. By folding $x[n]$, $x[-n]$ is obtained and represented in Fig. 1.23b.

3. $x[-n]$ is shifted to the left by one sample and $x[-n - 1]$ is obtained. This is represented in Fig. 1.23c.

1.6 Classification of Discrete Time Signals

Discrete time signals are classified as:

1. Periodic and non-periodic signals.
2. Odd and even signals.
3. Power and energy signals.

They are discussed below with suitable examples.

1.6.1 Periodic and Non-periodic DT Signals

A discrete time signal (sequence) $x[n]$ is said to be periodic with period N which is a positive integer if

$$x[n + N] = x[n] \quad \text{for all } n \quad (1.8)$$

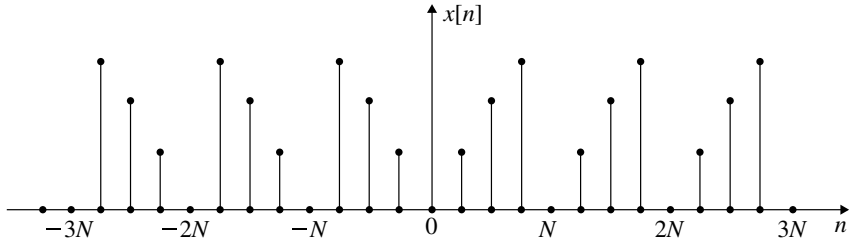


Fig. 1.24 Periodic sequence

Consider the DT sequence shown in Fig. 1.24. The signal gets repeated for every N . For Fig. 1.24, the following equation is written:

$$x[n + mN] = x[n] \quad \text{for all } n \quad (1.9)$$

where m is any integer. The smallest positive integer N in Eq.(1.9) is called the fundamental period N_0 . Any sequence which is not periodic is said to be non-periodic or aperiodic.

Example 1.10 Show that complex exponential sequence $x[n] = e^{j\omega_0 n}$ is periodic and find the fundamental period.

Solution

$$\begin{aligned} x[n] &= e^{j\omega_0 n} \\ x[n + N] &= e^{j\omega_0 (n+N)} \\ &= e^{j\omega_0 n} e^{j\omega_0 N} \\ &= e^{j\omega_0 n} \quad \text{if } e^{j\omega_0 N} = 1 \\ \omega_0 N &= m2\pi \quad \text{where } m \text{ is any integer.} \end{aligned}$$

$$N = m \frac{2\pi}{\omega_0}$$

or

$$\frac{\omega_0}{2\pi} = \frac{m}{N} = \text{rational number.}$$

Thus, $e^{j\omega_0 n}$ is periodic if $\frac{m}{N}$ is rational. For $m = 1$, $N = N_0$. The corresponding frequency $F_0 = \frac{1}{N_0}$ is the fundamental frequency. F_0 is expressed in cycles and not Hz. Similarly, ω_0 is expressed in radians and not in radians per second.

Example 1.11 Consider the following DT Signal:

$$x[n] = \sin(\omega_0 n + \phi)$$

Under what condition, the above signal is periodic.

Solution

$$\begin{aligned}
x[n] &= \sin(\omega_0 n + \phi) \\
x[n + N] &= \sin(\omega_0(n + N) + \phi) \\
&= \sin(\omega_0 n + \omega_0 N + \phi) \\
&= \sin(\omega_0 n + \phi) \quad \text{if } \omega_0 N = 2\pi m \quad \text{where } m \text{ is an integer} \\
&= x[n]
\end{aligned}$$

$\frac{\omega_0}{2\pi} = \frac{m}{N} = \text{rational}$

Example 1.12 If $x_1[n]$ and $x_2[n]$ are periodic, then show that the sum of the composite signal $x[n] = x_1[n] + x_2[n]$ is also periodic with the least common multiple (LCM) of the fundamental period of individual signal.

Solution Let N_1 and N_2 be the fundamental periods of $x_1[n]$ and $x_2[n]$, respectively. Since both $x_1[n]$ and $x_2[n]$ are periodic,

$$\begin{aligned}
x_1[n] &= x_1[n + mN_1] \\
x_2[n] &= x_2[n + kN_2] \\
x[n] &= x_1[n] + x_2[n] \\
&= x_1[n + mN_1] + x_2[n + kN_2]
\end{aligned}$$

For $x[n]$ to be periodic with period N ,

$$\begin{aligned}
x[n + N] &= x_1[n + N] + x_2[n + N] \\
x[n] &= x_1[n + N] \\
x_1[n + mN_1] + x_2[n + kN_2] &= x_1[n + N] + x_2[n + N]
\end{aligned}$$

The above equation is satisfied if

$$mN_1 = kN_2 = N$$

m and k which are integers are chosen to satisfy the above equation. It implies that N is the LCM of N_1 and N_2 .

On similar line it can be proved that if $x_1[n]$ and $x_2[n]$ are periodic signals with fundamental period N_1 and N_2 , respectively, then $x[n] = x_1[n]x_2[n]$ is periodic if

$$mN_1 = kN_2 = N$$

Example 1.13 Find whether the following signals are periodic. If periodic, determine the fundamental period

- (a) $x[n] = e^{j\pi n}$
 (b) $x[n] = \cos\left[\frac{n}{8} - \pi\right]$
 (c) $x[n] = \sin^2 \frac{\pi}{4}n$

Solution (a) $x[n] = e^{j\pi n}$

$$\omega_0 = \pi$$

$$N = \frac{2\pi}{\omega_0}m$$

$N = \frac{2\pi}{\pi} = 2 \quad \text{if } m = 1$

$x[n]$ is periodic with fundamental period 2.

(b) $x[n] = \cos\left[\frac{n}{8} - \pi\right]$

$$\omega_0 = \frac{1}{8}$$

$$N = \frac{2\pi}{\omega_0}m = 16\pi m$$

For any integer value of m , N is not integer. Hence, $x[n]$ is not periodic.

$x[n] \text{ is not periodic}$

(c) $x[n] = \sin^2 \frac{\pi}{4}n$

$$\begin{aligned}
 x[n] &= \sin^2 \frac{\pi}{4}n \\
 &= \frac{1}{2} - \frac{1}{2} \cos \frac{2\pi}{4}n \\
 &= x_1[n] + x_2[n] \\
 x_1[n] &= \frac{1}{2} = \frac{1}{2}(1)^n \text{ is periodic with } N_1 = 1 \\
 x_2[n] &= -\frac{1}{2} \cos \frac{\pi}{2}n \\
 \omega_0 &= \frac{\pi}{2}
 \end{aligned}$$

$$\begin{aligned}
 N_2 &= \frac{2\pi}{\omega_0} m = 4m = 4 \quad \text{for } m = 1 \\
 \frac{N_1}{N_2} &= \frac{1}{4} \\
 \text{or } 4N_1 &= N_2 = N
 \end{aligned}$$

$$\boxed{N = 4}$$

Example 1.14 Find the periodicity of the following DT signal:

$$x[n] = \sin \frac{2\pi}{3}n + \cos \frac{\pi}{2}n$$

Solution

$$\begin{aligned}
 x[n] &= \sin \frac{2\pi}{3}n + \cos \frac{\pi}{2}n \\
 &= x_1[n] + x_2[n] \\
 x_1[n] &= \sin \frac{2}{3}\pi n \\
 \omega_1 &= \frac{2}{3}\pi \\
 N_1 &= \frac{2\pi}{\omega_1} m_1 = \frac{2\pi}{2\pi} 3m_1 = 3 \quad \text{for } m_1 = 1 \\
 x_2[n] &= \cos \frac{\pi}{2}n \\
 \omega_2 &= \frac{\pi}{2} \\
 N_2 &= \frac{2\pi}{\omega_2} m_2 = \frac{2\pi}{\pi} 2m_2 = 4 \quad \text{for } m_2 = 1 \\
 \frac{N_1}{N_2} &= \frac{3}{4} \quad \text{or } 4N_1 = 3N_2 = N
 \end{aligned}$$

$$\boxed{N = 12}$$

Example 1.15 Determine whether the following signal is periodic. If periodic, find its fundamental period:

$$x[n] = \cos\left(\frac{n\pi}{2}\right) \cos\left(\frac{n\pi}{4}\right)$$

Solution

$$\begin{aligned}
 x[n] &= \cos\left(\frac{n\pi}{2}\right) \cos\left(\frac{n\pi}{4}\right) \\
 &= x_1[n]x_2[n] \\
 x_1[n] &= \cos\frac{n\pi}{2} \\
 \omega_1 &= \frac{\pi}{2} \\
 N_1 &= \frac{2\pi}{\omega_1}m_1 = \frac{2\pi}{\pi}2m_1 = 4 \quad \text{for } m_1 = 1
 \end{aligned}$$

$$\begin{aligned}
 x_2[n] &= \cos\frac{n\pi}{4} \\
 \omega_2 &= \frac{\pi}{4} \\
 N_2 &= \frac{2\pi}{\omega_2}m_2 = \frac{2\pi}{\pi}4m_2 = 8 \quad \text{for } m_2 = 1 \\
 \frac{N_1}{N_2} &= \frac{4}{8} = \frac{1}{2} \quad \text{or} \\
 2N_1 &= N_2 = N
 \end{aligned}$$

$$\boxed{N = 8}$$

The signal is periodic and the fundamental period $N = 8$.

Example 1.16 Test whether the following signals are periodic or not, and if periodic, calculate the fundamental period.

$$\begin{aligned}
 (a) \quad x[n] &= \cos\left(\frac{\pi}{2}n\right) + \sin\left(\frac{\pi}{8}n\right) + 3\cos\left(\frac{\pi}{4}n + \frac{\pi}{3}\right) \\
 (b) \quad x[n] &= e^{j\frac{2\pi}{3}n} + e^{j\frac{3\pi}{4}n}
 \end{aligned}$$

Solution (a)

$$\begin{aligned}
 x[n] &= \cos\left(\frac{\pi}{2}n\right) + \sin\left(\frac{\pi}{8}n\right) + 3\cos\left(\frac{\pi}{4}n + \frac{\pi}{3}\right) \\
 &= x_1[n] + x_2[n] + x_3[n] \\
 x_1[n] &= \cos\frac{\pi}{2}n \\
 \omega_1 &= \frac{\pi}{2}; \quad N_1 = \frac{2\pi}{\omega_1} = \frac{2\pi}{\pi} \cdot 2 \quad \text{for } m_1 = 1 \\
 N_1 &= 4
 \end{aligned}$$

$$\begin{aligned}
 x_2[n] &= \sin\left(\frac{\pi}{8}n\right) \\
 \omega_2 &= \frac{\pi}{8}; \quad N_2 = \frac{2\pi}{\omega_2}m_2 = \frac{2\pi}{\pi}8 \quad \text{for } m_2 = 1 \\
 N_2 &= 16
 \end{aligned}$$

$$\begin{aligned}
 x_3[n] &= 3 \cos\left(\frac{\pi}{4}n + \frac{\pi}{3}\right) \\
 \omega_3 &= \frac{\pi}{4}; \quad N_3 = \frac{2\pi}{\omega_3}m_3 = \frac{2\pi}{\pi}4 \quad \text{for } m_3 = 1 \\
 N_3 &= 8
 \end{aligned}$$

To find the LCM of N_1 , N_2 and N_3

$$\begin{array}{c|ccc}
 4 & 4, & 8, & 16 \\
 2 & 1, & 2, & 4 \\
 & 1, & 1, & 2
 \end{array}$$

$$\text{LCM} = 4 \times 2 \times 2 = 16$$

$$\boxed{N = 16}$$

The signal is periodic

(b)

$$\begin{aligned}
 x[n] &= e^{j\frac{2\pi}{3}n} + e^{j\frac{3\pi}{4}n} \\
 &= x_1[n] + x_2[n] \\
 x_1[n] &= e^{j\frac{2\pi}{3}n} \\
 \omega_1 &= \frac{2\pi}{3}; \quad N_1 = \frac{2\pi}{\omega_1}m_1 = \frac{2\pi}{2\pi}3 \quad \text{for } m_1 = 1 \\
 N_1 &= 3 \\
 x_2[n] &= e^{j\frac{3\pi}{4}n} \\
 \omega_2 &= \frac{3\pi}{4}; \quad N_2 = \frac{2\pi}{\omega_2}m_2 = \frac{2\pi}{3\pi}4m_2
 \end{aligned}$$

$$\begin{aligned}
 N_2 &= 8 \quad \text{for } m_2 = 3 \\
 \frac{N_1}{N_2} &= \frac{3}{8} \\
 8 N_1 &= 3 N_2 = N = 24
 \end{aligned}$$

$$\boxed{N = 24}$$

The signal is periodic with fundamental period $N = 24$.

1.6.2 Odd and Even DT Signals

DT signals are classified as odd and even signals. The relationships are analogous to CT signals.

A discrete time signal $x[n]$ is said to be an even signal if

$$x[-n] = x[n] \quad (1.10)$$

A discrete time signal $x[n]$ is said to be an odd signal if

$$x[-n] = -x[n] \quad (1.11)$$

The signal $x[n]$ can be expressed as the sum of odd and even signals as

$$x[n] = x_e[n] + x_o[n] \quad (1.12)$$

The even and odd components of $x[n]$ can be expressed as

$$x_e[n] = \frac{1}{2} [x[n] + x[-n]] \quad (1.13)$$

$$x_o[n] = \frac{1}{2} [x[n] - x[-n]] \quad (1.14)$$

It is to be noted that

- An even function has an odd part which is zero.
- An odd function has an even part which is zero.
- The product of two even signals or of two odd signals is an even signal.
- The product of an odd and an even signal is an odd signal.
- At $n = 0$, the odd signal is zero.

The even and odd signals are represented in Fig. 1.25a and b, respectively.

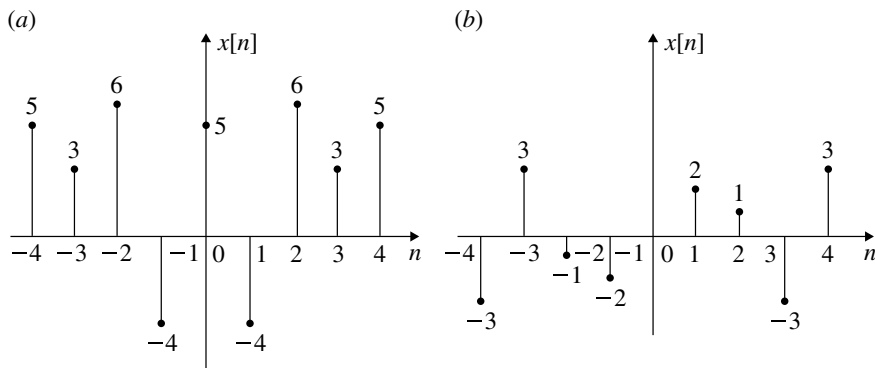


Fig. 1.25 **a** Even function and **b** Odd function

Example 1.17 Determine whether the following functions are odd or even:

(a) $x[n] = \sin 2\pi n$

(b) $x[n] = \cos 2\pi n$

Solution (a) $x[n] = \sin 2\pi n$

$$\begin{aligned} x[-n] &= \sin(-2\pi n) = -\sin 2\pi n \\ &= -x[n] \end{aligned}$$

This is an odd signal.

(b) $x[n] = \cos 2\pi n$

$$\begin{aligned} x[-n] &= \cos(-2\pi n) = \cos 2\pi n \\ &= x[n] \end{aligned}$$

This is an even signal.

Example 1.18 Find the even and odd components of DT signal given below. Verify the same by graphical method.

$$x[n] = \{-2, 1, 3, -5, 4\}$$

↑

Solution $x[-n]$ is obtained by folding $x[n]$. Thus,

$$\begin{array}{c} x[-n] = \{4, -5, 3, 1, -2\} \\ \uparrow \\ -x[-n] = \{-4, 5, -3, -1, 2\} \\ \uparrow \end{array}$$

$$\begin{aligned} x_e[n] &= \frac{1}{2} [x[n] + x[-n]] \\ &= \frac{1}{2} [\underbrace{\{-2, 1, 3, -5, 4\}}_{\uparrow} + \underbrace{\{4, -5, 3, 1, -2\}}_{\uparrow}] \\ &= \frac{1}{2} [(-2 + 4), (1 - 5), (3 + 3), (-5 + 1), (4 - 2)] \\ &\quad \uparrow \end{aligned}$$

$$x_e[n] = \{1, -2, 3, -2, 1\}$$

↑

$$\begin{aligned} x_o[n] &= \frac{1}{2} [x[n] - x[-n]] \\ &= \frac{1}{2} [\underbrace{\{-2, 1, 3, -5, 4\}}_{\uparrow} + \underbrace{\{-4, 5, -3, -1, 2\}}_{\uparrow}] \\ &= \frac{1}{2} [(-2 - 4), (1 + 5), (3 - 3), (-5 - 1), (4 + 2)] \\ &\quad \uparrow \end{aligned}$$

$$x_o[n] = \{-3, 3, 0, -3, 3\}$$

↑

Odd and even components by graphical method.

Solution 1. $x[n]$ is represented in Fig. 1.26a.

2. $x[-n]$ is obtained by folding $x[n]$ which is represented in Fig. 1.26b.

3. $-x[n]$ is obtained by inverting $x[-n]$ of Fig. 1.26b. This is represented in Fig. 1.26c.

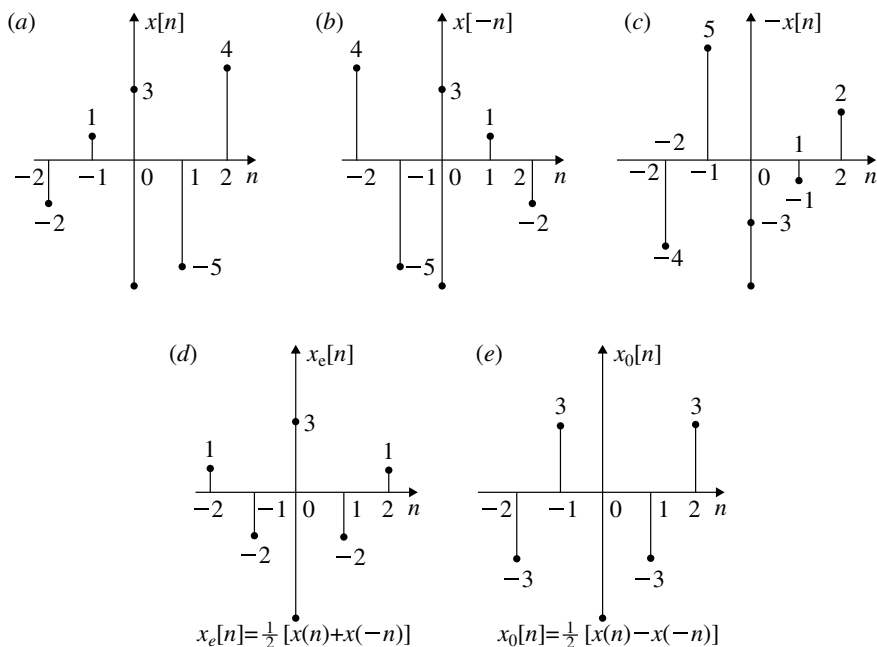


Fig. 1.26 Graphical determination of even and odd components from $x[n]$

4. $x_e[n] = \frac{1}{2} [x[n] + x[-n]]$. Figure 1.26a and b sample wise are added and their amplitudes are divided by the factor 2. This gives $x_e[n]$ and is represented in Fig. 1.26d.
5. $x_o[n] = \frac{1}{2} [x[n] - x[-n]]$. Figure 1.26a and c sample wise are added and their amplitudes are divided by a factor 2 to get $x_o[n]$. This is represented in Fig. 1.26e.

Example 1.19 Find the even and odd components of the following DT signal and sketch the same:

$$x[n] = \{-2, 1, 2, -1, 3\}$$

Solution

$$\begin{aligned}
 x[n] &= \{-2, 1, 2, -1, 3\} \\
 x[-n] &= \{3, -1, 2, 1, -2\} \\
 &\quad \uparrow \\
 x_e[n] &= \frac{1}{2} \{x[n] + x[-n]\} \\
 &= \frac{1}{2} \{-2, 1, 2, -1, 3\} + \{3, -1, 2, 1, -2\}
 \end{aligned}$$

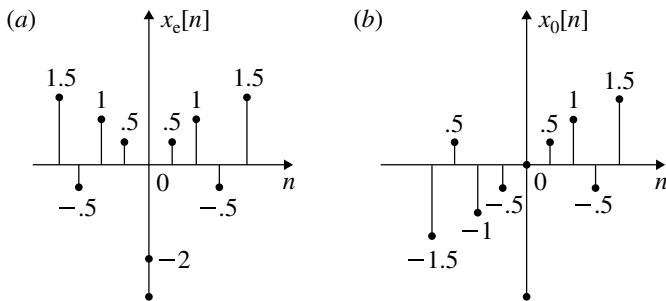


Fig. 1.27 **a** Even components and **b** Odd components

$$\begin{aligned}
 & \uparrow \qquad \qquad \qquad \uparrow \\
 & = \{1.5, -0.5, 1, 0.5, -2, 0.5, 1, -0.5, 1.5\} \\
 & \qquad \qquad \qquad \uparrow \\
 x_o[n] &= \frac{1}{2} [x[n] - x[-n]] \\
 &= \frac{1}{2} [\{-2, 1, 2, -1, 3\} - \{3, -1, 2, 1, -2\}] \\
 & \qquad \qquad \qquad \uparrow \qquad \qquad \qquad \uparrow \\
 x_o[n] &= \{-1.5, 0.5, -1, -0.5, 0, 0.5, 1, -0.5, 1.5\} \\
 & \qquad \qquad \qquad \uparrow
 \end{aligned}$$

Even and odd components of $x[n]$ are represented in Fig. 1.27a and b, respectively.

1.6.3 Energy and Power of DT Signals

For a discrete time signal $x[n]$, the total energy is defined as

$$E = \sum_{n=-\infty}^{\infty} |x[n]|^2 \quad (1.15)$$

The average power is defined as

$$P = \lim_{N \rightarrow \infty} \frac{1}{(2N+1)} \sum_{n=-N}^N |x[n]|^2 \quad (1.16)$$

From the definitions of energy and power, the following inferences are derived:

1. $x[n]$ is an energy sequence iff $0 < E < \infty$. For finite energy signal, the average power $P = 0$.
2. $x[n]$ is a power sequence iff $0 < P < \infty$. For a sequence with average power P being finite, the total energy $E = \infty$.
3. Periodic signal is a power signal and *vice versa* is not true. Here the energy of the signal per period is finite.
4. Signals which do not satisfy the definitions of total energy and average power, neither termed as a power signal nor an energy signal. The following summation formulae are very often used while evaluating the average power and total energy of the DT sequence.

1.

$$\sum_{n=0}^{N-1} a^n = \frac{(1 - a^N)}{(1 - a)} \quad a \neq 1 \quad (1.17)$$

$$= N \quad a = 1$$

2.

$$\sum_{n=0}^{\infty} a^n = \frac{1}{(1 - a)} \quad a < 1 \quad (1.18)$$

3.

$$\sum_{n=m}^{\infty} a^n = \frac{a^m}{(1 - a)} \quad a < 1 \quad (1.19)$$

4.

$$\sum_{n=0}^{\infty} na^n = \frac{a}{(1 - a)^2} \quad a < 1 \quad (1.20)$$

Example 1.20 Determine whether the following signals are energy signals or power signals:

- (a) $x[n] = A\delta[n]$
- (b) $x[n] = u[n]$
- (c) $x[n] = \text{ramp } n$
- (d) $x[n] = A$
- (e) $x[n] = 2e^{j(\pi n + \theta)}$
- (f) $x[n] = \cos \frac{\pi}{2}n$

Solution (a) $x[n] = A\delta[n]$

$$\begin{aligned} x[n] &= A\delta[n] \\ &= A \quad n = 0 \\ &= 0 \quad n \neq 0 \end{aligned}$$

$$\text{Energy } E = \sum_{n=0}^0 (A)^2$$

$$\boxed{E = A^2}$$

For unit impulse, $A = 1$ and $E = 1$.

(b) $x[n] = u[n]; n \geq 0$

$$\begin{aligned} P &= \lim_{N \rightarrow \infty} \frac{Lt}{(2N+1)} \sum_{n=0}^N |x(n)|^2 \\ &= \lim_{N \rightarrow \infty} \frac{Lt}{(2N+1)} \sum_{n=0}^N 1 \end{aligned}$$

But $\sum_{n=0}^N 1 = (N+1)$

$$\begin{aligned} P &= \lim_{N \rightarrow \infty} \frac{Lt}{(2N+1)} \frac{(N+1)}{1} \\ &= \lim_{N \rightarrow \infty} \frac{Lt}{N(2 + \frac{1}{N})} = \frac{1}{2} \end{aligned}$$

$$\boxed{\begin{aligned} P &= \frac{1}{2} \\ E &= \infty \end{aligned}}$$

(c) $x[n] = \text{ramp } n; n \geq 0$

$$\begin{aligned} P &= \lim_{N \rightarrow \infty} \frac{Lt}{(2N+1)} \sum_{n=0}^N |x[n]|^2 \\ P &= \lim_{N \rightarrow \infty} \frac{Lt}{(2N+1)} \sum_{n=0}^N n^2 \end{aligned}$$

But $\sum_{n=0}^N n^2 = \frac{N(N+1)(2N+1)}{6}$

$$P = \lim_{N \rightarrow \infty} \frac{N(N+1)(2N+1)}{(2N+1)6}$$

$$\boxed{P = \infty}$$

$$\begin{aligned} E &= \lim_{N \rightarrow \infty} \sum_{n=0}^N n^2 \\ &= \lim_{N \rightarrow \infty} \frac{N(N+1)(2N+1)}{6} = \infty \end{aligned}$$

$$\boxed{E = \infty}$$

The signal $x[n] = n$ is neither a power signal nor an energy signal.

(d) $x[n] = A$

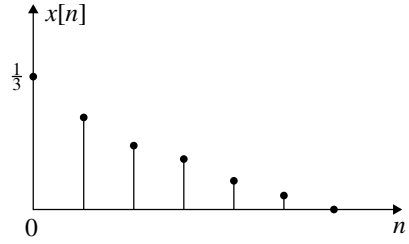
$$\begin{aligned} P &= \lim_{N \rightarrow \infty} \frac{1}{(2N+1)} \sum_{n=-\infty}^{\infty} A^2 \\ &= \lim_{N \rightarrow \infty} \frac{A^2}{(2N+1)} \quad \left[\sum_{n=-\infty}^{\infty} 1 = (2N+1) \right] \end{aligned}$$

$$\boxed{\begin{aligned} P &= A^2 \\ E &= \infty \end{aligned}}$$

(e) $x[n] = 2e^{j(\pi n + \theta)}$

$$\begin{aligned} P &= \lim_{N \rightarrow \infty} \frac{1}{(2N+1)} \sum_{-N}^N |2e^{j(n\pi + \theta)}|^2 \\ P &= \lim_{N \rightarrow \infty} \frac{1}{2N+1} 4 \sum_{-N}^N |e^{j(n\pi + \theta)}|^2 \end{aligned}$$

But $|e^{j(n\pi + \theta)}| = 1$ and $\sum_{-N}^N 1 = (2N+1)$

Fig. 1.28 $x[n] = \left(\frac{1}{3}\right)^2 u[n]$ 

$$P = \lim_{N \rightarrow \infty} \frac{Lt}{(2N+1)} 4 \frac{(2N+1)}{(2N+1)} = 4$$

$$\boxed{\begin{array}{l} P = 4 \\ E = \infty \end{array}}$$

(f) $x[n] = \cos \frac{\pi}{2} n$

$$P = \frac{1}{(2N+1)} \sum_{-N}^N \cos^2 \frac{\pi}{2} n$$

Since $\sum_{-N}^N \cos \pi n = 0$,

$$\begin{aligned} P &= \lim_{N \rightarrow \infty} \frac{Lt}{(2N+1)} \sum_{-N}^N \frac{(1 + \cos \pi n)}{2} \\ &= \frac{1}{2} \lim_{N \rightarrow \infty} \frac{Lt}{(2N+1)} (2N+1) \\ &= \frac{1}{2} \end{aligned}$$

$$\boxed{\begin{array}{l} P = \frac{1}{2} \\ E = \infty \end{array}}$$

Example 1.21 Determine the energy of the signal shown in Fig. 1.28 whose

$$x[n] = \left(\frac{1}{3}\right)^n u[n]$$

Solution

$$\begin{aligned}
 E &= \lim_{N \rightarrow \infty} \sum_{n=0}^N \left(\frac{1}{3}\right)^{2n} \\
 &= \lim_{N \rightarrow \infty} \sum_{n=0}^N \left(\frac{1}{9}\right)^n \\
 &= 1 + \frac{1}{9} + \left(\frac{1}{9}\right)^2 + \dots \\
 &= \frac{1}{1 - \frac{1}{9}}
 \end{aligned}$$

$$\boxed{
 \begin{aligned}
 E &= \frac{9}{8} \\
 P &= 0
 \end{aligned}
 }$$

Example 1.22 Find the energy of the following sequence:

$$x[n] = n \quad 0 \leq n \leq 4$$

Solution

$$\begin{aligned}
 x[n] &= n \\
 &= \{0, 1, 2, 3, 4\} \\
 E &= \sum_{n=0}^4 n^2 \\
 &= 0 + 1 + 4 + 9 + 16
 \end{aligned}$$

$$\boxed{E = 30}$$

Example 1.23 Determine the average power and the energy per period of the sequence shown in Fig. 1.29.

Solution The fundamental period N of the signal is 5. Hence, the average power per period is

$$P = \frac{1}{5} \sum_{n=0}^4 n^2 = \frac{1}{5} [0 + 1 + 4 + 9 + 16]$$

$$\boxed{P = 6}$$

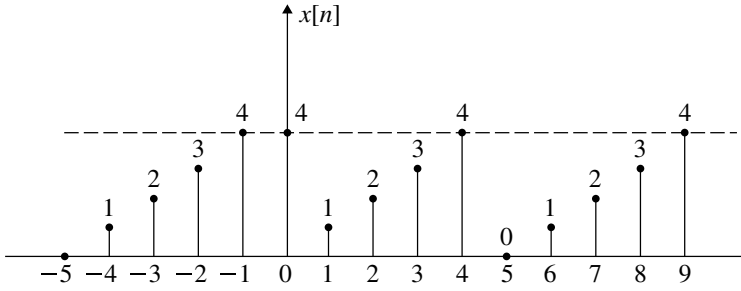


Fig. 1.29 $x[n]$ of Example 1.23

Average energy per period is

$$\begin{aligned}
 E &= \sum_{n=0}^4 n^2 \\
 &= [0 + 1 + 4 + 9 + 16]
 \end{aligned}$$

$$E = 30$$

Example 1.24 Find the energy and power of the following signal:

$$x[n] = a^n u[n]$$

for the following cases:

- (a) $|a| < 1$
- (b) $|a| = 1$
- (c) $|a| > 1$

Solution (a) $x[n] = a^n u[n]$ where $|a| < 1$ and $n \geq 0$

$$\begin{aligned}
 E &= \sum_{n=0}^{\infty} (a^n)^2 \\
 &= 1 + a^2 + a^4 + \dots
 \end{aligned}$$

$$E = \frac{1}{1 - |a|^2}$$

$$P = 0$$

(b) $x[n] = a^n u[n]$ where $|a| = 1$

$$E = \lim_{N \rightarrow \infty} \sum_0^N 1^n = \lim_{N \rightarrow \infty} (N + 1)$$

$$\boxed{E = \infty}$$

$$P = \lim_{N \rightarrow \infty} \frac{1}{(2N + 1)} \sum_0^N (1)^n$$

$$\begin{aligned} P &= \lim_{N \rightarrow \infty} \frac{(N + 1)}{(2N + 1)} \\ &= \lim_{N \rightarrow \infty} \frac{N(1 + \frac{1}{N})}{N(2 + \frac{1}{N})} \end{aligned}$$

$$\boxed{P = \frac{1}{2}}$$

(c) $x[n] = a^n u[n]$ where $|a| > 1$

$$\begin{aligned} E &= \lim_{N \rightarrow \infty} \sum_0^N a^n \\ &= 1 + a + a^2 + \dots + a^N \end{aligned}$$

$$\boxed{E = \infty}$$

$$\begin{aligned} P &= \lim_{N \rightarrow \infty} \frac{1}{(2N + 1)} \sum_{n=0}^N a^n \\ &= \lim_{N \rightarrow \infty} \frac{1}{(N + 1)} \frac{(1 - a^{N+1})}{(1 - a)} \end{aligned}$$

$$\boxed{P = \infty}$$

The signal is neither an energy signal nor a power signal.

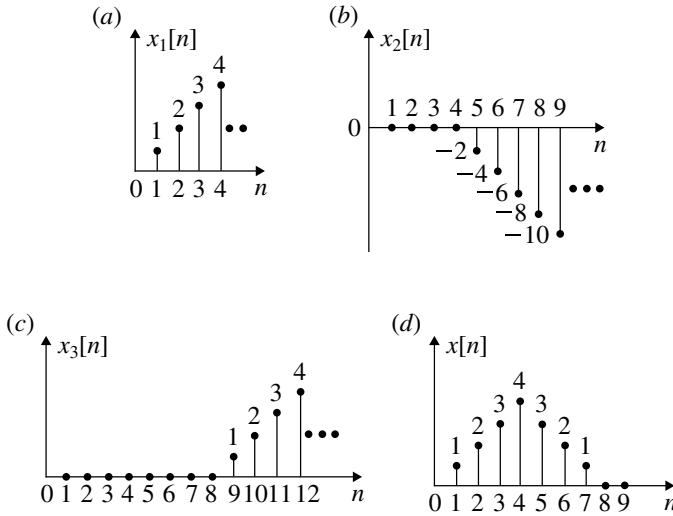


Fig. 1.30 DT energy signal of Example 1.25

Example 1.25 Find the energy of the following signal:

Solution $x[n] = \text{ramp}[n] - 2 \text{ramp}[n - 4] + \text{ramp}[n - 8]$

$$\begin{aligned} x[n] &= \text{ramp}[n] - 2 \text{ramp}[n - 4] + \text{ramp}[n - 8] \\ &= x_1[n] + x_2[n] + x_3[n] \end{aligned}$$

$x_1[n]$, $x_2[n]$ and $x_3[n]$ are shown in Fig. 1.30a, b and c, respectively. Figure 1.30d represents $x[n]$. From Fig. 1.30d, the energy of the signal $x[n]$ is obtained as

$$E = 1^2 + 2^2 + 3^2 + 4^2 + 3^2 + 2^2 + 1^2$$

$E = 44$

Example 1.26 Determine the value of power and energy of each of the following signals:

(a) $x[n] = e^{j(\frac{\pi n}{2} + \frac{\pi}{8})}$

(b) $x[n] = \left(\frac{1}{2}\right)^n u[n]$

Solution (a) $x[n] = e^{j(\frac{\pi n}{2} + \frac{\pi}{8})}$

$$\begin{aligned}
 P &= \lim_{N \rightarrow \infty} \frac{Lt}{2N+1} \sum_{-N}^N |e^{j(\frac{\pi n}{2} + \frac{\pi}{8})}|^2 \\
 &= \lim_{N \rightarrow \infty} \frac{Lt}{(2N+1)} \sum_{-N}^N 1 \\
 P &= \frac{(2N+1)}{(2N+1)} = 1
 \end{aligned}$$

$$\boxed{P = 1 \text{ and } E = \infty}$$

(b) $x[n] = \left[\frac{1}{2}\right]^n u[n]$

$$\begin{aligned}
 E &= \lim_{N \rightarrow \infty} \frac{Lt}{N} \sum_0^N \left(\frac{1}{2}\right)^{2n} \\
 &= \lim_{N \rightarrow \infty} \frac{Lt}{N} \sum_0^N \left(\frac{1}{4}\right)^n = \frac{1}{1 - \frac{1}{4}} = \frac{4}{3}
 \end{aligned}$$

$$\boxed{E = \frac{4}{3} \text{ and } P = 0}$$

Example 1.27 Find the energy of the following DT signal:

$$\begin{aligned}
 x[n] &= \left(\frac{1}{2}\right)^n & n \geq 0 \\
 &= 3^n & n < 0
 \end{aligned}$$

Solution

$$\begin{aligned}
 E &= \left[\sum_{-\infty}^{-1} (3)^{2n} + \sum_0^{\infty} \left(\frac{1}{2}\right)^{2n} \right] \\
 &= \left[\sum_{-\infty}^{-1} (9)^n + \sum_0^{\infty} \left(\frac{1}{4}\right)^n \right]
 \end{aligned}$$

$$\begin{aligned}
&= \left[\sum_1^{\infty} (9)^{-n} + \frac{1}{(1 - \frac{1}{4})} \right] \\
&= \left[\sum_1^{\infty} \left(\frac{1}{9} \right)^n + \frac{4}{3} \right] \\
&= \left[\frac{1}{9} + \frac{1}{9^2} + \frac{1}{9^3} + \cdots \right] + \frac{4}{3} \\
&= \frac{1}{9} \left[1 + \frac{1}{9} + \frac{1}{9^2} + \frac{1}{9^3} + \cdots \right] + \frac{4}{3} \\
&E = \frac{1}{9} \frac{1}{[1 - \frac{1}{9}]} + \frac{4}{3} \\
&= \frac{1}{8} + \frac{4}{3} \\
&\boxed{E = \frac{35}{24}}
\end{aligned}$$

1.7 Discrete Time System

The block diagram of a discrete time system is shown in Fig. 1.31. $x[n]$ is the excitation (input) signal and $y[n]$ is the response (output) signal of the DT system. H represents the functional relationship between the input and output which is described by the difference equation. The input–output signals appear at discrete intervals of time n where $n = 0, 1, 2 \dots$ which is an integer. n can also take the negative value of an integer.

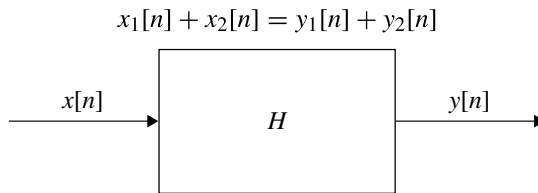


Fig. 1.31 Block diagram representation of discrete time system

1.8 Properties of Discrete Time System

Like CT systems, DT systems also possess similar properties which are given below:

1. Linear and Non-linear Systems.
2. Time Invariant and Time Varying DT Systems.
3. Causal and Non-causal DT Systems.
4. Stable and Unstable Systems.
5. Static and Dynamic Systems.
6. Invertible and Inverse Discrete Time Systems.

1.8.1 Linear and Non-linear Systems

A linear discrete time system obeys the property of superposition. As discussed for the CT system, the superposition property is composed of homogeneity and additivity. Let $x_1[n]$ excitation produce $y_1[n]$ response and $x_2[n]$ produce $y_2[n]$ response. According to additivity property of superposition theorem, if both $x_1[n]$ and $x_2[n]$ are applied simultaneously, then

$$x_1[n] + x_2[n] = y_1[n] + y_2[n]$$

Let $a_1x_1[n]$ and $a_2x_2[n]$ be the inputs. According to the homogeneity (scaling) property, when these signals are separately applied,

$$a_1x_1[n] = a_1y_1[n]$$

$$a_2x_2[n] = a_2y_2[n]$$

If $a_1x_1[n] + a_2x_2[n]$ are simultaneously applied, the output is obtained by applying superposition theorem as

$$a_1x_1[n] + a_2x_2[n] = a_1y_1[n] + a_2y_2[n]$$

In the above equation, $a_1x_1[n] + a_2x_2[n]$ is called the weighted sum of input and $a_1y_1[n] + a_2y_2[n]$ is called the weighted sum of the output. Therefore, the following procedure is followed to test the linearity of a DT system:

1. Express

$$y_1[n] = f(x_1[n])$$

$$y_2[n] = f(x_2[n])$$

2. Find the weighted sum of the output as

$$y_3[n] = a_1 y_1[n] + a_2 y_2[n]$$

3. Find the output $y_4[n]$ due to the weighted sum of input as

$$y_4[n] = f(a_1 x_1[n] + a_2 x_2[n])$$

4. If $y_3[n] = y_4[n]$, then the given DT system is linear. Otherwise, it is non-linear.

The following examples illustrate the method of testing a DT system for its linearity.

Example 1.28 Test whether the following DT systems are linear or not:

- (a) $y[n] = x^2[n]$
- (b) $y[n] = x[4n + 1]$
- (c) $y[n] = x[n] + \frac{1}{x[n + 1]}$
- (d) $y[n] = x[n^2]$
- (e) $y[n] = x[n] + nx[n + 1]$

Solution (a) $y[n] = x^2[n]$

$$\begin{aligned} y_1[n] &= x_1^2[n] \\ y_2[n] &= x_2^2[n] \end{aligned}$$

1. The weighted sum of the output $y_3[n]$ is

$$\begin{aligned} y_3[n] &= a_1 y_1[n] + a_2 y_2[n] \\ &= a_1 x_1^2[n] + a_2 x_2^2[n] \end{aligned}$$

2. The output due to the weighted sum of the input $y_4[n]$ is

$$\begin{aligned} y_4[n] &= [a_1 x_1[n] + a_2 x_2[n]]^2 \\ &= a_1^2 x_1^2[n] + a_2^2 x_2^2[n] + 2a_1 a_2 x_1[n] x_2[n] \end{aligned}$$

- 3.

$$y_3[n] \neq y_4[n]$$

The system is Non-linear.

(b) $y[n] = x[4n + 1]$

$$a_1 y_1[n] = a_1 x_1[4n + 1]$$

$$a_2 y_2[n] = a_2 x_2[4n + 1]$$

$$y_3[n] = a_1 y_1[n] + a_2 y_2[n]$$

1. The weighted sum of the output is

$$\begin{aligned} y_3[n] &= a_1 y_1[n] + a_2 y_2[n] \\ &= a_1 x_1[4n + 1] + a_2 x_2[4n + 1] \end{aligned}$$

2. The output due to the weighted sum of the input is

$$y_4[n] = a_1 x_1[4n + 1] + a_2 x_2[4n + 1]$$

3.

$$y_3[n] = y_4[n]$$

The system is Linear.

(c) $y[n] = x[n] + \frac{1}{x(n+1)}$

$$a_1 y_1[n] = a_1 \left[x_1[n] + \frac{1}{x_1(n+1)} \right]$$

$$a_2 y_2[n] = a_2 \left[x_2[n] + \frac{1}{x_2(n+1)} \right]$$

1. The weighted sum of the output $y_3[n]$ is,

$$\begin{aligned} y_3[n] &= a_1 y_1[n] + a_2 y_2[n] \\ &= a_1 \left[x_1[n] + \frac{1}{x_1(n+1)} \right] + a_2 \left[x_2[n] + \frac{1}{x_2(n+1)} \right] \end{aligned}$$

2. The output due to the weighted sum of the input is

$$\begin{aligned} y_4[n] &= f[a_1 x_1[n] + a_2 x_2[n]] \\ &= a_1 [x_1[n] + a_2 x_2[n]] + \left[\frac{1}{a_1 x_1[n+1] + a_2 x_2[n+1]} \right] \end{aligned}$$

3.

$$y_3[n] \neq y_4[n]$$

The system is Non-linear.

(d) $y[n] = x[n^2]$

$$a_1 y_1[n] = a_1 x_1[n^2]$$

$$a_2 y_2[n] = a_2 x_2[n^2]$$

1. The weighted sum of the output $y_3[n]$ is

$$\begin{aligned} y_3[n] &= a_1 y_1[n] + a_2 y_2[n] \\ &= a_1 x_1[n^2] + a_2 x_2[n^2] \end{aligned}$$

2. The output $y_4[n]$ due to the weighted sum of input is,

$$y_4[n] = a_1 x_1[n^2] + a_2 x_2[n^2]$$

- 3.

$$y_3[n] = y_4[n]$$

The system is Linear.

(e) $y[n] = x[n] + nx[n + 1]$

$$a_1 y_1[n] = a_1 [x_1[n] + nx_1[n + 1]]$$

$$a_2 y_2[n] = a_2 [x_2[n] + nx_2[n + 1]]$$

1. The weighted sum of the output is,

$$\begin{aligned} y_3[n] &= a_1 y_1[n] + a_2 y_2[n] \\ &= a_1 [x_1[n] + nx_1[n + 1]] + a_2 [x_2[n] + nx_2[n + 1]] \end{aligned}$$

2. The output due to the weighted sum of the input is

$$y_4[n] = a_1 x_1[n] + a_2 x_2[n] + a_1 nx_1[n + 1] + a_2 nx_2[n + 1]$$

- 3.

$$y_3[n] = y_4[n]$$

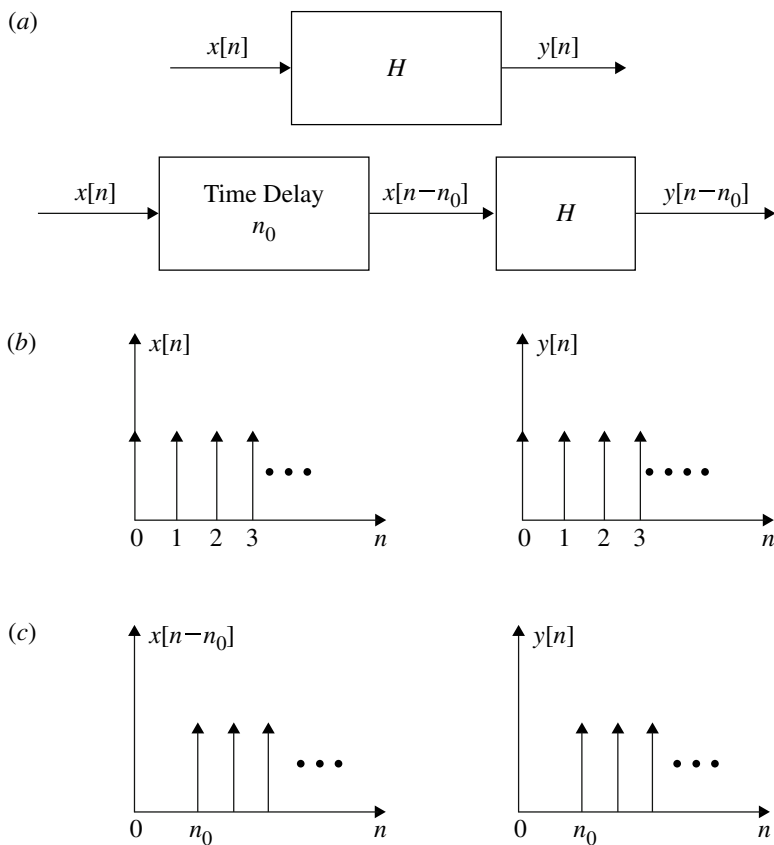


Fig. 1.32 Block diagram and signal representation to illustrate time invariancy of DT system

The system is Linear.

1.8.2 Time Invariant and Time Varying DT Systems

Consider the discrete time system represented in the block diagram of Fig. 1.32a. If the input is $x[n]$, then the output is $y[n]$. If the input is time delayed by n_0 , which becomes $x[n - n_0]$, the output becomes $y[n - n_0]$. The signal representation and the delayed signals are shown in Fig. 1.32b and c, respectively. Such systems are called time invariant.

If an arbitrary excitation $x[n]$ of a system causes a response $y[n]$ and the delayed excitation $x[n - n_0]$ where n_0 is any arbitrary integer causes $y[n - n_0]$ then the system is said to be time invariant.

Procedure to Check Time Invariance of DT Systems

1. For the delayed input $x[n - n_0]$ find the output $y[n, n_0]$.
2. Obtain the delayed output $y[n - n_0]$ by substituting $n = n - n_0$ in $y[n]$.
3. If $y[n, n_0] = y[n - n_0]$, the system is time invariant. If not, the system is time variant.

The following examples illustrate the method of testing the time invariance of DT systems.

Example 1.29 Determine whether the following systems are time invariant or not:

- (a) $y[n] = nx[n]$
- (b) $y[n] = x[2n]$
- (c) $y[n] = x[-n]$
- (d) $y[n] = \sin(x[n])$
- (e) $y[n] = x[n]x[n - 1]$

Solution (a) $y[n] = nx[n]$

1. The output for the delayed input $x[n - n_0]$ is

$$y[n, n_0] = nx[n - n_0]$$

2. The delayed output for the input $x[n]$ is

$$y[n - n_0] = (n - n_0)x[n - n_0]$$

- 3.

$$y[n, n_0] \neq y[n - n_0]$$

The system is Time Variant.

(b) $y[n] = x[2n]$

$$\begin{aligned} y[n, n_0] &= x[2n - n_0] \\ y[n - n_0] &= x[2(n - n_0)] \\ &= x[2n - 2n_0] \\ y[n, n_0] &\neq y[n - n_0] \end{aligned}$$

The system is Time Variant.

(c) $y[n] = x[-n]$

$$\begin{aligned} y[n, n_0] &= x[-n - n_0] \\ y[n - n_0] &= x[-(n - n_0)] \\ &= x[-n + n_0] \\ y[n, n_0] &\neq y[n - n_0] \end{aligned}$$

The system is Time Variant.

(d) $y[n] = \sin(x[n])$

$$\begin{aligned} y[n, n_0] &= \sin(x[n - n_0]) \\ y[n - n_0] &= \sin(x[n - n_0]) \\ y[n, n_0] &= y[n - n_0] \end{aligned}$$

The system is Time Invariant.

(e) $y[n] = x[n]x[n - 1]$

$$\begin{aligned} y[n, n_0] &= x[n - n_0]x[n - n_0 - 1] \\ y[n - n_0] &= x[n - n_0]x[n - n_0 - 1] \\ y[n, n_0] &= y[n - n_0] \end{aligned}$$

The system is Time Invariant.

1.8.3 Causal and Non-causal DT Systems

A discrete time system is said to be causal if the response of the system depends on the present or the past inputs applied. The system is non-causal if the output depends on the future input.

The following examples illustrate the method of identifying causal and non-causal systems.

Example 1.30 Determine whether the following systems are causal or not:

- (a) $y[n] = x[n - 1]$
- (b) $y[n] = x[n] + x[n - 1]$
- (c) $y[n - 1] = x[n]$
- (d) $y[n] = \sin(x[n])$
- (e) $y[n] = \sum_{k=-\infty}^{n+4} x(k)$
- (f) $y[n] = \sum_{k=0}^{-3} x(k)$

Solution (a) $y[n] = x[n - 1]$

$$y[0] = x[-1]$$

$$y[1] = x[0]$$

The output depends on the past value of $x[n]$. Hence

The system is Causal.

(b) $y[n] = x[n] + x[n - 1]$

$$y[0] = x[0] + x[-1]$$

$$y[1] = x[1] + x[0]$$

here $x[1]$ is present value and $x[0]$ is past value. The output depends on the present and past inputs. Hence

The system is Causal.

(c) $y[n - 1] = x[n]$

$$y[-1] = x[0] \quad (\text{Future input})$$

$$y[0] = x[1] \quad (\text{Future input})$$

The output depends on the future inputs. Hence

The system is Non-causal.

(d) $y[n] = \sin x[n]$

$$y[0] = \sin x[0]$$

$$y[-1] = \sin x[-1]$$

The output depends on the present input. Hence

The system is Causal.

$$(e) \ y[n] = \sum_{k=-\infty}^{n+4} x[k]$$

$$y[0] = \sum_{-\infty}^4 x[k]$$

$$= x[-\infty] + x[-\infty + 1] + \cdots + x[-1] + x[0] + x[1] + x[2] + x[3] + x[4]$$

$$x[-\infty] + x[-\infty + 1], \dots, x[-1] = \text{Past input}$$

$$x[0] = \text{Present input}$$

$$x[0], x[1], x[2], x[3] \text{ and } x[4] = \text{Future input}$$

The output depends on past, present and future input. Hence

The system is Non-causal.

$$(f) \ y[n] = \sum_{k=0}^{n-3} x[k]$$

$$y[0] = \sum_{k=0}^{-3} x[k]$$

$$= x[0] + x[-1] + x[-2] + x[-3]$$

$$x[0] = \text{Present input}$$

$$x[-1] = \text{Past input}$$

The output depends on the present and past input. Hence

The system is Causal.

1.8.4 Stable and Unstable Systems

A discrete time system is said to be stable, if for any bounded input, it produces a bounded output. This implies that the impulse response

$$y[n] = \sum_{-\infty}^{\infty} |h[n]| < \infty$$

is absolutely summable.

For a bounded input

$$|x[n]| \leq M_x < \infty$$

the output

$$|y[n]| \leq M_y < \infty$$

From the above two conditions, it can be obtained that

$$y[n] = \sum_{-\infty}^{\infty} |h[n]| < \infty$$

The following examples illustrate the above procedure.

Example 1.31 Check whether the DT systems described by the following equations are stable or not:

$$(a) \quad y[n] = \sin x[n]$$

$$(b) \quad y[n] = \sum_{k=0}^{n+1} x[k]$$

$$(c) \quad y[n] = e^{x[n]}$$

$$(d) \quad h[n] = 3^n u[n + 3]$$

$$(e) \quad y[n] = \delta[-n - 3]$$

$$(f) \quad y[n] = \delta[n - 1] + \delta[n] + \delta[n + 1]$$

$$(g) \quad h[n] = e^{-|n|}$$

$$(h) \quad h[n] = n u[n]$$

$$(i) \quad h[n] = 3^n u[n - 3]$$

$$(j) \quad h[n] = 2^n u[-n]$$

Solution (a) $y[n] = \sin x[n]$

If $x[n]$ is bounded, then $\sin x[n]$ is also bounded and so $y[n]$ is also bounded

The system is Stable.

(b) $y[n] = \sum_{k=0}^{n+1} x[k]$

Here as $n \rightarrow \infty$, $y[n] \rightarrow \infty$ and the output is unbounded. For bounded input, n should be a finite number. In that case, $y[n]$ is bounded and the system is stable.

The system is Stable.	for $n = \text{finite}$
The system is Unstable.	for $n = \infty$

(c) $y[n] = e^{x[n]}$

For $|x[n]|$ bounded, $e^{|x[n]|}$ is bounded and the system is stable.

The system is Stable.

(d) $h[n] = 3^n u[n + 3]$

$$\begin{aligned}
 |y[n]| &= \sum_{n=-3}^{\infty} 3^n \\
 &= (3)^{-3} + (3)^{-2} + (3)^{-1} + (3)^0 + (3)^1 + \dots + (3)^{\infty} \\
 &= \infty
 \end{aligned}$$

The output is unbounded.

The system is Unstable.

(e) $y[n] = \delta[-n - 3]$

$$\begin{aligned}
 y[n] &= \delta[-n - 3] \\
 &= 1 \quad n = -3 \\
 &= 0 \quad \text{otherwise}
 \end{aligned}$$

The system is Stable.

(f) $y[n] = \delta[n - 1] + \delta[n] + \delta[n + 1]$

$$\begin{aligned}
 y[0] &= \delta[-1] + \delta[0] + \delta[1] = 0 + 1 + 0 = 1 \\
 y[1] &= \delta[0] + \delta[1] + \delta[2] = 1 + 0 + 0 = 1 \\
 y[-1] &= \delta[-2] + \delta[-1] + \delta[0] = 0 + 0 + 1 = 1 \\
 y[-2] &= \delta[1] + \delta[2] + \delta[3] = 0 + 0 + 0 = 0 \\
 y[2] &= \delta[1] + \delta[2] + \delta[3] = 0 + 0 + 0 = 0 \\
 y[n] &= \sum_{k=-\infty}^{\infty} |h[k]| = 1 + 1 + 1 = 3 < \infty
 \end{aligned}$$

The system is Stable.

(g) $h[n] = e^{-|n|}$

$$\begin{aligned}
 y[n] &= \sum_{-\infty}^{\infty} e^{|n|} = \sum_{-\infty}^{-1} e^{|n|} + \sum_0^{\infty} e^{-|n|} \\
 &= \sum_1^{\infty} e^{-n} + \sum_0^{\infty} e^{-n} \\
 &= e^{-1} + e^{-2} + \cdots + 1 + e^{-1} + e^{-2} + \cdots \\
 &= e^{-1}[1 + e^{-1} + e^{-2} + \cdots] + 1 + e^{-1} + e^{-2} + \cdots \\
 &= e^{-1} \frac{1}{[1 - e^{-1}]} + \frac{1}{[1 - e^{-1}]} \\
 &= \frac{e^{-1}}{(1 - e^{-1})} + \frac{1}{(1 - e^{-1})} \\
 &= \frac{[1 + e^{-1}]}{[1 - e^{-1}]} < \infty
 \end{aligned}$$

The system is Stable.

(h) $h[n] = n u[n]$

$$y[n] = \sum_0^{\infty} n = 1 + 2 + \cdots + \infty = \infty$$

The system is Unstable.

(i) $h[n] = 3^n u[n - 3]$

$$y[n] = \sum_3^{\infty} 3^n = 3^3 + 3^2 + \cdots + \infty = \infty$$

The system is Unstable.

(j) $h[n] = 2^n u[-n]$

$$y[n] = \sum_{-\infty}^{-1} 2^n = \sum_1^{\infty} \left(\frac{1}{2}\right)^n$$

$$\begin{aligned}
&= \frac{1}{2} + \left(\frac{1}{2}\right)^2 + \dots \\
&= \frac{1}{2} \left[1 + \frac{1}{2} + \left(\frac{1}{2}\right)^2 + \dots \right] \\
&= \frac{1}{2} \left[\frac{1}{1 - \frac{1}{2}} \right] = 1 < \infty
\end{aligned}$$

The system is Stable.

1.8.5 Static and Dynamic Systems

A discrete time system is said to be static (memoryless or instantaneous) if the output response depends on the present value only and not on the past and future values of excitation. Discrete systems described by difference equations require memory and hence they are dynamic systems.

The following examples illustrate the method of identifying static and dynamic discrete systems.

Example 1.32 Identify whether the following systems are static or dynamic:

- (a) $y[n] = x[3n]$
- (b) $y[n] = \sin(x[n])$
- (c) $y[n-1] + y[n] = x[n]$
- (d) $y[n] = \text{sgn}|x[n]|$

Solution (a) $y[n] = x[3n]$

$$\begin{aligned}
y[1] &= x[3] \\
y[-1] &= x[-3]
\end{aligned}$$

The output $y[1]$ and $y[-1]$ depends on the future value $x[3]$ and the past input $x[-3]$, respectively. Hence,

The system is Dynamic.

(b) $y[n] = \sin(x[n])$

$$\begin{aligned}
y[0] &= \sin(x[0]) \\
y[1] &= \sin(x[1])
\end{aligned}$$

The output depends on the present input at all times. Hence,

The system is Static.

(c) $y[n - 1] + y[n] = x[n]$

The system is described by first order difference equation which require memory.
Hence,

The system is Dynamic.

(d) $y[n] = \text{sgn}|x[n]|$

$$\begin{aligned} \text{sgn}|x[n]| &= 1 && \text{for } n > 0 \\ &= -1 && \text{for } n < 0 \\ y[1] &= x[1] = 1 \\ y[-1] &= x[-1] = -1 \end{aligned}$$

The output depends on the present value of the input. Hence,

The system is Static.

1.8.6 Invertible and Inverse Discrete Time Systems

A discrete time system is said to be invertible if distinct input leads to distinct output. If a system is invertible then an inverse system exists.

Consider the system shown in Fig. 1.33. The input $x[n]$ produces the output $y[n]$. This system is in cascade with its inverse system. The output of this system is nothing but the difference of the two successive inputs $y[n] - y[n - 1]$. This is the input to the original system. Thus, by connecting an inverse system in cascade with the original system, the excitation signal $x[n]$ is re-established provided the original system is invertible. The concept of invertibility is very widely used in communications.

Example 1.33 Determine whether the following systems are static, causal, time invariant, linear and stable.

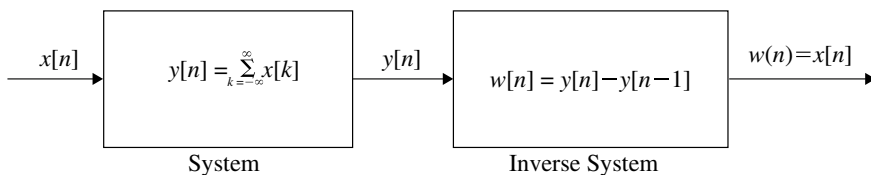


Fig. 1.33 Inverse discrete time system

- (a) $y[n] = x[4n + 1]$
- (b) $y[n] = x[n] + n x[n + 1]$
- (c) $y[n] = x[n]u[n]$
- (d) $y[n] = \log_{10} x[n]$
- (e) $y[n] = x^2[n]$

Solution (a) $y[n] = x[4n + 1]$

1.

$$y[0] = x[1]$$

The output depends on future input. Hence,

The system is Dynamic and Non-causal.

2. The output due to the delayed input is,

$$y[n, n_0] = x[4n - n_0 + 1]$$

The delayed output due to the input is,

$$\begin{aligned} y[n - n_0] &= x[4(n - n_0) + 1] \\ &= x[4n - 4n_0 + 1] \\ y[n, n_0] &\neq y[n - n_0] \end{aligned}$$

The system is Time Variant.

3.

$$\begin{aligned} a_1 y_1[n] &= a_1 x_1[4n + 1] \\ a_2 y_2[n] &= a_2 x_2[4n + 1] \\ y_3[n] &= a_1 y_1[n] + a_2 y_2[n] \\ &= a_1 x_1[4n + 1] + a_2 x_2[4n + 1] \\ y_4[n] &= a_1 x_1[4n + 1] + a_2 x_2[4n + 1] \\ y_3[n] &= y_4[n] \end{aligned}$$

The system is Linear.

4. The input is time shifted and time compressed signal. As long as the input is bounded the output is also bounded.

The system is Stable.

The system is

(1) Dynamic, (2) Non-causal, (3) Time Variant, (4) Linear and (5) Stable.

(b) $y[n] = x[n] + nx[n + 1]$

1.

$$y[0] = x[0] + 0 \times x[1]$$

The output depends on present and future inputs.

The system is Dynamic and Non-causal.

2. The output due to the delayed input is,

$$y[n, n_0] = x[n - n_0] + nx[n - n_0 + 1]$$

The delayed output due to the input is

$$y[n - n_0] = x[n - n_0] + (n - n_0)x[n - n_0 + 1]$$

$$y[n, n_0] \neq y[n - n_0]$$

The system is Time Variant.

3. The weighted sum of the output due to the input is

$$\begin{aligned} y_3[n] &= a_1 y_1[n] + a_2 y_2[n] = a_1 x_1[n] + a_1 n x_1[n + 1] + a_2 x_2[n] \\ &\quad + a_2 n x_2[n + 1] \end{aligned}$$

The output due to the weighted sum of the input is

$$y_4[n] = a_1 \{x_1[n] + nx_1[n + 1]\} + a_2 \{x_2[n] + nx_2[n + 1]\}$$

$$y_3[n] = y_4[n]$$

The system is Linear.

4. As long as $x[n]$ is bounded, $y[n]$ increases as n increases and the system is unstable.

The system is Unstable.

The system is

(1) Dynamic, (2) Non-causal, (3) Time Variant, (4) Linear and (5) Unstable.

(c) $y[n] = x[n]u[n]$

1.

$$y[0] = x[0]u[0]$$

$$y[1] = x[1]u[1]$$

The output depends on the present input only.

The system is Static and Causal.

2. For a causal signal $n \geq 0$. The weighted sum of the output due to input is,

$$\begin{aligned} y_3[n] &= a_1y_1[n] + a_2y_2[n] \\ &= \{a_1x_1[n] + a_2x_2[n]\}u[n] \end{aligned}$$

The output due to the weighted sum of input is,

$$\begin{aligned} y_4[n] &= \{a_1x_1[n] + a_2x_2[n]\}u[n] \\ y_3[n] &= y_4[n] \end{aligned}$$

The system is Linear.

3. The output due to the delayed input is

$$y[n, n_0] = x_1[n - n_0]u[n]$$

The delayed output due to the input is

$$\begin{aligned} y[n - n_0] &= x_1[n - n_0]u[n - n_0] \\ y[n, n_0] &\neq y[n - n_0] \end{aligned}$$

The system is Time Variant.

4. As long as $x[n]$ is bounded, $y[n]$ is also bounded.

The system is Stable.

The system is

(1) Static, (2) Causal, (3) Linear, (4) Time Variant and (5) Stable.

(d) $y[n] = \log_{10} x[n]$

1.

$$\begin{aligned} y[0] &= \log_{10} x[0] \\ y[1] &= \log_{10} x[1] \\ y[-1] &= \log_{10} x[-1] \end{aligned}$$

The output depends on the present input only.

The system is Static and Causal.

2. The weighted sum of the output due to input is

$$\begin{aligned} y_3[n] &= a_1 y_1[n] + a_2 y_2[n] \\ &= a_1 \log_{10} x_1[n] + a_2 \log_{10} x_2[n] \end{aligned}$$

The output due to the weighted sum of input is

$$\begin{aligned} y_4[n] &= \log_{10}(a_1 x_1[n] + a_2 x_2[n]) \\ y_3[n] &\neq y_4[n] \end{aligned}$$

The system is Non-linear.

3. The output due to the delayed input is,

$$y[n, n_0] = \log_{10}[n - n_0]$$

The delayed output due to input is

$$\begin{aligned} y[n - n_0] &= \log_{10}[n - n_0] \\ y[n, n_0] &= y[n - n_0] \end{aligned}$$

The system is Time Invariant.

4. As long as $x[n]$ is bounded, $\log_{10} x[n]$ is bounded and $y[n]$ is also bounded.

The system is Stable.

The system is

(1) Static, (2) Causal, (3) Non-linear, (4) Time invariant and (5) Stable.

(e) $y[n] = x^2[n]$

1.

$$y[0] = x^2[0]$$

$$y[1] = x^2[1]$$

The output depends on the present input only.

The system is Static and Causal.

2. The weighted sum of the output due to input is,

$$y_3[n] = a_1 y_1[n] + a_2 y_2[n]$$

$$= a_1 x_1^2[n] + a_2 x_2^2[n]$$

The output due to weighted sum of input is

$$\begin{aligned} y_4[n] &= \{a_1 x_1[n] + a_2 x_2[n]\}^2 \\ &= a_1^2 x_1^2[n] + a_2^2 x_2^2[n] + 2a_1 a_2 x_1[n] x_2[n] \\ y_3[n] &\neq y_4[n] \end{aligned}$$

The system is Non-linear.

3. The output due to the delayed input is

$$y[n, n_0] = x^2[n - n_0]$$

The delayed output due to the input is

$$y[n - n_0] = x^2[n - n_0]$$

$$y[n, n_0] = y[n - n_0]$$

The system is Time Invariant.

4. If $x[n]$ is bounded, $x^2[n]$ is bounded and $y[n]$ is also bounded.

The system is Stable.

The system is

(1) Static, (2) Causal, (3) Non-linear, (4) Time invariant and (5) Stable.

1.9 Sampling

1.9.1 Sampling Process

Sampling is a process of converting a continuous time signal into discrete time signal. After sampling, the signal is defined at discrete instants of time and the time interval between two subsequent sampling instants is called sampling interval. In this process of sampling, one of the important factors that we have to consider is that the sampling rate must be kept sufficiently high so that the original signal can be reconstructed from its samples.

The sampling operation can be represented by a switch as shown in Fig. 1.34. The switch is closed for a very short interval of time τ , during which the signal is available at the out. Therefore, if the input is $x(t)$, then the sampled output is $x(nT)$, $n = 0, \pm 1, \pm 2, \dots$ and $x(nT)$ is called the sampled sequence of $x(t)$ where T is the time interval between successive samples and the sampling frequency is given by $f_s = \frac{1}{T}$ Hz. Therefore, the discrete time signal

$$x(nT) = x_s(t) = x(t)\delta_T(t) \quad (1.21)$$

where

$$\delta_T(t) = \sum_{n=-\infty}^{\infty} \delta(t - nT) = \frac{1}{T} \sum_{m=-\infty}^{\infty} e^{jm\Omega_0 t} \quad (1.22)$$

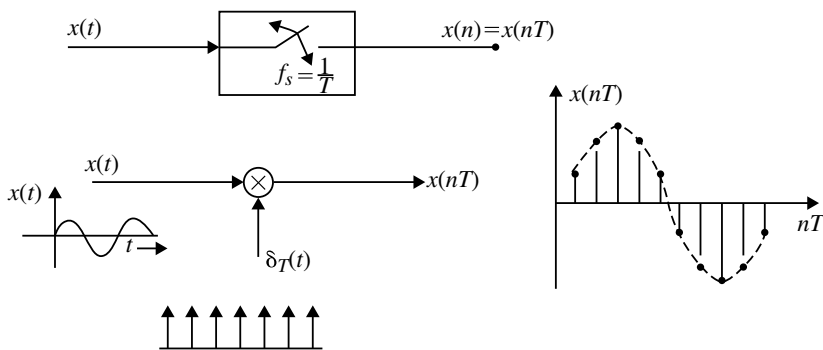
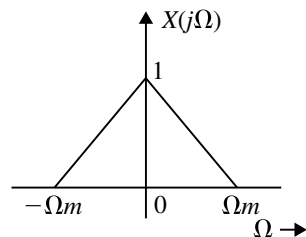


Fig. 1.34 Sampling process

Fig. 1.35 Spectrum of continuous time signal



Equation (1.22) is a Fourier series representation of impulse, $[c_k = \frac{1}{T}]$

$$x_s(t) = x(t) \left[\frac{1}{T} \sum_{m=-\infty}^{\infty} e^{jm\Omega_0 t} \right]$$

$$x_s(t) = \frac{1}{T} \sum_{m=-\infty}^{\infty} x(t) e^{jm\Omega_0 t} \quad (1.23)$$

$$F[x_s(t)] = \frac{1}{T} \sum_{m=-\infty}^{\infty} F[x(t) e^{jm\Omega_0 t}] \quad (1.24)$$

If $F[x(t)] = X(j\Omega)$ then

$$F[x(t) e^{jm\Omega_0 t}] = X(j(\Omega - m\Omega_0)) \quad (1.25)$$

$$\therefore F[x_s(t)] = \frac{1}{T} \sum_{m=-\infty}^{\infty} X(j\Omega - m\Omega_0) \quad (1.26)$$

$$X_s(j\Omega) = F[x_s(t)] = \frac{1}{T} \sum_{m=-\infty}^{\infty} X \left[j \left(\Omega - \frac{2\pi}{T} m \right) \right] \quad (1.27)$$

Thus, the Fourier transform of the sampled signal is given by an infinite sum of shifted replicas of the Fourier transform of the original signal.

Consider the signal $x(t)$ is band limited to f_m , i.e. the highest frequency component of $x(t)$ is f_m . Then $X(j\Omega) = 0$ for $|\Omega| > \Omega_m$ is shown in Fig. 1.35.

In Eq. (1.27), the term $X[j(\Omega - \frac{2\pi m}{T})]$ is the shifting of $X(j\Omega)$ from $\Omega = 0$ to $\Omega = \frac{2\pi m}{T}$ where $m = 0 \pm 1, \pm 2, \dots$

Case (i) If $\frac{\pi}{T} = \Omega_m$, the spectrum of the sampled signal is shown in Fig. 1.36. The replicas will not overlap.

As a result, the frequency spectrum of $T X_s(j\Omega)$ in the frequency range $(-\frac{\pi}{T}, \frac{\pi}{T})$ is identical to $X(j\Omega)$. The frequency spectrum $X(j\Omega)$ can be recovered from $X_s(j\Omega)$, by using a lowpass filter which has a sharp cut off at $\Omega = \frac{\pi}{T}$.

Case (ii) If $\frac{\pi}{T} > \Omega_m$, the spectrum of the sampled signal is shown in Fig. 1.37. The replicas will not overlap.

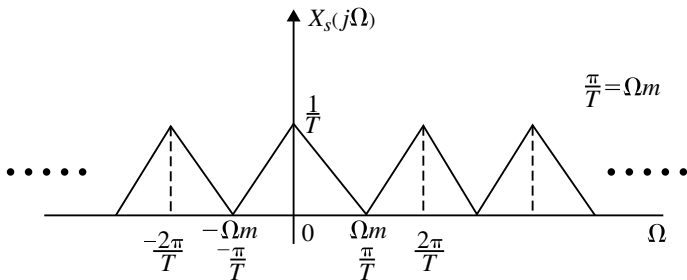


Fig. 1.36 Spectrum of sampled signal for $\frac{\pi}{T} = \Omega m$

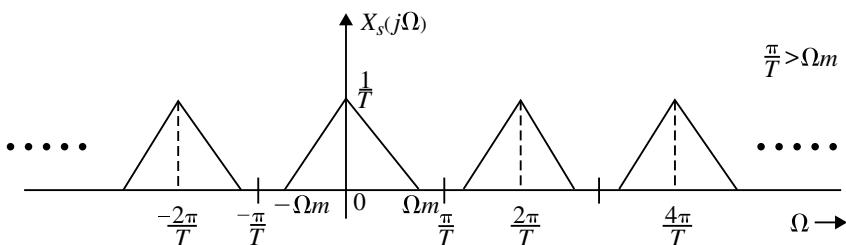


Fig. 1.37 Spectrum of sampled signal for $\frac{\pi}{T} > \Omega m$

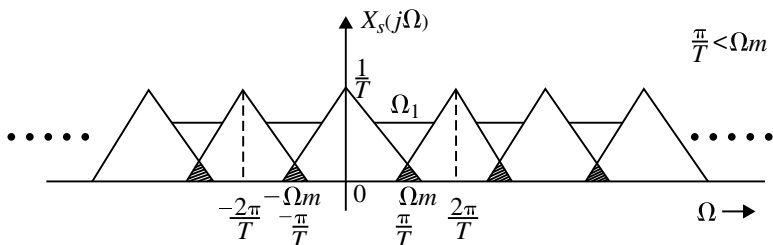


Fig. 1.38 Spectrum of sampled signal for $\frac{\pi}{T} < \Omega m$

The frequency spectrum $X(j\Omega)$ can be recovered from $X_s(j\Omega)$.

Case (iii) If $\frac{\pi}{T} < \Omega m$, the overlapping of successive frequency replicas is shown in Fig. 1.38.

As a result, the frequency spectrum $X(j\Omega)$ will not be recovered from the frequency spectrum of $T X(j\Omega)$.

For example, consider an arbitrary point at Ω_1 , which falls in the region of the overlap. The frequency spectrum at $\Omega = \Omega_1$ is the sum of two components, one component is $X_s(j\Omega_1)$ and the other component is $X_s[j(\frac{2\pi}{T} - \Omega_1)]$. This high frequency component ($\frac{2\pi}{T} - \Omega_1$) is folded in about the folding frequency $\frac{\pi}{T}$ and appears as a low frequency at Ω_1 . The super imposition of high frequency component on the low frequency is known as “frequency aliasing”. Because of aliasing, the spectrum

$X(j\Omega)$ is no longer recoverable from the spectrum of $X_s(j\Omega)$. The aliasing error can be prevented if the highest frequency component Ω_m in the signal $x(t)$ is less than or equal to $\frac{\pi}{T}$. That is

$$\frac{\pi}{T} \geq \Omega_m \quad (1.28)$$

If $f_s = \frac{1}{T}$. Then

$$\pi f_s \geq \Omega_m \quad (1.29)$$

$$\pi f_s \geq 2\pi f_m \quad (1.30)$$

$$\boxed{f_s \geq 2f_m} \quad (1.31)$$

To avoid aliasing, the sampling frequency must be greater than twice the highest frequency present in the signal.

1.9.2 Sampling Theorem

A band limited signal $x(t)$ with $X(j\Omega) = 0$ for $|\Omega| > \Omega_m$ is uniquely determined from its sample $x(nT)$, if the sampling frequency $f_s \geq 2f_m$, i.e. sampling frequency must be at least twice the highest frequency present in the signal.

1.9.3 Nyquist Rate

The minimum rate at which a signal can be sampled and still be reconstructed from its samples is called the Nyquist rate. It is always equal to $2f_m$, where f_m is the maximum frequency component present in the signal.

$$\boxed{\text{Nyquist rate} = 2f_m} \quad (1.32)$$

A signal sampled at greater than the Nyquist rate is said to be over sampled and a signal sampled at less than the Nyquist rate is said to be under sampled.

1.9.4 Anti-aliasing Filter

Sampling theorem states that a signal can be perfectly reconstructed if it is band limited. In practice, communication signals have frequency spectra consisting of low frequency components as well as high frequency noise components. When a signal

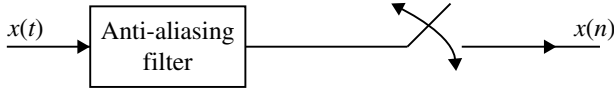


Fig. 1.39 Anti-aliasing filter before the sampler

is sampled, with sampling frequency f_s , all signals with frequency range higher than f_s appear as a signal frequency between 0 and $\frac{\Omega_s}{2}$ creating aliasing.

Therefore, to avoid aliasing errors caused by the undesired high frequency signals, an analog lowpass filter called an anti-aliasing filter is used prior to the sampler.

A filter that is used to reject high frequency signals before it is sampled to reduce the aliasing is called an anti-aliasing filter and is shown in Fig. 1.39.

1.9.5 Signal Reconstruction

The band limited signal $x(t)$ can be reconstructed from its samples if the sampling rate is above the Nyquist rate.

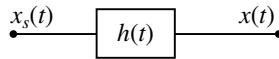
Let the sampled signal be $x_s(t)$.

$$x_s(t) = \sum_{n=-\infty}^{\infty} x(nT)\delta(t - nT) \quad (1.33)$$

$$F[x_s(t)] = \sum_{n=-\infty}^{\infty} F[x(nT)\delta(t - nT)] \quad (1.34)$$

$$X(j\omega) = \frac{1}{T} \sum_{m=-\infty}^{\infty} X\left(j\left(\Omega - \frac{2\pi m}{T}\right)\right) \quad (1.35)$$

To get the original spectrum, the signal $x_s(t)$ is to be passed through a reconstruction (lowpass) filter with an impulse response $h(t)$.

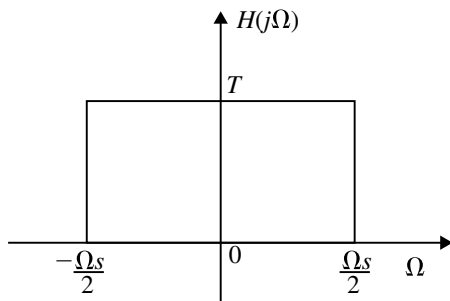


The original signal can be obtained by convolving the sampled signal $x_s(t)$ and the impulse response $h(t)$ of the filter

$$x(t) = x_s(t) * h(t) \quad (1.36)$$

$$= \left[\sum_{n=-\infty}^{\infty} x(nT)\delta(t - nT) \right] * h(t) \quad (1.37)$$

Fig. 1.40 Spectrum of lowpass filter



$$x(t) = \int_{-\infty}^{\infty} \sum_{n=-\infty}^{\infty} x(nT) \delta(\lambda - nT) h(t - \lambda) d\lambda \quad (1.38)$$

$$= \sum_{n=-\infty}^{\infty} x(nT) \int_{-\infty}^{\infty} \delta(\lambda - nT) h(t - \lambda) d\lambda \quad (1.39)$$

$$\therefore \delta(\lambda - nT) = \begin{cases} 1, & \lambda = nT \\ 0, & \lambda \neq nT \end{cases}$$

$$x(t) = \sum_{n=-\infty}^{\infty} x(nT) h(t - nT) \quad (1.40)$$

The frequency response of the lowpass filter can be expressed by

$$H(j\Omega) = \begin{cases} T, & \text{for } |\Omega| \leq \frac{\Omega_s}{2} \\ 0, & \text{for } |\Omega| > \frac{\Omega_s}{2} \end{cases} \quad (1.41)$$

The spectrum of ideal lowpass filter is shown in Fig. 1.40.

The impulse response of the lowpass filter is

$$\begin{aligned} h(t) &= F^{-1}[H(j\Omega)] \\ &= \frac{1}{2\pi} \int_{-\frac{\Omega_s}{2}}^{\frac{\Omega_s}{2}} H(j\Omega) e^{j\Omega t} d\Omega \\ &= \frac{T}{2\pi} \int_{-\frac{\Omega_s}{2}}^{\frac{\Omega_s}{2}} e^{j\Omega t} d\Omega = \frac{T}{\pi t} \sin \frac{\Omega_s t}{2} \\ &= \frac{T}{\pi T} \sin \frac{2\pi f_s t}{2} = \frac{T}{\pi t} \sin \frac{\pi t}{T} \end{aligned} \quad (1.42)$$

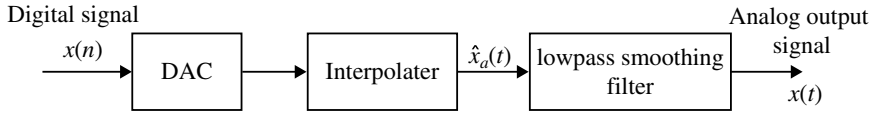


Fig. 1.41 Digital to analog conversion

$$h(t) = \frac{T}{\pi} \frac{\sin \frac{\pi t}{T}}{t} \quad (1.43)$$

$$h(t - nT) = \frac{T}{\pi} \frac{\sin \frac{\pi}{T}(t - nT)}{t - nT} \quad (1.44)$$

Substitute Eq. (1.44) in Eq. (1.39)

$$x(t) = \sum_{n=-\infty}^{\infty} x(nT) \times \frac{\sin \frac{\pi}{T}(t - nT)}{\frac{\pi}{T}(t - nT)} \quad (1.45)$$

$$x(t) = \sum_{n=-\infty}^{\infty} x(nT) \sin c \frac{\pi}{T}(t - nT) \quad (1.46)$$

The response $x(t)$ given by the formula is suitable for practical implementation. The $\sin c$ function spreads over the entire time axis. To reconstruct the signal at the time $t = t_0$, we must know all the sample values including those for $nT > t_0$.

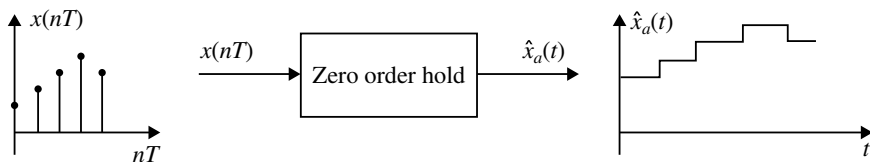
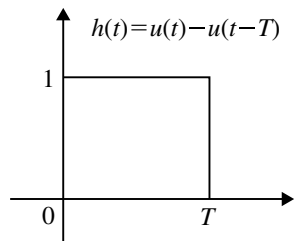
That is, $x(t_0)$ requires the knowledge of future values of $x(nT)$ so it represents a non-casual system. Hence, this type of reconstruction is not suitable for real time applications. Therefore, we need a different approach to convert digital signal to analog signal which is shown in Fig. 1.41.

The digital to analog converter produces an output voltage that is proportional to the value of the binary word. The output of DAC is applied to an interpolator which converts the output samples of DAC into an analog signal. There are several interpolators. One of the interpolators is zero order hold.

1.9.6 Sampling with Zero Order Hold

One of the most widely used interpolators is zero order hold. The block diagram and input-output of zero order hold is shown in Fig. 1.42. In this technique, a given sample is held for an interval until the next sample is received (Fig. 1.43).

$$\hat{x}_a(t) = x(n) \quad \text{for } nT \leq t \leq (n+1)T \quad n = 0, 1, 2, \dots \quad (1.47)$$

**Fig. 1.42** Input-output of zero order hold**Fig. 1.43** Output of a zero order hold

$$\begin{aligned}
 \hat{x}_a(t) &= x(0) \quad \text{for } 0 \leq t < T \\
 &= x(T) \quad \text{for } T \leq t < 2T \\
 &= x(2T) \quad \text{for } 2T \leq t < 3T
 \end{aligned}$$

The impulse response of zero order hold is given by

$$h(t) = \begin{cases} 1, & 0 \leq t \leq T \\ 0, & \text{otherwise} \end{cases} \quad (1.48)$$

The impulse response function of a zero order hold circuit is shown in Fig. 1.43.

1.9.6.1 Transfer Function of a Zero Order Hold

The output of zero order hold is expressed in terms of $h(t)$. That is the output $\hat{x}_a(t)$ is the convolution of $x(nT)$ and $h(t)$

$$\hat{x}_a(t) = \sum_{n=-\infty}^{\infty} x(nT)h(t - nT) \quad (1.49)$$

$h(t)$ can be expressed as

$$h(t) = u(t) - u(t - T) \quad (1.50)$$

Taking Laplace transform on both sides, we get

$$H(s) = U(s) - U(s)e^{-st}$$

$$\boxed{H(s) = \frac{1}{s}[1 - e^{-st}]}$$
 (1.51)

where $U(s) = \frac{1}{s}$.

The output of zero order hold consists of higher order harmonics. To remove these harmonics, the output of zero order hold is applied to a lowpass filter. This filter is often called a smoothing filter because it tends to smooth the corners on the step approximations generated by ZOH.

Example 1.34 A signal having a spectrum ranging from DC to 10 kHz is to be sampled and converted into a discrete form. What is the minimum number of samples per second that must be taken to ensure recovery?

Solution Given $f_m = 10$ kHz. From the Nyquist rate, the minimum number of samples per second that must be taken to ensure recovery is

$$\begin{aligned} f_s &= 2f_m \\ f_s &= 20,000 \text{ samples/s} \end{aligned}$$

Example 1.35 Consider the following analog signal:

$$x_a(t) = 3 \cos 100\pi t$$

- Determine the minimum sampling rate required to avoid aliasing.
- Suppose that the signal is sampled at the rate $F_s = 200$ Hz, what is the discrete time signal obtained after sampling?
- Repeat (b) if $F_s = 75$ Hz.
- What is the frequency $0 < F < \frac{F_s}{2}$ of a sinusoid that yields samples identical to those obtained in part (c).

Solution (a) Given

$$\begin{aligned} \Omega_m &= 100\pi \\ 2\pi f_m &= 100\pi \\ f_m &= 100/2 \end{aligned}$$

Hence, the minimum sampling rate required to avoid aliasing $f_s = 2f_m = 100$ Hz.

- If the signal is sampled at $f_s = 200$ Hz, the discrete time signal is $3 \cos 100\pi nT$

$$\begin{aligned}
 x(n) &= x(t)|_{t=nT} \\
 x(n) &= 3 \cos \frac{100\pi}{200} n \\
 x(n) &= 3 \cos \frac{n\pi}{2}
 \end{aligned}$$

(c) If the signal is sampled at $F_s = 75$ Hz, the discrete time signal is

$$\begin{aligned}
 x(n) &= x(t)|_{t=nT} \\
 &= 3 \cos 100\pi nT \\
 &= 3 \cos \frac{100\pi}{75} n \\
 &= 3 \cos \left(\frac{4\pi}{3} \right) n \\
 &= 3 \cos \left(2\pi - \frac{2\pi}{3} \right) n \\
 x(n) &= 3 \cos \frac{4\pi}{3} n
 \end{aligned}$$

(d) For the sampling rate of $F_s = 75$ Hz, we have

$$\begin{aligned}
 F &= \frac{F_0}{F_s} \\
 F_0 &= F \cdot F_s
 \end{aligned}$$

Therefore, in general, the sampling of CT sinusoidal signal $x(t) = A \cos(2\pi F_0 t + \theta)$ with sampling rate $F_s = (1/T)$ results in a discrete time signal $x(n) = A \cos(2\pi f_0 n + \theta)$. Therefore, relative frequency $f_0 = (F_0/F_s)$.

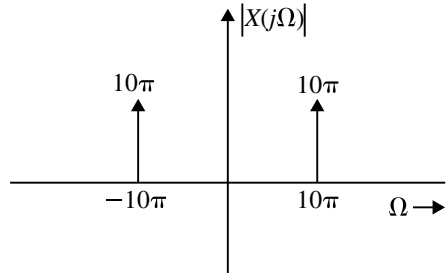
The frequency of the sinusoidal in part (c) is

$$\begin{aligned}
 F &= \frac{1}{3} \\
 \omega &= \left(\frac{2\pi}{3} \right) \\
 T &= \left(\frac{2\pi}{\omega} \right) m \\
 T &= \left(\frac{2\pi}{2\pi/3} \right) m = 3m = 3
 \end{aligned}$$

Hence

$$\begin{aligned}
 F_0 &= \frac{1}{3} \cdot 75 \\
 F_0 &= 25 \text{ Hz}
 \end{aligned}$$

Fig. 1.44 Amplitude spectrum of continuous time signal



Clearly, the sinusoidal signal

$$y_a(t) = 3 \cos 2\pi F_0 t$$

$$y_a(t) = 3 \cos 50\pi t$$

Example 1.36 The signal $x(t) = 10 \cos(10\pi t)$ is sampled at a rate of 8 samples per second. Plot the amplitude spectrum for $|\Omega| \leq 30\pi$. Can the original signal be recovered from samples? Explain.

Solution Given

$$x(t) = 10 \cos(10\pi t)$$

$$\Omega_m = 10\pi$$

$$F[x(t)] = X(j\Omega) = 10[\pi[\delta(\Omega + \Omega_0) + \delta(\Omega - \Omega_0)]]$$

$$X(j\Omega) = 10\pi[\delta(\Omega + 10\pi) + \delta(\Omega - 10\pi)]$$

The amplitude spectrum of $x(t)$ is shown in Fig. 1.44.

$$\text{The sampling rate } f_s = 8 \text{ Hz} \Rightarrow T = \frac{1}{8}$$

$$\Omega_s = 2\pi f_s = 16\pi$$

$$\Omega_m = 10\pi = 2\pi f_m$$

$$f_m = 5 \text{ Hz}$$

$$\text{Nyquist rate} = 2f_m = 10 \text{ Hz} \quad (\text{Here } f_s < 2f_m)$$

Therefore, the original signal cannot be recovered from the samples. The frequency spectrum of sampled signal $x(t)$ is given by

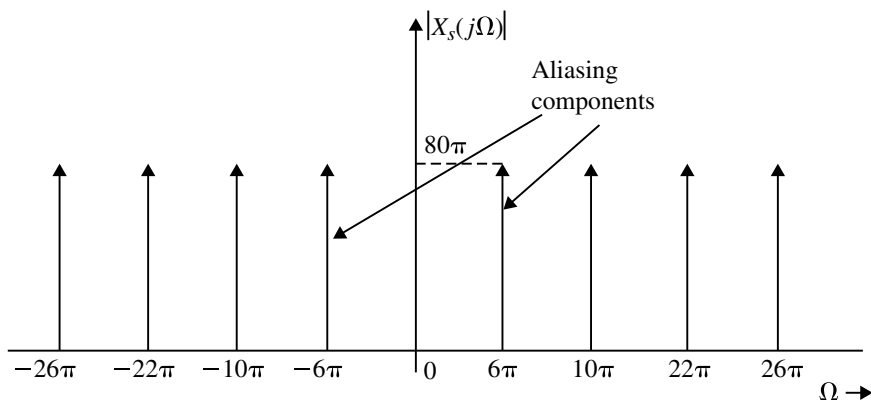


Fig. 1.45 Amplitude spectrum of sampled signal

$$\begin{aligned}
 X_s(j\Omega) &= \frac{1}{T} \sum_{n=-\infty}^{\infty} X \left[j \left(\Omega - \frac{2\pi n}{T} \right) \right] \\
 &= \frac{1}{T} \sum_{n=-\infty}^{\infty} 10\pi \left[\delta \left(\Omega + 10\pi - \frac{2\pi}{T}n \right) + \delta \left(\Omega - 10\pi - \frac{2\pi}{T}n \right) \right] \\
 X_s(j\Omega) &= 80\pi \sum_{n=-\infty}^{\infty} [\delta(\Omega + 10\pi - 16\pi n) + \delta(\Omega - 10\pi - 16\pi n)]
 \end{aligned}$$

The plot of amplitude spectrum for $|\Omega| \leq 30\pi$ is shown in Fig. 1.45.

Example 1.37 A signal $x(t) = \sin c(150\pi t)$ is sampled at a rate of

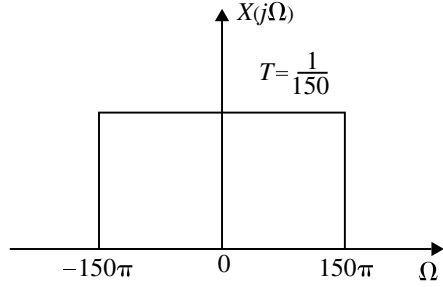
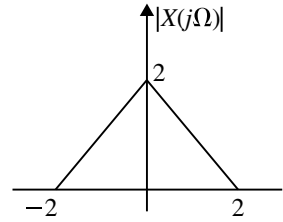
- (a) 100 Hz;
- (b) 200 Hz and
- (c) 300 Hz.

For each of these three cases, can you recover the signal $x(t)$ from the sampled signal.

Solution Given

$$\begin{aligned}
 x(t) &= \sin c(150\pi t) \\
 \Omega_m &= 150\pi \\
 2\pi f_m &= 150\pi \\
 f_m &= 75 \\
 f_s = 2f_m &\Rightarrow f_s = 150, \quad T = \frac{1}{150}
 \end{aligned}$$

The spectrum of the signal $x(t)$ is a rectangular pulse with a band width (maximum frequency components of 150π rad/s. is shown in Fig. 1.46.

Fig. 1.46 Spectrum of $x(t)$ **Fig. 1.47** Spectrum of $x(t)$ for Example 1.38

(a) In the first case, the sampling rate is 100 Hz which is less than the Nyquist rate. Therefore, $x(t)$ can be recovered from its samples (Fig. 1.47).

$$f_s = 100, \quad 2f_m = 150 \Rightarrow f_s < 2f_m$$

For (b) and (c) cases, the sampling rate is greater than the Nyquist rate. Therefore, $x(t)$ cannot be recovered from its samples. For $f_s = 200$ and $f_s = 300$

$$2f_m = 150 \quad \text{and} \quad f_s > 2f_m$$

Example 1.38 Draw $|X_s(i\Omega)|$ for the following cases when $x_s(t) = x(t)\delta_T(t)$ with sampling period T , where

$$\delta_T(t) = \sum_{n=-\infty}^{\infty} \delta(t - nT)$$

The spectrum of its signal for Example 1.38 is shown in Fig. 1.47.

- (a) $T = \frac{\pi}{3}$ s
- (b) $T = \frac{\pi}{2}$
- (c) $T = \frac{2\pi}{3}$

Solution In general, for band limited signal, the amplitude spectrum is shown in Fig. 1.48.

$$\therefore \Omega_m = 2\text{rad/s.}$$

$$2\pi f_m = 2$$

$$f_m = \frac{1}{\pi} \text{ Hz}$$

$$\begin{aligned} \text{Nyquist rate} &= 2f_m \\ &= \frac{2}{\pi} \text{ Hz} \end{aligned}$$

(a) Given

$$T = \frac{\pi}{3} \text{ s}$$

$$\begin{aligned} f_s &= \frac{1}{T} \\ &= \frac{3}{\pi} \text{ Hz} \end{aligned}$$

$$\begin{aligned} \Omega_s &= 2\pi f_s \\ &= 6 \text{ rad/s.} \end{aligned}$$

$$f_s > 2f_m \left[\because \text{since } \frac{2}{\pi} > \frac{3}{\pi} \right]$$

Therefore, the spectrum $|X_s(j\Omega)|$ is free from aliasing and is shown in Fig. 1.49.

(b) Given

$$T = \frac{\pi}{2} \text{ s}$$

$$f_s = \frac{2}{\pi} \text{ Hz}$$

$$\Omega_s = 2\pi f_s = 4 \text{ rad/s.}$$

$$2f_m = \frac{2}{\pi} \text{ Hz}$$

$$f_s = 2f_m.$$

Therefore, the spectrum $|X_s(j\Omega)|$ is free from aliasing and is shown in Fig. 1.50.

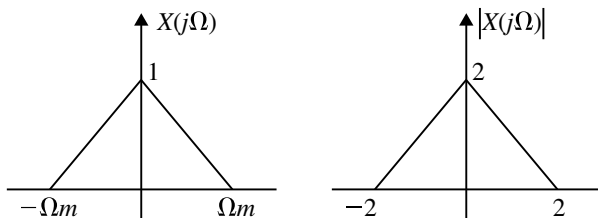


Fig. 1.48 Spectrum of bandlimited signal

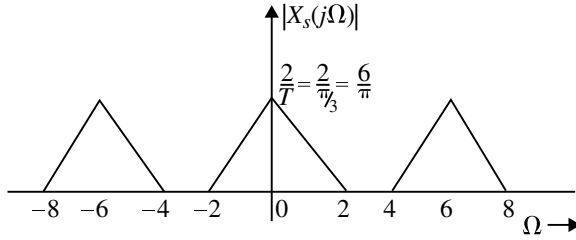


Fig. 1.49 Spectrum of sampled signal for $T = \pi/3$ s

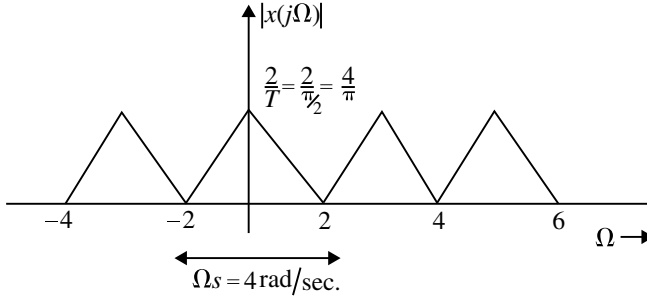
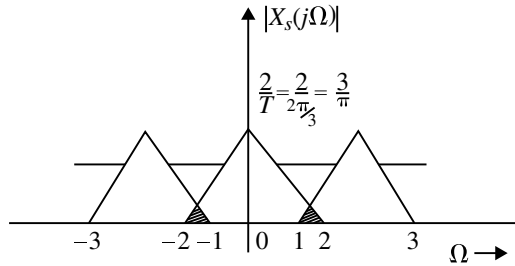


Fig. 1.50 Spectrum of sampled signal for $T = \pi/2$ s

Fig. 1.51 Spectrum of sampled signal for $T = \frac{2\pi}{3}$ s



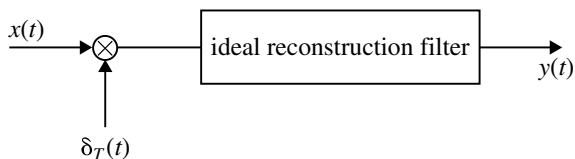
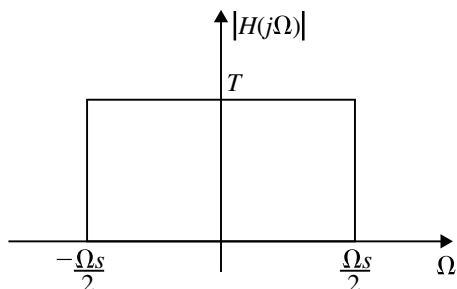
(c) Given

$$T = \frac{2\pi}{3}, \quad f_s = \frac{3}{2\pi} \text{ s}, \quad \Omega_s = 3 \text{ rad/s.}$$

$$f_s < 2f_m$$

The sampling frequency is less than the Nyquist rate. The spectra $X(j\Omega)$ repeats for every 3 rad/s and the successive spectrum overlaps. Therefore, $x(t)$ cannot be recovered from its samples. The spectrum of sampled signal for $T = 2\pi/3$ is shown in Fig. 1.51.

Example 1.39 Consider the following sampling and reconstruction block as shown in Fig. 1.52.

Fig. 1.52 Sampling and reconstruction block**Fig. 1.53** Frequency response of lowpass filter

The output of the ideal reconstruction filter can be found by sending signal $x_s(t)$ through an ideal lowpass filter having characteristics as shown in Fig. 1.53.

For the following signals, draw the spectrum of $|X_s(j\Omega)|$, and find expression for $x(nT)$ and $y(t)$:

- (i) $x(t) = 2 + \cos(100\pi t)$ for $T = 0.0125$ s.
- (ii) $x(t) = 2 + \cos(100\pi t)$ for $T = \frac{1}{150}$ s.
- (iii) $x(t) = 1 + \cos(10\pi t) + \cos 30\pi t$ for $T = 0.04$ s.
- (iv) If $X(j\Omega) = \frac{1}{2+j\Omega}$ and $T = 2$ s.

Draw $|X_s(j\Omega)|$. Test for aliasing.

Solution (i) $x(t) = 2 + \cos(100\pi t)$ for $T = 0.0125$ s.

$$\begin{aligned}
 x(t) &= 2 + \cos(100\pi t) \\
 &= 2 + \frac{1}{2} [e^{j100\pi t} + e^{-j100\pi t}] \\
 X(j\Omega) &= 4\pi\delta(\Omega) + \pi [\delta(\Omega + 100\pi) + \delta(\Omega - 100\pi)] \\
 \therefore F[1] &= 2\pi\delta(\Omega) \\
 F[e^{j\Omega_0 t}] &= 2\pi\delta(\Omega + \Omega_0)
 \end{aligned}$$

The sketch of the amplitude spectrum of $x(t)$ is shown in Fig. 1.54. The sampling time

$$\begin{aligned}
 T &= 0.0125 \\
 f_s &= \frac{1}{T}
 \end{aligned}$$

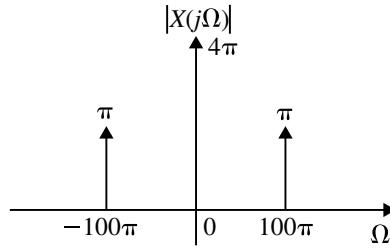
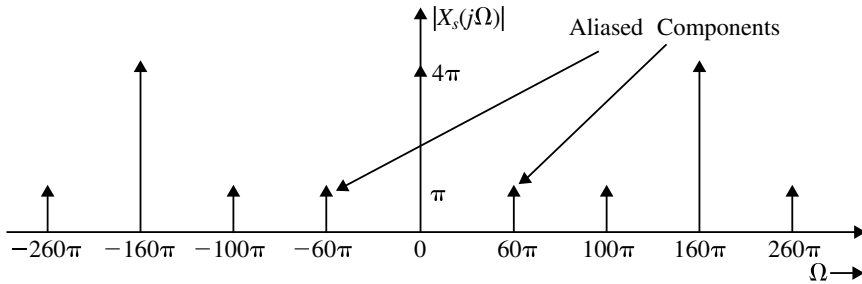
Fig. 1.54 Spectrum of $x(t)$ 

Fig. 1.55 Spectrum of sampled signal

$$\begin{aligned}
 &= \frac{1}{0.0125} \\
 &= 80 \text{ Hz} \\
 \Omega_s &= 2\pi f_s \\
 &= 160 \text{ rad/s} \\
 \Omega_m &= 100\pi
 \end{aligned}$$

Given

$$f_m = 50 \text{ Hz}$$

Nyquist rate is $2f_m = 100 \text{ Hz}$

$$f_s < 2f_m$$

Therefore, the successive spectrum $X_s(j\Omega)$ overlap and the signal $x(t)$ cannot be recovered from its samples.

The spectrum $X_s(j\Omega)$ of the sampled signal $x(t)$ is a periodic repetition of $X(j\Omega)$ for every $160\pi \text{ rad/s}$. (80 Hz) is shown in Fig. 1.55.

The frequency characteristics of the reconstruction filter (LPF) is

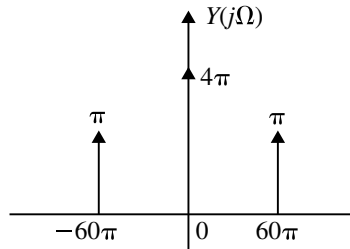
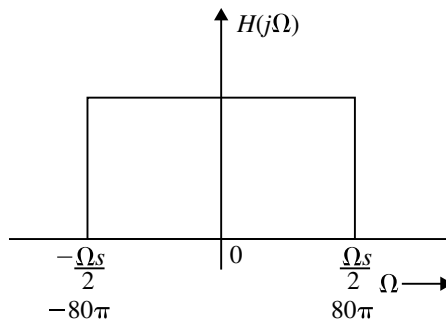


Fig. 1.56 Spectrum of output $y(t)$



When the spectrum $|X_s(j\Omega)|$ is passed through the reconstruction filter (LPF), the output of the filter consists of the frequency components given by

$$Y(j\Omega) = 4\pi\delta(\Omega) + \pi[\delta(\Omega + 60\pi) + \delta(\Omega - 60\pi)]$$

The spectrum of output $y(t)$ is shown in Fig. 1.56.

$$y(t) = 2 + \cos 60\pi t$$

$$x(nT) = x(t)|_{t=nT}$$

$$x(nT) = 2 + \cos 100\pi t|_{t=0.0125n}$$

$$= 2 + \cos(1.25)n\pi$$

$$x(nT) = 2 + \cos(0.75n\pi)$$

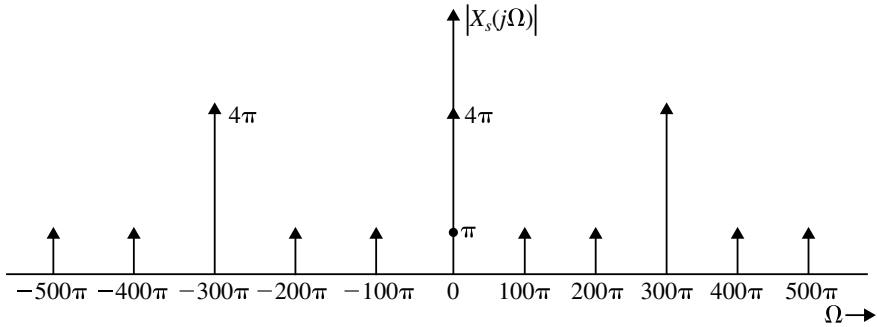


Fig. 1.57 Spectrum of sampled signal

(ii) Given $x(t) = 2 + \cos(100\pi t)$

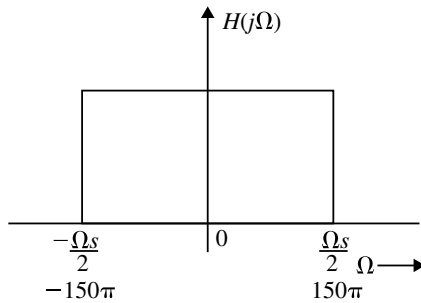
$$T = \frac{1}{150} \text{ s}$$

$$f_s = \frac{1}{T} = 150 \text{ Hz}$$

$$\Omega_s = 2\pi f_s = 300\pi \text{ rad/s.}$$

The Amplitude spectrum ($|X(j\Omega)|$) of the signal is same as Fig. 1.57. The spectrum of sampled signal of $x(t)$, is a periodic extension of $X(j\Omega)$ for every 300π (rad/s) and is shown in Fig. 1.55.

The frequency characteristics of a reconstruction filter are shown below.



The resultant output spectrum is shown in Fig. 1.55.

$$Y(j\Omega) = 4\pi\delta(\Omega) + \pi[\delta(\Omega - 100\pi) + \delta(\Omega + 100\pi)]$$

$$y(t) = 2 + \cos 100\pi t$$

$$\begin{aligned} x(nT) &= x(t) \Big|_{t=nT} \\ &= 2 + \cos 100\pi t \Big|_{t=\frac{1}{150}n} \\ x(nT) &= 2 + \cos(0.666n\pi) \end{aligned}$$

(iii) Given

$$x(t) = 1 + (\cos 10\pi t) + (\cos 30\pi t); \quad T = 0.04 \text{ s}$$

$$x(nT) = x(t) \Big|_{t=nT} = 0.04n$$

$$x(nT) = 1 + (\cos 0.4\pi t) + (\cos 0.2\pi t)$$

$$x(nT) = 1 + (\cos 0.4n\pi) + (\cos 1.2n\pi)$$

$$f_s = \frac{1}{T} = 25 \text{ Hz}, \quad \Omega_s = 2\pi f_s = 50\pi \text{ rad/s.}$$

$$\Omega_m = 2\pi f_m = 30\pi$$

$$f_m = 15 \text{ Hz}$$

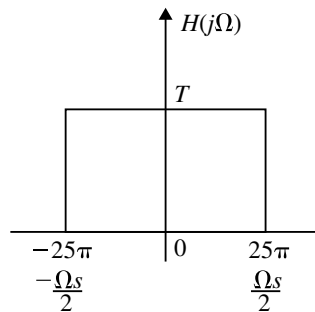
$$2f_m = 30 \text{ Hz} \quad (\because f_s < 2f_m)$$

Therefore, the successive spectrum of $X(j\Omega)$ in $X_s(j\Omega)$ overlaps and $x(t)$ cannot be recovered from $x(nT)$ (Fig. 1.58)

$$\begin{aligned} X(j\Omega) &= 2\pi\delta(\Omega) + \pi[\delta(\Omega + 10\pi) + \delta(\Omega - 10\pi)] \\ &\quad + \pi[8(\Omega + 30\pi) + (\Omega - 30\pi)] \end{aligned}$$

The sketch of the spectrum of the sampled signal is shown in Fig. 1.59.

The frequency spectrum of the lowpass filter is given as



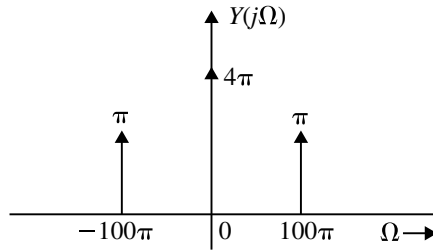


Fig. 1.58 Spectrum of output $y(t)$

The spectrum of $y(\Omega)$ is shown in Fig. 1.58

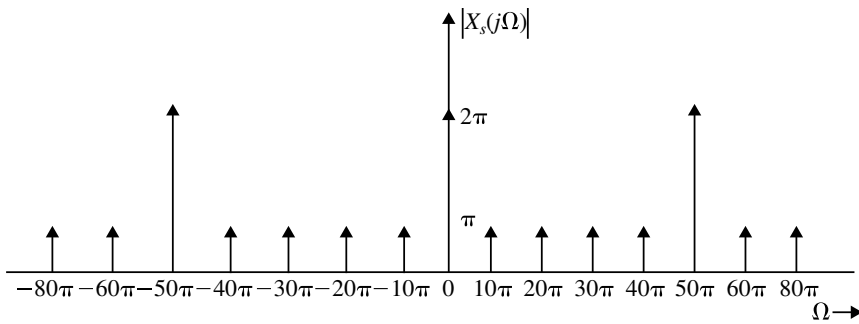
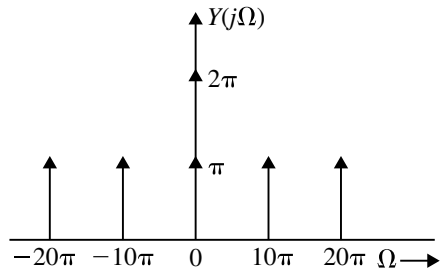


Fig. 1.59 Spectrum of sampled signal

Fig. 1.60 Spectrum of output $y(t)$



The output spectrum is shown in Fig. 1.60. From that, we can obtain

$$\begin{aligned}
 Y(j\Omega) &= 2\pi\delta(\Omega) + \pi[\delta(\Omega + 10\pi) + \delta(\Omega - 10\pi)] \\
 &\quad + \pi[\delta(\Omega + 20\pi) + \delta(\Omega - 20\pi)] \\
 y(t) &= 1 + \cos(10\pi t) + \cos(20\pi t)
 \end{aligned}$$

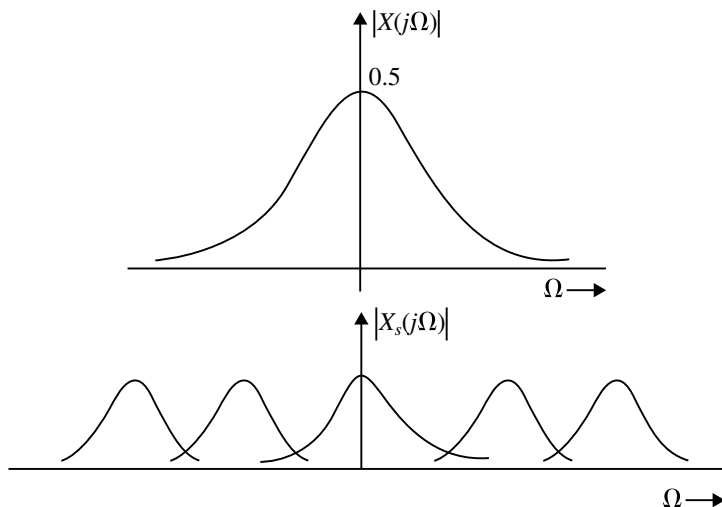


Fig. 1.61 Spectrum of sampled signal

(iv) Given $T = 2$ s, $f_s = 0.5$ Hz

$$\begin{aligned}\Omega_s &= \pi \text{ rad/s} \\ X(j\Omega) &= \frac{1}{2 + j\Omega} \\ |X(j\Omega)| &= \frac{1}{\sqrt{4 + \Omega^2}}\end{aligned}$$

The spectrum of $|X(j\Omega)|$ is not band limited. Therefore, aliasing of successive frequency components occurs even Ω is large. The spectra $|X_s(j\Omega)|$ is shown in Fig. 1.61.

Example 1.40 Determine the Nyquist sampling rate and Nyquist sampling intervals for the following signals:

- (i) $\sin^2(200\pi t)$
- (ii) $0.5 \sin^2(200\pi t)$
- (iii) $\sin^2 c(200\pi t) + 3 \sin^2 c(120\pi t)$
- (iv) $\sin c(100\pi t) \sin c(200\pi t)$

Solution The Fourier transform of the triangular function is the square of $\sin c$ function.

Similarly, Fourier transform of a rectangular pulse is a $\sin c$ function. This is shown in Figs. 1.62 and 1.63, respectively.

(i) Given,

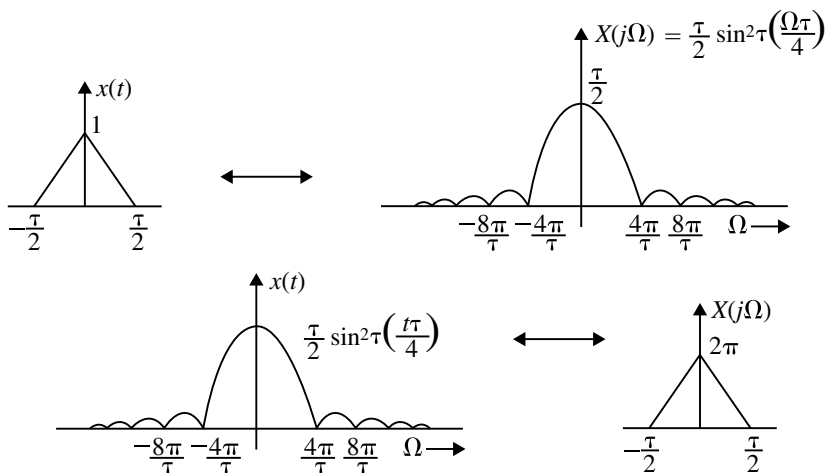


Fig. 1.62 a Fourier transform of triangular function. b Dual property of figure a

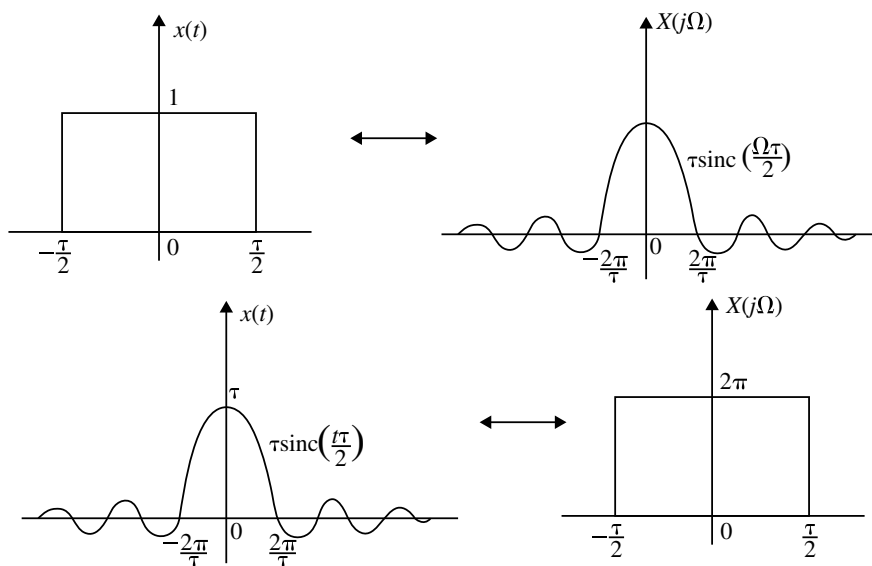


Fig. 1.63 a Fourier transform of rectangular pulse. b Dual property of figure a

$$x(t) = \sin c^2(200\pi t)$$

Comparing the signal $x(t)$ with the above signal, we get

$$\begin{aligned} x(t) &= \frac{\tau}{2} \sin c^2\left(\frac{t\tau}{4}\right) \\ \frac{t\tau}{4} &= 200\pi t \\ \tau &= 800\pi \\ \frac{\tau}{2} &= 400\pi \end{aligned}$$

That is the maximum frequency component present in the signal is 400π rad/s (or) the signal is band limited to 400π rad/s.

$$\Omega_m = 400\pi \quad \text{and} \quad f_m = 200 \text{ Hz}$$

$$\text{Nyquist rate } f_s = 2f_m = 400 \text{ Hz.}$$

Sampling interval $T = \frac{1}{f_s} = \frac{1}{400} = 2.5 \text{ ms.}$

(ii) Same values as above.

(iii) Given

$$x(t) = \sin c(200\pi t) + 3 \sin c^2(120\pi t)$$

Let

$$x_1(t) = \sin c(200\pi t)$$

$$x_2(t) = 3 \sin^2 c(120\pi t)$$

Compare the signal $x_1(t)$ with the signal $x(t) = \tau \sin c\left(\frac{\tau t}{2}\right)$

$$\begin{aligned} \frac{\tau t}{2} &= 200\pi t \\ \frac{\tau}{2} &= 200\pi \text{ rad/s} \\ \therefore \Omega m_1 &= 200 \text{ rad/s} \\ f m_1 &= 100 \text{ Hz} \end{aligned}$$

Compare the signal $x_2(t)$ with the signal $x(t) = \frac{\tau}{2} \sin c^2\left(\frac{\tau t}{4}\right)$,

$$\begin{aligned}
 \frac{\tau t}{4} &= 120\pi t \\
 \frac{\tau}{4} &= 120\pi \\
 \frac{\tau}{2} &= 240\pi \\
 \therefore \Omega m_2 &= 240\pi \text{ rad/s.}
 \end{aligned}$$

When both signals are added, the maximum frequency components is

$$\begin{aligned}
 \Omega_m &= \Omega m_2 = 240\pi \text{ rad/s} \\
 fm &= 120 \text{ Hz} \\
 \text{Nyquist rate } f_s &= 2fm = 240 \text{ Hz} \\
 \text{Nyquist interval} &= \frac{1}{f_s} = 4.167 \text{ m s.}
 \end{aligned}$$

(iv) $x(t) = \sin c(100\pi t) \sin c(200\pi t)$ Let

$$x_1(t) = \sin c(100\pi t); x_2(t) = \sin c(200\pi t)$$

The signal $x_1(t)$ is band limited to frequency 100π rad/s and the signal $x_2(t)$ is bandlimited to frequency 200π rad/s and are shown in Fig. 1.64.

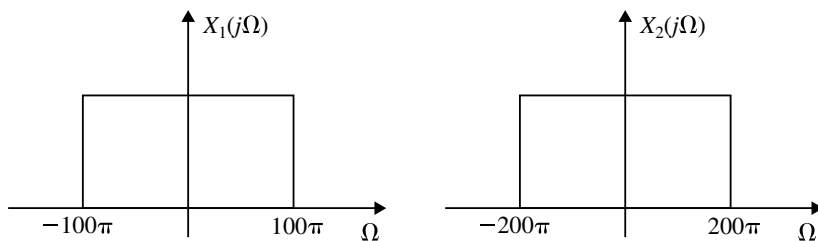
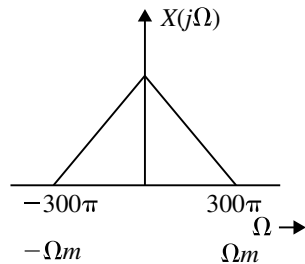
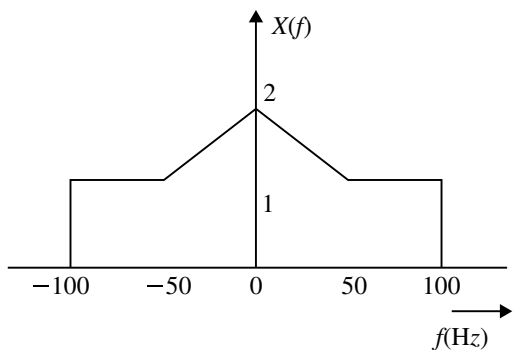
$$F[x_1(t) \cdot x_2(t)] = X_1(j\Omega) * X_2(j\Omega)$$

The convolution of $X_1(j\Omega)$ and $X_2(j\Omega)$ results is a triangular shape frequency spectra with $\Omega_m = 300\pi$ rad/s and is shown in Fig. 1.65.

$$\begin{aligned}
 fm &= \frac{\Omega_m}{2\pi} \\
 fm &= 150 \text{ Hz}
 \end{aligned}$$

$$\begin{aligned}
 \text{Nyquist rate} &= 2fm = 300 \text{ Hz} \\
 \text{Nyquist interval} &= \frac{1}{300} = 3.33 \text{ m s.}
 \end{aligned}$$

Example 1.41 A signal $x(t)$, whose spectrum is shown in Fig. 1.66, is sampled at a rate of 300 samples/s. What is the spectrum of the sampled discrete time signal?

**Fig. 1.64** Spectrum of $x_1(t)$ and $x_2(t)$ **Fig. 1.65** Spectrum of $x(t)$ **Fig. 1.66** Spectrum of $x(t)$ **Solution**

$$f_m = 100 \text{ Hz}$$

$$\text{Nyquist rate} = 2f_m = 200 \text{ Hz}$$

$$\text{Sampling frequency } (f_s) = 300 \text{ Hz}$$

$$f_s > 2f_m.$$

Therefore, there is no aliasing. The sketch of the spectrum of sampled signal is shown in Fig. 1.67.

1.10 Analog to Digital Conversion

A signal that is specified for a continuum of values of time is a continuous time signal and a signal that is specified only at a discrete value of t is a discrete time signal. Continuous time signals are even though often referred to as analog signals, it is not strictly true. Similarly, discrete time signals are not the same as the digital signals. A signal whose amplitude can take on any value in a continuous range is an analog signal. This means an analog signal amplitude can take on an infinite number of values. On the other hand, a digital signal is one whose amplitude can take on only a finite number of values. The signals which are associated with a microprocessor or a digital signal processor or a digital computer are digital signals and they take on only two values (binary). The terms continuous time and discrete time indicate the nature of a signal along the time axis, whereas the terms analog and digital indicates the nature of the amplitude of the signal which is represented in the vertical axis. Thus, the signals are classified as analog continuous time, digital continuous time, analog discrete time and digital discrete time signal. The necessity of converting an analog signal into a digital signal is explained as given below.

The block diagram of a typical DSP system in its simplest form is shown in Fig. 1.68. The analog signal $x(t)$ is applied to ADC that converts into $x(n)$ which is in digital form. The digital signal $x(n)$ is processed by DSP and the output $y(n)$ is obtained. The DAC normally has a built in sample and hold circuit. The DAC processes the signal back into analog form as $y(t)$, which is applied to a continuous time system. Signal processing using a digital processor implies that the input signal $x(n)$ must be in digital form before it can be processed. The analog to digital conversion requires that the signal is first sampled and converted into a discrete time continuous time amplitude signal. The amplitude of each sample is then quantized into one of 2^b levels where b is the number of bits used to represent a sample in

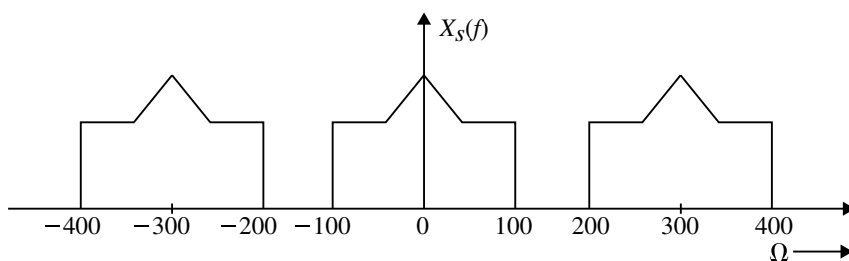


Fig. 1.67 Spectrum of sampled signal



Fig. 1.68 Block diagram of digital signal processing system

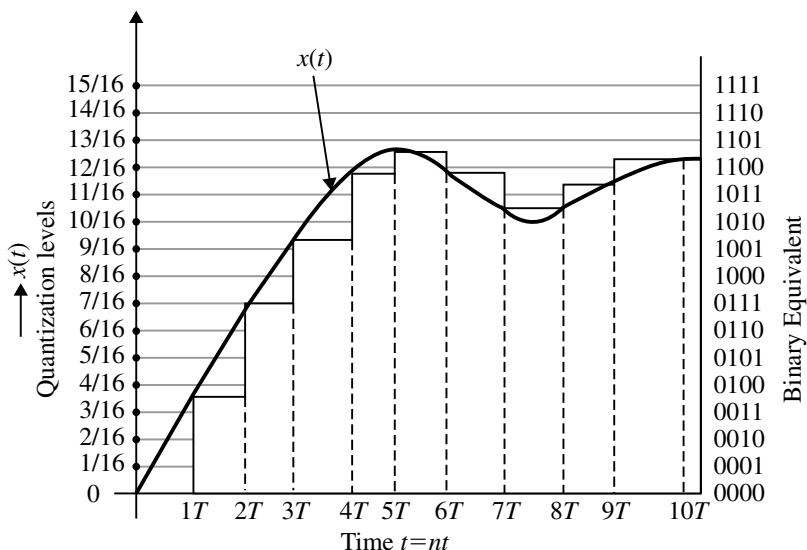


Fig. 1.69 Quantizing analog signal $x(t)$

ADC. The discrete amplitude levels are represented or encoded into distinct binary words each of length ' b ' bits.

1.11 Quantization

Quantization is the process of converting analog signal to digital signal. The signal $x(t)$ is converted to a sampled signal at $t = nT$, where n is an integer and T is the sampling period. The sampling rate is at least twice the bandwidth of the signal which is called the Nyquist sampling rate or else analog signal recovery after it was processed by a digital computer or DSP is not possible without distortion. Figure 1.69 shows the continuous signal $x(t)$ which is to be digitized and quantized. The signal $x(t)$ is sampled at discrete time intervals $t = nT$ and held over the sampling interval by a device called zero order hold circuit (ZOH) that yields staircase approximation to the analog signal. The ZOH circuit keeps its output level equal to the magnitude of an input pulse and then resets itself when a new pulse arrives. On the arrival of the new pulse, the signal jumps to the new height of the continuous signal and holds the output at that level. After sampling and holding, the analog to digital converter converts the sampled signal to a digital number as explained below.

The analog voltage of $x(t)$ is divided into equal discrete levels and each level is assigned a digital number. These levels are called quantization levels. A 16 level discretization is shown in Fig. 1.69. A four bit digital number can represent each of the 16 levels. If v is the maximum value of the analog signal, then each level will represent

$\frac{v}{16}$ v. In general, the difference between any two quantization levels is $\frac{v}{2^b}$ v, where b is the binary bit used for analog to digital conversion. However, the quantization level has to be converted to binary bits because the digital computer or the digital signal processor will accept only binary bits as the input for processing the signal. Thus, it necessitates that each sample amplitude is approximated to the nearest quantization level. Each sample can be represented by pulses. Dealing with a large number of distinct pulses is difficult, and therefore, the use of smallest possible number of distinct pulses is preferable, the number two being the best choice. Thus, we need to assign a distinct electrical pulse to each of the two binary states. In practice, a positive pulse is used to represent the binary 1 and a negative pulse is used to represent a binary 0. Therefore, the sample is represented by a group of four binary impulses for 16 levels of quantization, and thus the binary 0110 represents quantization six.

1.11.1 Quantization Error

The analog signal $x(t)$ is sampled at regular interval to get $x(n)$ at $t = nT$, where $n = 0, 1, 2, \dots$. The amplitude of the sample should be assigned to the nearest quantization level to express it in terms of numeric equivalent $x_q(n)$. This is shown in Fig. 1.70. The difference between $x(n)$, amplitude of the sample signal and $x_q(n)$ the quantized value of $x(n)$ is expressed as

$$e(n) = x_q(n) - x(n)$$

$e(n)$ is called quantization error.

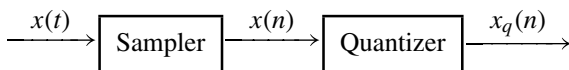
Let a sinusoidal signal amplitude vary from $+1$ to -1 v. Let the ADC use $(b + 1)$ bits which include sign bit. The number of levels available for quantization of $x(n)$ is 2^{b+1} . Therefore, the interval between any quantization level is given by

$$q = \frac{2}{2^{b+1}} = 2^{-b}$$

For $b = 4$, $q = 0.0625$. The sampled signal $x(n)$ is quantized as $x_q(n)$ by the following methods:

- (a) Truncation.
- (b) Rounding.

By truncation, the sampled signal is approximated to the nearest higher quantization level. By rounding the b bit, sampled signal is rounded to the original number unrounded.

Fig. 1.70 Block diagram of sampler and quantizer

1.12 Energy Spectral Density of CT Signals

Consider two energy signals $x(t)$ and $y(t)$. The convolution of these two signals is written as

$$q(t) = x(t) * y(t)$$

Taking Fourier transform on both sides, we get

$$Q(\omega) = X(\omega)Y(\omega)$$

From equation for cross-correlation, we get

$$R_{xy}(\tau) = [x(t) * y(-t)]$$

Taking Fourier transform on both sides, we get

$$\begin{aligned} F[R_{xy}(\tau)] &= F[x(t) * y(-t)] \\ &= X(\omega)Y(-\omega) \end{aligned}$$

For auto-correlation, the above equation becomes

$$\begin{aligned} F[R_{xx}(\tau)] &= X(\omega)X(-\omega) \\ &= |X(\omega)|^2 \end{aligned} \tag{1.52}$$

Equation (1.52) represents Parseval's theorem for Fourier transform which gives the energy of the signal. $|X(\omega)|^2$ gives the energy spectral density and is denoted by $S_{xx}(\omega)$. It gives the energy density of $x(t)$ as the frequency ω is varied. Thus, from Eq. (1.52), the energy spectral density of a CT signal is nothing but the Fourier transform of its auto-correlation function. Thus,

$$\boxed{R_{xx}(\tau) \xleftrightarrow{\text{FT}} S_{xx}(\omega)} \tag{1.53}$$

Equation (1.53) implies that by taking Fourier transform of the auto-correlation function, one can get energy spectral density of $x(t)$. Similarly by taking inverse Fourier transform of energy spectral density, the auto-correlation of $x(t)$ is obtained.

1.13 Power Spectral Density of CT Signals

Periodical signals do not have finite energy for the interval $-\infty < t < \infty$. However, they can be considered as energy signal over a period. Consider the auto-correlation of a signal $x(t)$ whose time period is T .

$$R_{xx}(\tau) = \lim_{T \rightarrow \infty} \frac{1}{T} \int_{-\infty}^{\infty} x(t)x(t-\tau)dt \quad (1.54)$$

Taking Fourier transform on both sides, we get

$$\begin{aligned} F[R_{xx}(\tau)] &= F\left[\lim_{T \rightarrow \infty} \frac{1}{T} \int_{-\infty}^{\infty} x(t)x(t-\tau)dt\right] \\ &= \lim_{T \rightarrow \infty} F\left[\frac{1}{T} \int_{-\infty}^{\infty} x(t)x(t-\tau)dt\right] \\ &= \lim_{T \rightarrow \infty} \frac{1}{T} F[R_{xx}(\tau)] \\ &= \lim_{T \rightarrow \infty} \frac{1}{T} |X(\omega)|^2 \\ &= P_{xx}(\omega) \end{aligned}$$

(1.55)

The power spectrum density which is a function of ω is obtained by taking the Fourier transform of the auto-correlation function of $x(t)$ which is periodical. Conversely, by taking inverse Fourier transform of power spectral density, the auto-correlation function of $x(t)$ is obtained.

Example 1.42 Find the Energy Spectral Density (ESD) and hence the auto-correlation function for the following signal:

$$x(t) = e^{-2t}u(t)$$

Solution

$$\begin{aligned} x(t) &= e^{-2t}u(t) \\ X(\omega) &= F[x(t)] = \frac{1}{j\omega + 2} \\ |X(\omega)| &= \frac{1}{\sqrt{\omega^2 + 4}} \end{aligned}$$

Energy spectral density is given by the following equation:

$$S_{xx} = |X(\omega)|^2$$

$$S_{xx} = \frac{1}{\omega^2 + 4}$$

$$\begin{aligned} R_{xx}(\tau) &= X(\omega)X(-\omega) \\ &= F \left[\frac{1}{(j\omega + 2)} \frac{1}{(-j\omega + 2)} \right] \\ &= F^{-1} \frac{1}{4} \left[\frac{1}{j\omega + 2} + \frac{1}{2 - j\omega} \right] \\ &= F^{-1} \frac{1}{4} \left[\frac{1}{j\omega + 2} - \frac{1}{j\omega - 2} \right] \end{aligned}$$

$$R_{xx}(\tau) = \frac{1}{4} [e^{-2\tau} u(\tau) + e^{2\tau} u(-\tau)]$$

Example 1.43 Find the energy spectral density if the auto-correlation function is given by

$$R_{xx}(\tau) = e^{-|\tau|} \quad \text{for } -\infty < \tau < \infty$$

Solution The energy density spectrum and auto-correlation are related by the following equation:

$$\begin{aligned} S_{xx} &= \int_{-\infty}^{\infty} R_{xx}(\tau) e^{-j\omega\tau} d\tau \\ &= \int_{-\infty}^{\infty} e^{-4|\tau|} e^{-j\omega\tau} d\tau \\ &= \int_{-\infty}^0 e^{+4\tau} e^{-j\omega\tau} d\tau + \int_0^{\infty} e^{-4\tau} e^{-j\omega\tau} d\tau \\ &= \int_{-\infty}^0 e^{(4-j\omega)\tau} d\tau + \int_0^{\infty} e^{-(4+j\omega)\tau} d\tau \\ &= \frac{1}{(4-j\omega)} [e^{(4-j\omega)\tau}]_{-\infty}^0 - \frac{1}{(4+j\omega)} [e^{-(4+j\omega)\tau}]_0^{\infty} \\ &= \frac{1}{(4-j\omega)} + \frac{1}{(4+j\omega)} \end{aligned}$$

$$S_{xx} = \frac{8}{(\omega^2 + 16)}$$

Example 1.44 Find the energy density spectrum of the following discrete time sequence:

$$x[n] = \left(\frac{1}{2}\right)^n u[n]$$

Solution The energy density spectrum of $x[n]$ is the Fourier transform of its auto-correlation function and is related by the following equation:

$$S_{xx} = |X(j\omega)|^2$$

where $X(j\omega)$ is the F.T. of $x[n]$

$$X(j\omega) = \sum_{n=-\infty}^{\infty} \left(\frac{1}{2}\right)^n u[n] e^{-j\omega n}$$

$$\begin{aligned} X(j\omega) &= \sum_{n=0}^{\infty} \left(\frac{1}{2}\right)^n e^{j\omega n} \\ &= \sum_{n=0}^{\infty} \left(\frac{1}{2} e^{j\omega}\right)^n \end{aligned}$$

Using the summation formula, we get

$$\begin{aligned} X(j\omega) &= \frac{1}{\left(1 - \frac{1}{2} e^{j\omega}\right)} \\ &= \frac{1}{(1 - 0.5 \cos \omega - 0.5j \sin \omega)} \\ |X(j\omega)|^2 &= \frac{1}{(1 - 0.5 \cos \omega)^2 + 0.25 \sin^2 \omega} \\ &= \frac{1}{(1.25 - \cos \omega)} \end{aligned}$$

Energy density spectrum

$$S_{xx} = \frac{1}{(1.25 - \cos \omega)}$$

Example 1.45 Consider the following discrete time sequence:

$$x[n] = \cos \frac{n\pi}{3}$$

Determine the power density spectrum.

Solution Given

$$x[n] = \cos \frac{n\pi}{3}$$

where $\omega_0 = \frac{\pi}{3}$. The fundamental period of $x[n]$ is

$$\begin{aligned} N_0 &= \frac{2\pi}{\omega_0} m \\ &= \frac{2\pi}{\pi/3} \\ &= 6 \quad \text{for } m = 1 \end{aligned}$$

The discrete time signal $x[n]$ can be written in the following form using Euler's formula:

$$\cos \frac{n\pi}{3} = \frac{1}{2} e^{j(n\pi/3)} + \frac{1}{2} e^{-j(n\pi/3)} \quad (a)$$

The Fourier series representation of $x[n]$ is given by the following formula:

$$\begin{aligned} x[n] &= \sum_{k=(N_0)} C_k e^{(j2\pi kn)/N_0} \\ &= \sum_{k=-2}^3 C_k e^{(j2\pi kn)/N_0} \\ &= \sum_{k=-2}^3 C_k e^{(j\pi kn)/3} \end{aligned}$$

$$x[n] = C_{-2} e^{-(2j\pi n)/3} + C_{-1} e^{-(j\pi n)/3} + C_0 + C_1 e^{(j\pi n)/3} + C_2 e^{(2j\pi n)/3} + C_3 e^{j\pi n} \quad (b)$$

Comparing equations (a) and (b), we get

$$C_1 = \frac{1}{2} \quad \text{and} \quad C_{-1} = \frac{1}{2}$$

with other coefficients being zero. Power spectral density is given by

$$P_{ss} = \sum_{k=0}^{N_0-1} |C_k|^2 = \left(\frac{1}{2}\right)^2 + \left(\frac{1}{2}\right)^2$$

$$\boxed{P_{xx} = \frac{1}{2}}$$

Example 1.46 Consider the following signal:

$$x[n] = \cos \frac{\pi}{5}n + \sin \frac{\pi}{6}n$$

Determine the power density spectrum.

Solution Given

$$x[n] = \cos \frac{\pi}{5}n + \sin \frac{\pi}{6}n$$

where

$$\begin{aligned}\omega_{01} &= \frac{\pi}{5} \\ \omega_{02} &= \frac{\pi}{6} \\ N_{01} &= \frac{2\pi}{\omega_{01}} = 10 \\ N_{02} &= \frac{2\pi}{\omega_{02}} = 12\end{aligned}$$

The periodicity of the combined signal is obtained by taking LCM of 10 and 12, which is 60.

$$\begin{aligned}N_0 &= 60 \\ \omega_0 &= \frac{2\pi}{N_0} = \frac{\pi}{30}\end{aligned}$$

Using Euler's formula, the given signal is expressed as

$$x[n] = \frac{1}{2} [e^{j(\pi/5)n} + e^{-j(\pi/5)n}] + \frac{1}{2j} [e^{j(\pi/6)n} - e^{-j(\pi/6)n}]$$

The above sequence can be expressed in terms of fundamental frequency as

$$x[n] = \frac{1}{2} [e^{j6\omega_0 n} + e^{-j6\omega_0 n}] + \frac{1}{2j} [e^{j5\omega_0 n} - e^{-j5\omega_0 n}]$$

From the above equations, the Fourier series coefficients are obtained as

$$D_6 = \frac{1}{2}; \quad D_{-6} = \frac{1}{2}; \quad D_5 = \frac{1}{2j}; \quad D_{-5} = \frac{1}{2j}$$

The power spectrum density is given by

$$\begin{aligned}P_{xx} &= |D_6|^2 + |D_{-6}|^2 + |D_5|^2 + |D_{-5}|^2 \\ &= \frac{1}{4} + \frac{1}{4} + \frac{1}{4} + \frac{1}{4}\end{aligned}$$

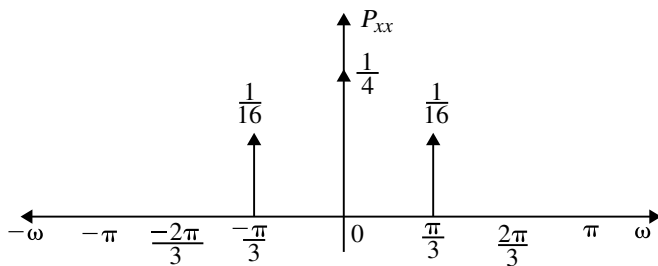


Fig. 1.71 Frequency spectrum of Example 1.47

$$P_{xx} = 1$$

Example 1.47 Find the power spectrum density for the following DT signal:

$$x[n] = \sin^2\left(\frac{\pi}{6}n\right)$$

Sketch the frequency spectrum (Fig. 1.71).

Solution

$$\begin{aligned}\sin^2\left(\frac{\pi}{6}n\right) &= \frac{1}{2} - \frac{1}{2} \cos \frac{\pi n}{3} \\ \omega_0 &= \frac{\pi}{3} \\ N_0 &= \frac{2\pi m}{6} \\ N_0 &= 6 \quad \text{for } m = 1\end{aligned}$$

Using Euler's formula, we can write

$$\begin{aligned}\left[\sin \frac{\pi}{6}n\right]^2 &= \left[\frac{1}{2j} (e^{j(\pi/6)n} - e^{-j(\pi/6)n})\right]^2 \\ &= \frac{1}{2} - \frac{1}{4} [e^{j(\pi/3)n} + e^{-j(\pi/3)n}] \\ &= \frac{1}{2} - \frac{1}{4} [e^{j\omega_0 n} + e^{-j\omega_0 n}] \\ D_0 &= \frac{1}{2} \\ D_1 &= -\frac{1}{4} \\ D_{-1} &= D_{-1+6} = D_5 = -\frac{1}{4}\end{aligned}$$

The frequency spectrum of Example 1.47 is shown in Fig. 1.71

The power density function is given by

$$\begin{aligned} P_{xx} &= |D_0|^2 + |D_1|^2 + |D_5|^2 \\ &= \frac{1}{4} + \frac{1}{16} + \frac{1}{16} \\ &= \frac{3}{8} \end{aligned}$$

1.13.1 Properties of Power Spectral Density

1. The total area under the power spectrum density curve equals the average value of the signal $x(t)$.

Proof: From Eq. (1.53) we have,

$$\begin{aligned} R_{xx}(\tau) &\xleftrightarrow{\text{FT}} P_{xx}(\omega) \\ F^{-1}[P_{xx}(\omega)] &= R_{xx}(\tau) \\ &= \int_{-\infty}^{\infty} P_{xx}(\omega) e^{j\omega\tau} d\omega \end{aligned}$$

For $\tau = 0$,

$$R_{xx}(0) = \int_{-\infty}^{\infty} P_{xx}(\omega) d\omega$$

From Eq. (1.54), the average power is obtained for $\tau = 0$, as

$$\begin{aligned} R_{xx}(0) &= \lim_{T \rightarrow \infty} \frac{1}{T} \int_{-\infty}^{\infty} |x(t)|^2 dt \\ &= \text{average power} \\ \int_{-\infty}^{\infty} P_{xx}(\omega) d\omega &= R_{xx}(0) \\ &= \text{Area under the power spectrum density curve} \end{aligned}$$

2. The power spectral density of a real valued power signal $x(t)$ is an even function of frequency which satisfies

$$P_{xx}(-\omega) = P_{xx}(\omega)$$

Proof: Let $x_T(t)$ be the segment of $x(t)$ which is periodic with period T . It is defined as follows:

$$x_T(t) = \begin{cases} x(t), & |t| \geq T/2 \\ 0, & \text{otherwise} \end{cases}$$

The power density of a periodical signal is given by

$$P_{xx}(\omega) = \lim_{T \rightarrow \infty} \left[\frac{|x_T(\omega)|^2}{T} \right]$$

$$P_{xx}(-\omega) = \lim_{T \rightarrow \infty} \left[\frac{|x_T(-\omega)|^2}{T} \right]$$

$$P_{xx}(-\omega) = \lim_{T \rightarrow \infty} \left[\frac{|x_T(\omega)|^2}{T} \right]$$

$$= P_{xx}(\omega)$$

Hence, the power density function is an even function.

- 3. Power density** $P_{xx}(\omega)$ of a signal $x(t)$ is the Fourier transform of the auto-correlation function of $x(t)$.

Proof: The proof is given in Eq. (1.55).

- 4. Power spectral density** $P_{xx}(\omega)$ of a signal $x(t)$ is always non-negative.

Proof: From Eq. (1.55), we get

$$P_{xx}(\omega) = \lim_{T \rightarrow \infty} \frac{1}{T} |X(\omega)|^2$$

Since $|X(\omega)|^2$ is always positive, $P_{xx}(\omega)$ is non-negative.

1.14 Recursive Systems

A discrete time system with the N th order linear constant coefficient can be described by the following difference equation:

$$\sum_{k=0}^N a_k y(n-k) = \sum_{k=0}^M b_k x(n-k) \quad (1.56)$$

the above equation is written in the following form as:

$$y(n) = \frac{1}{a_0} \left[\sum_{k=0}^M b_k x(n-k) - \sum_{k=1}^N a_k y(n-k) \right] \quad (1.57)$$

Equation (1.57) gives the value of the output $y(n)$ at any time n in terms of the previous values of the input and output. A discrete time system described by dif-

ference equation of the type given in Eq. (1.57), where the output $y(n)$ is fed back and compared with the input, is called a **Recursive System**. In such systems, $y(n-1)$, $y(n-2)$, $\dots$, $y(n-N)$ need to be known to calculate $y(n)$. Therefore, if the input for all n together with a set of auxiliary conditions $y(-N)$, $y(-N+1)$, $\dots$, $y(-1)$, Eq. (1.57) can be solved for $y(n)$. Equation (1.57) is called a recursive equation because it requires a recursive procedure for determining the output in terms of the input and the previous outputs.

Now consider the case when $N = 0$. Equation (1.57) reduces to the following form:

$$y(n) = \sum_{k=0}^M \left(\frac{b_k}{a_0} \right) x(n-k) \quad (1.58)$$

Equation (1.58) gives the output $y(n)$ in terms of the present and previous values of the input. Discrete time systems described by difference equation (1.58) is called **Non-Recursive System**. Here we do not use the previously computed values of the output recursively, the system described by Eq. (1.58) is called a non-recursive system. Further, for a non-recursive system, we do not require auxiliary conditions to determine the output $y(n)$ at any time n .

Equation (1.58) also describes the LTI system whose impulse response is given as

$$h(n) = \begin{cases} \frac{b_n}{a_0}, & 0 \leq n \leq M \\ 0, & \text{otherwise} \end{cases} \quad (1.59)$$

From Eq. (1.59), it is evident that the impulse response is non-zero only over a finite interval. Hence system described by Eq. (1.58) is called **Finite Impulse Response (FIR) System**. On the other hand, the impulse response of equation (1.57) is infinite length and hence it is also called **Infinite Impulse Response (IIR) System**.

Example 1.48 Determine the output of the recursive system described by the following difference equation:

$$y(n+2) + 3y(n+1) + 2y(n) = u(n)$$

where $y(0) = 0$; $y(1) = 1$ and $u(n)$ is unit step response.

Solution

$$y(n+2) + 3y(n+1) + 2y(n) = u(n)$$

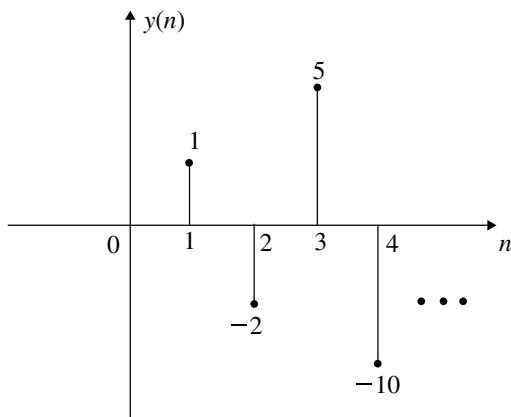
For $n = 0$,

$$y(2) + 3y(1) + 2y(0) = u(0)$$

$$y(2) + 3 = 1$$

$$y(2) = -2$$

Fig. 1.72 Output response of Example 1.48



For $n = 1$,

$$y(3) + 3y(2) + 2y(1) = u(1)$$

$$y(3) = 5$$

For $n = 2$,

$$y(4) + 3y(3) + 2y(2) = u(2)$$

$$y(4) = -10$$

the output response $y(n)$ is sketched in Fig. 1.72.

Summary

- Signals are broadly classified as continuous time (CT) and discrete time (DT) signals. They are further classified as deterministic and stochastic, periodic and non-periodic, odd and even and energy and power signals.
- Basic DT signals include impulse, step, ramp, parabolic, rectangular pulse, triangular pulse, signum function, sinc function, sinusoid, real and complex exponentials.
- Basic operations on DT signals include addition, multiplication, amplitude scaling, time scaling, time shifting, reflection or folding and amplitude inverted signals.
- In time shifting of DT signal, for $x(n + n_0)$ and $x(-n - n_0)$, the time shift is made to the left of $x(n)$ and $x(-n)$ respectively by n_0 . For $x(n - n_0)$ and $x(-n + n_0)$, the time shift is made to the right of the $x(n)$ and $x(-n)$ respectively by n_0 .

- To plot DT signals, the operation performed is in the following sequence. The signal is folded (if necessary), time shifted, time scaled, amplitude scaled and inverted.
- Signals are classified as even signals and odd signals. Even signals are **sym-metric** about the vertical axis, whereas odd signals are **anti-symmetric** about the time origin. Odd signals pass through the origin. The product of two even signals or two odd signals is an even signal. The product of an even and an odd signal is an odd signal.
- A DT signal which repeats itself every N sequence is called a periodic signal. If the signal is not periodic, it is called an aperiodic or non-periodic signal. The necessary condition for the composite of two or more signals to be periodic is that the individual signal should be periodic.
- A signal is an energy signal if the total energy of the signal satisfies the condition $0 < E < \infty$. A signal is called a power signal if the average power of the signal satisfies the condition $0 < P < \infty$. If the energy of a signal is finite, the average power is zero. If the power of the signal is finite, the signal has infinite energy. All periodic signals are power signals. However, all power signals need not be periodic. Signals which are deterministic and non-periodic are usually energy signals. Some signals are neither energy signals nor power signals.
- The system is broadly classified as a continuous and discrete time system.
- The DT systems are further classified based on the property of causality, linearity, time invariance, invertibility, memory and stability.
- A discrete time system is said to be causal if the impulse response $h(n) = 0$ for $n < 0$.
- If the impulse response of a discrete time system is absolutely summable, then the system is said to be BIBO stable.
- Step response $s(n)$ is obtained from impulse response $h(n)$ using the mathematical expression

$$s(n) = \sum_{k=0}^n h(k)$$

Short Questions and Answers

1. **How are signals classified?** signals are generally classified as CT and DT signals. They are further classified as deterministic and non-deterministic, odd and even, periodic and non-periodic and power and energy signals.
2. **What are odd and even signals?**
A continuous CT signal is said to be an even signal if it satisfies the condition $x(-t) = x(t)$ for all t . It is said to be an odd signal if $x(-t) = -x(t)$ for all t . For a DT signal if $x[-n] = x[n]$ condition is satisfied, it is an even sequence (signal). If $x[-n] = -x[n]$, the sequence is called an odd sequence.
3. **How even and odd components of a signal are mathematically expressed?**

$$x_e[n] = \frac{1}{2}\{x[n] + x[-n]\}$$

$$x_o[n] = \frac{1}{2}\{x[n] - x[-n]\}$$

4. What are periodic and non-periodic signals?

A discrete time signal is said to be a period signal if it satisfies the condition $x[n] = x[n + N]$ for all n . A signal which is not periodic is said to be non-periodic.

5. What is the fundamental period of a periodic signal? What is the fundamental frequency?

A DT signal is said to be periodic if it satisfies the condition $x(n) = x(n + N)$. If this condition is satisfied for $N = N_0$, it is also satisfied for $N = 2N_0, 3N_0, \dots$. The smallest value of N that satisfies the above condition is called the fundamental period. The fundamental frequency $f_0 = \frac{1}{N_0}$ Hz. It is also expressed as $\omega_0 = \frac{2\pi}{N_0}$ rad.

6. What are power and energy signals?

For a DT signal $x[n]$, the total energy is defined as

$$E = \sum_{n=-\infty}^{\infty} x^2[n]$$

The average power is defined as

$$P = \lim_{T \rightarrow \infty} \frac{1}{2N + 1} \sum_{n=-N}^N x^2[n]$$

7. Determine whether the signal $x[n] = \cos[0.1\pi n]$ is periodic?

The signal $x[n]$ is periodic with fundamental period $N_0 = 20$.

8. Find whether the signal $x[n] = 5 \cos[6\pi n]$ is periodic?

The signal is periodic with fundamental period $N_0 = 1$.

9. Find the average power of the signal.

$$x[n] = u[n] - u[n - N]$$

The average power $P = 1$.

10. Find the total energy of

$$x[n] = \{1, 1, 1\}$$

↑

The total energy $E = 3$.

11. If the discrete time signal $x[n] = \{0, 0, 0, 3, 2, 1, -1, -7, 6\}$ then find $y[n] = x[2n - 3]$?

$$y[n] = \{0, 0, 0, 3, 1, -7\}$$

12. What is the energy of the signal $x[n] = u[n] - u[n - 6]$?

$$E = 6$$

13. What are the properties of systems?

Systems are generally classified as continuous and discrete time systems. Further classifications of these systems are done based on their properties, which include

- (a) linear and non-linear,
- (b) time invariant and time variant,
- (c) static and dynamic,
- (d) causal and non-causal,
- (e) stable and unstable and
- (f) Invertible and non-invertible.

14. Define system. What is linear system?

A system is defined as the interconnection of objects with a definite relationship between objects and attributes.

A system is said to be linear if the weighted sum of several inputs produces a weighted sum of outputs. In other words, the system should satisfy the homogeneity and additivity of the superposition theorem if it is to be linear. Otherwise, it is a non-linear system.

15. What is time invariant and time variant system?

A system is said to be time invariant if the output due to the delayed input is the same as the delayed output due to the input. If the continuous time system is described by the differential equation, its coefficients should be time independent for the system to be time invariant. In the case of discrete time system, the coefficients of the difference equation describing the system should be time independent (constant) for the system to be time invariant. If the above conditions are not satisfied, the system (CT as well as DT) is said to be time variant.

16. What are static and dynamic systems?

If the output of the system depends only on the present input, the system is said to be static or instantaneous. If the output of the system depends on the past and future input, the system is not static and it is called a dynamic system. A static system does not require memory, and so it is called a memoryless system. A dynamic system requires memory, and hence it is called a system with memory. Systems which are described by differential and difference equations are dynamic systems.

17. What are causal and non-causal systems?

If the system output depends on present and on past inputs, it is called a causal

system. If the system output depends on future input it is called a non-causal system.

18. What are stable and unstable systems?

If the input is bounded and output is also bounded, the system is called BIBO stable system. If the input is bounded and the output is unbounded, the system is unstable. A system whose impulse response curve has a finite area is also called a stable system.

19. What are invertible and non-invertible systems?

A system is said to be invertible if the distinct inputs give distinct outputs.

20. State the condition for a discrete time LTI system to be causal and stable.

A discrete time LTI system is said to be causal and stable if the poles of the transfer function all lie in the left half s -plane and the Region of Convergence (ROC) is to the right of the right-most pole.

21. A certain LTID time system has the following impulse response.

$$h[n] = \left(\frac{1}{3}\right)^n u[n - 2]$$

Is the system both causal and stable?

The response depends on the past input $u[n - 2]$, and hence it is causal.

$$\begin{aligned} h[n] &= \sum_{n=2}^{\infty} \left(\frac{1}{3}\right)^n \\ &= \frac{1}{6} < \infty \end{aligned}$$

The system is stable. The system is both causal and stable.

22. What is convolution sum?

If $h(n)$ is the impulse response of a linear time invariant discrete time system and if $x(n)$ is input, then the output $y(n)$ is obtained from the following equation:

$$y(n) = \sum_{k=-\infty}^{\infty} x(k)h(n - k)$$

The above equation is called convolution sum. Symbolically, it is represented as

$$y(n) = x(n) * h(n)$$

23. Evaluate $x(n) * \delta(n)$ and $x(n) * \delta(n - n_0)$?

$$\begin{aligned} x(n) * \delta(n) &= x(n) \\ x(n) * \delta(n - n_0) &= x(n - n_0) \end{aligned}$$

24. If $x(n) = \delta(n) - 2\delta(n-1) + 3\delta(n-2) - \delta(n+1)$. Express $x(n-1)$ in terms of sequences?

$$x(n-1) = \delta(n-1) - 2\delta(n-2) + 3\delta(n-3) - \delta(n)$$

25. $h(n) = \left(\frac{1}{2}\right)^n u(n)$. Find A so that $h(n) - Ah(n-1) = \delta(n)$?

$$h(n) - h(n-1) = \left(\frac{1}{2}\right)^n u(n) - A \left(\frac{1}{2}\right)^{n-1} u(n-1)$$

If $A = 1/2$

$$h(n) - h(n-1) = \left(\frac{1}{2}\right)^n u(n) - \left(\frac{1}{2}\right)^n u(n-1) = \delta(n)$$

26. An LTID system has the following impulse response

$$h(n) = \left(\frac{1}{2}\right)^n u(n+1)$$

Is the system causal and BIBO stable?

The system is non-zero for $n = -1$. Hence, it is non-causal

$$\begin{aligned} \sum_{-\infty}^{\infty} h(n) &= \sum_{-1}^{\infty} \left(\frac{1}{2}\right)^n \\ &= \left(\frac{1}{2}\right)^{-1} + \sum_0^{\infty} \left(\frac{1}{2}\right)^n \\ &= 2 + \frac{1}{1 - 1/2} = 4 < \infty \end{aligned}$$

The system is BIBO stable.

Long Answer Type Questions

1. For the following DT signal, find even and odd components:

$$x[n] = \{1, -3, 2, 5, 4\}$$

$$x_e[n] = \{2, 2.5, 1, -1.5, 1, -1.5, 1, 2.5, 2\}$$

↑

$$x_0[n] = \{-2, -2.5, -1, 1.5, 0, -1.5, 1, 2.5, 2\}$$

↑

2. **Find whether the following signals are periodic. If periodic, find the fundamental period:** (a) $x[n] = \cos(\frac{n}{8} - \pi)$; (b) $x[n] = \cos(\frac{\pi}{8} + \frac{\pi}{2}) + \cos(\frac{\pi}{6} - \frac{\pi}{2})$; (c) $x[n] = \cos(\frac{5\pi N}{12} + \frac{\pi}{2}) + \sin \frac{10\pi n}{8}$; (d) $x[n] = e^{j3n} + e^{j4\pi n}$.
 (a) Not periodic. (b) Periodic with fundamental period $N_0 = 48$ samples/s. (c) Periodic with fundamental period $N_0 = 24$ samples/s. (d) Non-periodic.
3. **Given** $x[n]$ and $y[n]$

$$\begin{aligned} x[-1] &= 2 \\ x[n] &= 1 \quad 1 \leq n \leq 5 \\ x[6] &= \frac{1}{2} \\ &= 0 \quad \text{for other } n \end{aligned}$$

Plot (a) $x[\frac{n}{2}]$ and (b) $E_v x[n]$.

(a)

$$\begin{aligned} x[n] &= \{2, 1, 1, 1, 1, 1, 1, 0.5\} \\ &\quad \uparrow \\ x\left[\frac{n}{2}\right] &= \{2, 0, 1, 0, 1, 0, 1, 0, 1, 0, 1, 0, 1, 0.5\} \\ &\quad \uparrow \end{aligned}$$

(b)

$$\begin{aligned} x_e[n] &= \{0.25, 0.5, 0.5, 0.5, 0.5, 0.5, 1, 0.5, 0.5, 0.5, 0.5, 0.5, 0.25\} \\ &\quad \uparrow \end{aligned}$$

4. **Find whether the following signal is periodic. If periodic, find the fundamental period**

$$x[n] = \cos\left(2\pi n + \frac{\pi}{2}\right) \sin\left(5\pi n - \frac{\pi}{4}\right) \sin\left(8\pi n + \frac{\pi}{2}\right)$$

The signal is periodic. Their fundamental period $N_0 = 2$ samples/s.

5. **For the systems given below, determine whether each of them is** (a) static, (b) causal, (c) time invariant, (d) linear and (e) stable.

$$y[n] = x[5n]$$

- (a) The system response depends on present, past and future inputs. Hence, it is dynamic.
 (b) Non-causal.
 (c) The output due to the delayed input is not the same as the delayed output. Hence, it is time variant.

- (d) The weighted sum of the output is the same as the output due to the weighted sum of the input. The system is linear.
- (e) If the input $x[5n]$ is bounded, the output $y[n]$ is also bounded. The system is stable.

6.

$$y[k + 2] + 3y[k + 1] + 4y[k] = x[k]$$

- (a) The system is dynamic.
- (b) The system is causal.
- (c) The system is time invariant.
- (d) The system is linear.
- (e) The system is stable.

7.

$$y[n] = 5x[3^n]$$

- (a) The system is dynamic.
- (b) The system is non-causal.
- (c) The system is time invariant.
- (d) The system is linear.
- (e) The system is stable.

8.

$$y[n] = \sin(2\pi x[n]) + x[n + 1]$$

- (a) The system is dynamic.
- (b) The system is non-causal.
- (c) The system is time invariant.
- (d) The system is non-linear.
- (e) The system is stable.

9. $x[n] = u[n - 4] - u[n - 10]$ and $h[n] = u[n - 5] - u[n - 16]$.
Find $y[n] = x[n] * h[n]$?

$$\begin{aligned} y[n] &= (n - 8) & 9 \leq n \leq 13 \\ &= 6 & 10 \leq n \leq 19 \\ &= (25 - n) & 20 \leq n \leq 24 \\ &= 0 & n > 24 \end{aligned}$$

10. $x[n] = 4^n u[-n - 2]$ and $h[n] = u[n - 2]$. Find $y[n] = x[n] * h[n]$?

$$\begin{aligned}
 y[n] &= \frac{1}{3} \left[\frac{1}{4} \right]^{n-1} & n < 0 \\
 &= \frac{1}{12} & n > 0
 \end{aligned}$$

11. Determine whether the following LTID time systems whose impulse response given below are stable. (a) $h[n] = n \sin 2\pi n u[n]$, (b) $h[n] = 5^n u[-n]$ and (c) $h[n] = 2^{-n} u[n - 5]$?

(a) $y[n] = \infty$ B.I.B.O. unstable.

(b) $y[n] = \frac{1}{4} < \infty$ B.I.B.O. stable.

(c) $y[n] = \frac{1}{6} < \infty$ B.I.B.O. stable.

12. $x[n] = e^{-2n} u[n]$ and $h[n] = 3^{-n} u[n]$. Find $y[n] = x[n] * h[n]$?

$$y[n] = \frac{[3e^2 - (3e^2)^{-n}]u[n]}{[3e^2 - 1]}$$

13. $x[n] = (0.5)^n u[n]$ and $h[n] = (0.9)^n u[n]$. Find $y[n] = x[n] * h[n]$?

$$y[n] = 2.5[0.9^{n+1} - 0.5^{n+1}]u[n]$$

14. Determine the power and energy of the following signals:

(a) $x[n] = \left(\frac{1}{3}\right)^n$.

(b) $x[n] = e^{j\left(\frac{n\pi}{2} + \frac{\pi}{4}\right)}$.

(c) $x[n] = \sin\left(\frac{n\pi}{4}\right)$.

(d) $x[n] = e^{2n} u(n)$.

Ans:

- (a) $E = \frac{9}{8} J; P = 0.$
 (b) $E = \infty; P = 1 W.$
 (c) $E = \infty; P = \frac{1}{2} W.$
 (d) $E = \infty; P = \infty$ (Neither power nor energy).

15. Determine whether each of the following signals is periodic. Find the fundamental period

- (a) $x[n] = e^{j6\pi n}.$
 (b) $x[n] = e^{j\frac{3}{5}(n+\frac{1}{2})}.$
 (c) $x[n] = \cos\left(\frac{2\pi}{3}\right)n.$
 (d) $x[n] = \cos \frac{n\pi}{3} + \cos \frac{3n\pi}{4}.$

Ans:

- (a) Periodic. $N = 1$ sample
 (b) Not Periodic.
 (c) Periodic. $N = 3$ samples
 (d) Periodic. $N = 24$ samples

16. Find the odd and even components of the following sequence.

$$x(n) = \{2, 3, 4, 5, 6\}$$

↑

Ans:

$$x_e(n) = \{4, 4, 4, 4, 4\}$$

↑

$$x_o(n) = \{-2, -1, 0, 1, 2\}$$

17. Test whether the following systems are time variant or time invariant:

- (a) $y(n) = x(n) - x(n - 1).$
 (b) $y(n) = nx(n).$
 (c) $y(n) = x(-n).$

- (d) $y(n) = x(n) \cos \omega_0 n$.
- (e) $y(n) = x(n) - bx(n-1)$.
- (f) $y(n) = x(n) + c$.
- (g) $y(n) = e^{x(n)}$.
- (h) $y(n) = nx^2(n)$.
- (i) $y(n) = \sum_{k=0}^m a_k(n-k) - \sum_{k=1}^m b_k y(n-k)$.

Ans:

- (a) Time invariant.
- (b) Time varying.
- (c) Time varying.
- (d) Time varying.
- (e) Time varying.
- (f) Time invariant.
- (g) Time invariant.
- (h) Time varying.
- (i) Time invariant.

18. Test the linearity of the following systems:

- (a) $y(n) = nx(n)$.
- (b) $y(n) = x(n^2)$.
- (c) $y(n) = x^2(n)$.
- (d) $y(n) = Ax(n) + B$.
- (e) $y(n) = e^{x(n)}$.

Ans:

- (a) Linear.
- (b) Linear.
- (c) Non-linear.
- (d) Non-linear.
- (e) Non-linear.

19. Test the causality of the following systems:

- (a) $y(n) = x(n) - x(n-1)$.
- (b) $y(n) = \sum_{k=-\infty}^n x(k)$.
- (c) $y(n) = nx(n)$.

Ans:

- (a) Causal.
- (b) Causal.
- (c) Causal.

20. Test whether the following systems are stable or not:

- (a) $y(n) = x(-n-2)$.
- (b) $y(n) = x(n) + by(n-1)$.

$$(c) \ y(n) = \begin{cases} a^n & n \leq 0, \\ b^n & n < 0. \end{cases}$$

Ans:

(a) Stable.

(b) Stable if $|b| < 1$.

(c) Stable if $|a| < 1$ and $|b| > 1$.

21. **Determine the response of the relaxed system characterized by the impulse response $h(n) = (\frac{1}{2})^n u(n)$ to the input signal $x(n) = 2^n u(n)$.**

Ans:

$$y(n) = \left(\frac{1}{2}\right)^n \left(\frac{4^{n+1} - 1}{3}\right) u(n)$$